

# Global Strategy Brief – 2019

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## Global Commercial Aircraft Engines Manufacturers

Strategy Focus,  
Key Strategies &  
Plans, Trends &  
Growth  
Opportunities,  
Market Outlook

Noealt Corporate  
Services

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February 2019



Strategic Perspective..Decisive Edge..



A close-up, high-contrast photograph of a GE9X aircraft engine fan. The fan is circular with numerous blades radiating from a central hub. The lighting creates strong highlights and shadows, emphasizing the metallic texture and the complex geometry of the blades. The background is dark, making the engine the central focus.

# Global Strategy Brief – 2019 - Global Commercial Aircraft Engine Manufacturers – Strategy Focus, Key Strategies & Plans, Trends & Growth Opportunities, Market Outlook

# GE9X

The world's next great engine.

February 2019

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GE Aviation

Scheduled to enter service in 2020 with a backlog of 700 engines, it will also be the most fuel-efficient engine GE has ever produced on a per-pounds-of-thrust basis.

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Strategy Focus & Priorities

First engine tested in March 2016

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Business Outlook for 2019



The largest fan diameter in commercial aviation



8db  
margin to Stage 5  
noise regulations

5% better  
specific fuel  
consumption (SFC)  
than any other twin-aisle  
engine in service in 2020

30% NOx  
margin to CAEP  
8 regulations

10%  
lower fuel burn  
than the GE90-115B

27:1 compressor  
pressure ratio  
The highest ever in commercial aviation



Safran SA

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# Strategy Focus



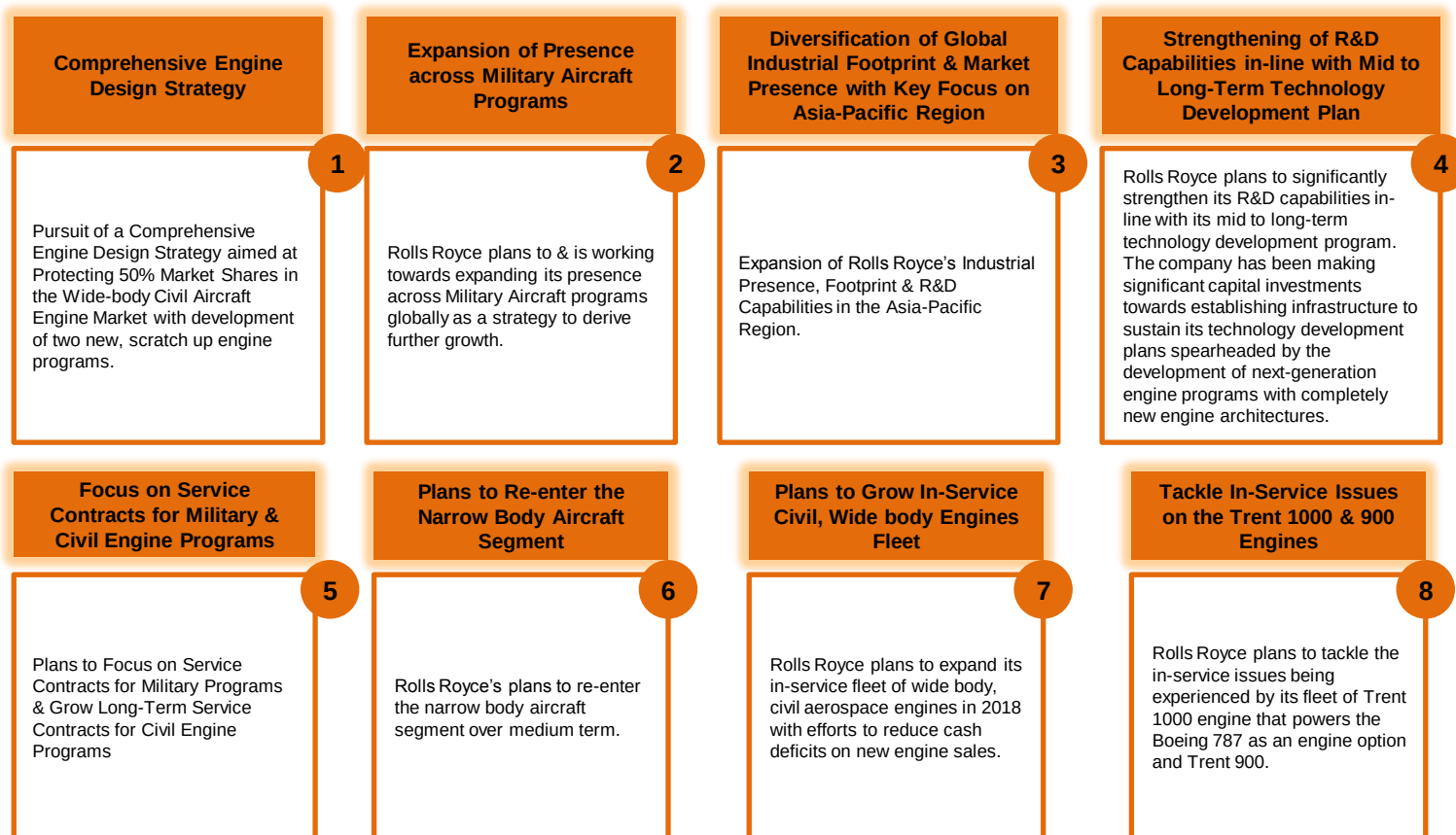
## Part 1.1



# Rolls Royce – Overall Strategy Focus - 1



Rolls-Royce®



7  
Enhancing of Focus on Core Strengths, namely,

- Engineering Excellence
- Operational Excellence
- Capturing Aftermarket Value leveraging Installed Engine Base, especially for the Defense Aerospace Segment.

8  
Ensuring Profitable & Cash Generative Growth over Long Term.

9  
Laying the Foundation for Strong, Long Term Growth Prospects through the Pursuit of Structural Business Transformation Program.

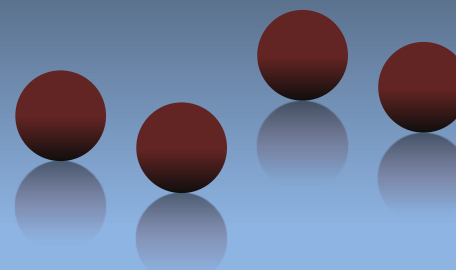
10  
Ramp-Up of Civil Engine Production to Deliver on Growing Backlog and in light of Visible Long-Term Positive Demand Growth Trend (with Global RPK projected to double over next 15 years).

11  
Strong R&D Focus & Investments geared towards Development of Products for Next Generation Transport, Patrol & Combat Aircraft Platforms in the Defense Aerospace segment.

12  
Strengthening of Global MRO Network and Expansion of Services Portfolio for the Defense Aerospace Segment aimed at Capturing Aftermarket Value.



# Key Strategies & Plans



## ■ Part 1.2

## Pursuit of a Comprehensive Restructuring Plan aimed at Achieving Reduction in Fixed Costs

Rolls Royce is pursuing a comprehensive restructuring plan aimed at achieving significant reduction in fixed costs amid ongoing significant pressures on profitability. The restructuring plan is spearheaded by the new CEO Warren East who has lambasted Rolls Royce's high fixed cost base & the company's inflexibility to industry & demand downturns leading to high sensitivity to market led revenue changes. The plan has been outlined as part of the CEO's review of operations. The latest move follows on to the group's earlier outlined 4C strategy focusing on Customers, Concentration, Cost & Cash. The company plans to implement latest set of restructuring measures under the comprehensive restructuring programme which has been in effect since 2016. The restructuring programme aims to generate significant cost savings to the tune of £150-200 million per annum. Rolls Royce also initiated a new business unit simplification & restructuring programme in January 2018 aimed at further rationalization of cost base and managing profitability (civil aerospace segment registered an operating loss of GBP 330 million for 2017).

Key areas & measures covered & outlined in the restructuring programme include:

- Implementation of a major programme aimed at organisational redesign with focus on simplification of organisation model
- Streamlining of management structure with core focus on senior management
- Implementation of further measures to reduce fixed costs which have included reduction of headcount by over 3,600 announced over the past 18 months
- Addition of greater pace & accountability to overall decision-making processes
- Plans to streamline & enhance effectiveness of organisational management information, forecasting & business systems

The restructuring programme is aimed at industrial & fundamental transformation targeted at paving the way for achieving long-term growth. Rolls Royce's focus as part of the restructuring programme remains on:-

- Improving financials
- Building resilience for short term demand fluctuations
- Building in flexibility to capitalize on growth opportunities

The new, realigned business structure being pursued in 2018 (with a 3 unit structured business portfolio with plans for exit from commercial marine business) is aimed at driving long term, sustainable free cash flow with Rolls Royce targeting a free cash flow generation target of GBP 1 billion by 2020 with further growth being targeted over the early part of next decade.

### **Tackling In-Service Issues on Trent 900 & Trent 1000 a Key Priority for Rolls Royce over Near Term as Pressures on Profitability Loom**

Rolls Royce has been cornered in a kind of tough spot with the coming to surface of in-service engine issues over the Trent 900 & Trent 1000 programs over lesser than expected durability of some engine parts as the British aerospace giant is entangled in frantic efforts & faces pressure to reduce cost base and boost profitability. The Trent 1000 engine, which powers the Boeing 787 as an engine option, faces performance issues due to faster than expected wear and tear of some engine parts leading to disruption in service across operators. Rolls Royce has created its short term intervention strategy with on wing support and shop visits. The same has cost Rolls Royce around GBP 400 million in 2017 towards accelerated maintenance activity over these two engine programs with the same cash cost likely to double in 2018 with maintenance activity spiking up. Rolls Royce plans to complete this unscheduled maintenance activity by 2022 through redesigning of engine parts which are going to impact cash flow positions & profitability over this period. For H1 2018, Rolls Royce booked charges worth £554 million towards the maintenance activity over these two engine programs with Rolls Royce booking a £1.3 billion loss for the first half of 2018 as compared against a net profit of £1.4 billion for H1 2017.

Rolls Royce has outlined in its latest estimates that the unexpected maintenance activity over these two engine programs is likely to cost the company over £1 billion over the next 3 years with the costs estimated at £450 million for 2018 and 2019 each and pegged at £350 million in 2020. The same is likely to take a toll on Rolls Royce's overall turnaround plan as well. However, with the sell off of the fuel injection and commercial marine businesses, pursuit of major cost cutting initiatives including downsizing with elimination of 4,600 administrative & managerial positions and improved performance of other business segments; Rolls Royce expects profitability for 2018 to be in the £400 to £500 million.



## **Rolls Royce's Acquisition of ITP Aero for \$847.5 Million**

Rolls Royce has acquired the Spanish aircraft engine and component manufacturer, ITP Aero, for \$847.5 million with the company receiving regulatory approvals pertaining to the transaction in December 2017. Rolls Royce bought the 53.1% outstanding stake in the JV it operated along with Sener Grupo de Ingeniería. The acquisition of ITP Aero further strengthens Rolls Royce's aerospace propulsion business given that ITP is the ninth largest global manufacturer of aircraft engines and components with its global industrial base well spread across regions with industrial facilities located in Spain, India, Malta, Mexico, the U.K. and the U.S.

The acquisition also opens additional growth opportunities for Rolls Royce in military aerospace market given that ITP is a leading provider of turboshaft engines to Airbus heavy military helicopters. Rolls Royce plans to operate ITP as a separate business unit as per the agreement it signed with the Spanish regulator. Rolls Royce reported a double digit revenues growth for ITP for the year 2017 with sharp increase in delivery volumes on civil engine programmes.

## Expansion of European Aerospace Industrial Footprint with a Series of JVs

Rolls Royce has been working towards expanding its Europe based Aerospace industrial footprint with the signing up of a series of JVs signed with Liebherr & Safran earlier. The JVs are aimed at establishing manufacturing facilities & expanding overall production capacity of gearboxes & transmissions for Rolls Royce's upcoming next generation, geared turbofan aircraft engines. The details of the JVs include:-

**Aerospace Transmission Technologies GmbH** - Rolls Royce's 50:50 owned JV with Liebherr Aerospace was set up as per an agreement signed in June 2015. The JV aims to produce powered gearboxes for Rolls Royce's next generation UltraFan engine which is likely to be commercially available in 2025. The operations of the company are based at Friedrichshafen, Germany; where Liebherr has its industrial presence & operations. The facility commenced operations in early November 2015. As part of the agreement, Rolls Royce will lead the design definition, integration & testing of gearboxes and the company finalized the design of the first test gear unit for the UltraFan engine in May 2015. Rolls Royce has also been simultaneously working towards expanding its research site based at Dahlewitz, Germany where the gearboxes will be tested with the company investing over EUR 80 million towards site expansion. Liebherr currently manufactures large, multi-stage planetary gearboxes for wind turbines and establishment of the JV with Rolls Royce will enable Liebherr to enter aerospace engine transmissions manufacturing thereby expand its overall scope of operations.

**Aero Gearbox International (AGI)**: AGI is the second key JV established by Rolls Royce of-late with reference to its upcoming next generation engine programs. AGI is a JV between Rolls Royce & Hispano-Suiza, a Safran group company, for manufacturing power transmissions for the UltraFan engine. The agreement for establishment of the JV was signed by the two companies in October 2014 and entails setting up of a new manufacturing facility to be based at Podkarpackie Province in southeast Poland which will design, develop & manufacture accessory drive trains (ADT) for Rolls Royce's civil aircraft engines portfolio ranging from business jets to large widebody aircrafts. The Trent 7000 will be the first engine program for which AGI will produce accessory drive trains. The JV is based at Colombes near Paris and has its operations located across Colombes in France, Derby in U.K. & Dahlewitz in Germany. The construction of the new Poland based production facility is scheduled to commence from the end of 2015 and the first set of accessory drive trains are scheduled to be delivered by the facility in 2017.

## **Plans to Develop Bristol as Research Hub for Developing Carbon Fiber Composite based Engine Fan Blades, Cases & Innovative Manufacturing Technologies**

Rolls Royce plans to establish a new research centre to be based in Bristol, England. The centre will develop carbon fibre composite based engine fan blades & cases as part of its latest product & technology strategy which is focusing significantly on cutting edge, innovative technologies. The carbon fibre based blades & cases will be in production over latest Rolls Royce turbofan engine platforms from 2019. The facility has been dubbed as a pre-production facility by Rolls Royce and will eventually have a headcount of 120 by 2019. The Carbon-Titanium (CTi) based fan blades will be a crucial & integral part of upcoming Advance engine design with these CTi based blades & engine casings enabling Rolls Royce to reduce overall engine weight by up to 1,500 lb that will translate into an expansion of overall aircraft passenger capacity by 7. In addition to this research centre, Rolls Royce is also setting up a new facility based in Bristol for producing carbon-fiber based electrical harness rafts. Rolls Royce is effectively setting up Bristol as a hub for conducting research into carbon fiber based manufacturing technologies with research also being carried out currently in partnership with National Composites Centre based in Bristol & at the Rolls-Royce University Technology Centre at the University of Bristol. Rolls Royce also plans to relocate CTi manufacturing capabilities & over 40 employees to Bristol in 2017 from a composites manufacturing center based at present on the Isle of Wight. The relocation is likely to translate into opening of over 80 new job positions by the company in Bristol. The U.K. Government has played a key role in establishment of the Rolls Royce's pre-production facility & the Isle of Wight facility with provision of financial assistance to the tune of GBP 7.4 Million.



## Establishment of Specialized Centres of Competence for Production of Critical Engine Parts with Overall Focus on Innovative, Cutting Edge Technologies

Rolls Royce is aiming to expand its existing industrial base with the establishment of new facilities for manufacturing engine components. The company has been focusing on expansion & transformation of its industrial base significantly with investments to the tune of £687 million made towards its global facilities & equipment in 2014 itself. Key initiatives taken by the company include:-

**Establishment of a Engine Centre of Competence & Mainline Engine Shaft Manufacturing at Inchinnan near Glasgow:** Rolls Royce plans to establish a new centre of competence for manufacturing specific types of airfoils, in addition, to setting up a production centre for manufacturing mainline engine shafts for Rolls Royce's entire civil aircraft engine portfolio aimed at expanding overall production capacity. The company plans to establish the same at the existing Inchinnan facility near Glasgow, Scotland at an estimated investment of £60 million. The Inchinnan facility currently produces pressure blades & high as well as intermediate pressure shrouds for the Trent 900 turbofan engines. The latest plan entails the production of airfoils for relatively lower volume engine programs, which include, Adour & BR715 which are installed on some fighter & regional jet programs respectively. The expansion of the facility's scope of operations will involve addition of over 130 new job positions at the site as well. The move is a part of Rolls Royce's overall strategy which has focused on transformation of its global industrial base with focus on innovations & high-end technologies through establishment of specialized centres of competence for critical engine parts, which include, aerospace discs, fan blades & engine casings besides pursuit of operational restructuring & efficiency enhancement measures.

**Manufacturing Facility for Aerospace Discs for Trent XWB Engine Program:** Rolls Royce opened a new advanced machining centre in July 2014 for producing aerospace discs for the company's Trent engine series with special focus on the XWB program. The 18,000 square meter machining centre is based at Washington town in Northeast England and has been set up at an estimated investment of £100 million. The production at the facility is being scaled up steadily with full annual production capacity of 2,500 fan & turbine discs likely to be reached in 2016. The plant features cutting edge technologies, which include, robotics & a range of other automation technologies. For carrying out an array of operational activities. The facility also features advanced processes for machining, grinding, broaching & inspection activities, in addition, to incorporating advanced manufacturing methods & processes developed by the Rotherham, England based Advanced Manufacturing Research Centre (AMRC) which together cut down overall production time for discs by almost 50%.

**Rolls Royce has also become the first industrial lead & key founder member of an Advanced Manufacturing Technology Centre (AMTC) to be based in Turkey.** The move is part of Rolls Royce strategy of setting up a global network of Advanced Manufacturing Research Centers, as per Dr. Hamid Mughal, Director of Global Manufacturing at Rolls Royce. The establishment of AMTC in Turkey was announced in early October 2015 and forms a key element of Turkey's plans to develop advanced core industrial capabilities and develop its domestic aerospace & defense industrial base with plans to establish the new AMTC as a leading regional center of excellence in manufacturing technology.

**Rolls Royce also opened a new, scratch up Advanced Blade Casting Facility (ABCF) in February 2015** based at Rotherham. The facility has been established by Rolls Royce at an estimated investment of £110 million. The 150,000 square feet facility will produce high & intermediate pressure single crystal turbine blades for the Trent engine family and will have an annual production capacity of 100,000 blades once it becomes fully operational in 2017.

### Continued Investments into U.K. based Industrial Footprint

Rolls Royce is continuing to invest in its U.K. based industrial footprint to support its planned ramp up of wide body engines production which is already underway with Rolls Royce delivering 483 wide body aircraft engines in 2017 as against 357 for 2016.

Rolls Royce announced investments of GBP 150 million in June 2017 with a majority of the same towards the new testing facilities for large engines based in Derby.

Rolls Royce also opened its new assembly line for its Trent XWB engines based at Dahlewitz in June 2017. The Dahlewitz site houses an employee base of around 2,500. The new assembly line complements the existing assembly line based at Derby. The new facility has been established to meet delivery schedules for the much in demand Trent XWB engine program which powers the Airbus A350 XWB aircraft program. With the second assembly line in place, Rolls Royce will be able to achieve its plan of delivering 7 engines per week to customers by mid-2018. Rolls Royce has a strong order book position for the Trent XWB which is a key element of Rolls Royce's strategy of holding a 50% market share of the global wide body aircraft engine market by 2020.

Rolls Royce has also opened its new fleet support facilities based in Tyne and Wear with the consolidation of operations leading to planned closure of an older facility in 2018. Further, Rolls Royce closed its legacy supply chain facilities based in Ansty and Sunderland in 2017.

## **Rolls Royce to Supply Trent 700 Engines to Airbus for Beluga XL Transporter Planes in a \$700 Million Contract as Part of Airbus' Latest Air Transport Strategy**

Rolls Royce has won a contract from Airbus to supply Trent 700 engines for the 5 Beluga XL Transport planes along with long term TotalCare engine service support service. The Beluga XL air transport planes are based on the Airbus A330 design and replace Airbus' existing Beluga aircraft fleet which is powered by engines supplied by an industry competitor. The Beluga XL program was launched by Airbus in November 2014 as per the group's latest air transport strategy and part of its initiative to ramp up Airbus A350XWB production & to expand transport capacity requirements across other aircraft programs. The Beluga XL program will expand Airbus' overall transportation capacity by around 30% as compared to previous generation Belugas. The selection of Trent 700 for the Beluga XL program further extends the leadership of Trent 700 program which has won over 60% of new orders over the past 3 years and has a market share of 90% on the Airbus A330 Freighters. The Trent 700 is the largest in-service engine from the Trent series with over 1,500 engines in-service or on order with Rolls Royce.

The first Airbus Beluga XL Transport plane powered by the Rolls Royce Trent 700 engines made its first flight from Toulouse, France in July 2018.

## Significant Growth Avenues for Helicopter Engines Segment with Signing up of Key, Long-Term Engine Supply Contracts

Rolls Royce has opened up significant growth avenues for its helicopter engine segment with the signing up of long-term, engine supply contracts with OEMs. Key contracts won by Rolls Royce include:-

**Rolls Royce to Supply 1,000 RR300 engines to Robinson Helicopter Company over the next 10 Years** as per an agreement signed by the two companies in March 2015. The RR300 is the exclusive power-plant on the R66 and Rolls Royce has supplied over 800 RR300 engines for the R66 program. The RR300 engine program has been optimized for the 240-300 shp power output range and was originally developed by Rolls Royce based on the M250 engine program and is intended as a replacement for piston engines used in light helicopters & general aviation aircrafts.

**RR300 Engine to Power Composite Helicopter's KC630 Helicopter Program:** Rolls Royce signed an agreement with New Zealand based Composite Helicopters in March 2015 to supply the RR300 engine for its latest KC630 helicopter program. The two companies also plan to work together towards incorporating Rolls Royce's M250 engine variants into future helicopter platforms to be developed by Composite Helicopters.

**Upgraded M250 Engine for Bell 407GXP:** Rolls Royce will provide upgraded M250 engines for the Bell 407GXP helicopter program as per a long term agreement signed between Rolls Royce & Bell in March 2015. The upgraded M250 variant; the M250-C47B/8 engines, will feature a 8% improvement in power output for hot & high conditions while delivering a 2% enhancement in overall fuel efficiency of the Bell 407GXP through a pre-installed Value Improvement Package. The installation of the package will reduce operating costs for operators while expanding mission range & overall payload capacity. The package had been available as an upgrade offering for operators since 2012, however, this is the first time that Rolls Royce has made it available as a standard, pre-installed offering based on a 10 year agreement signed between Rolls Royce & Bell Helicopters.



## **Enhanced Customer Focus with Pursuit of a Customer Centric Regional Product Support Strategy focusing on Proximity & High Responsiveness**

Rolls Royce is pursuing a customer centric product support strategy for its large civil aircraft engine customers. The strategy focuses on enhancing the level of customer service by ensuring proximity to customers & increased responsiveness through a network of region focused Customer Service Centers for civil large engines. Rolls Royce plans to support its airline customers through a network of 5 Regional Customer Teams based across Asia-Pacific, Americas, Europe, Greater China and Middle East & Africa. This will be complemented by the establishment of a sixth Lessors team that will support the company's leasing corporate customers globally. The company opened its first Customer Service Centre based in Singapore in early February 2015 that will serve the Asia-Pacific region. Rolls Royce plans to house a team of 50 executives at the Singapore based center by the end of 2015. These customer service centers will play a key role in:-

- Coordinating operational planning
- Delivering analytics based data services
- Customer account management
- Spearheading regional sales campaigns
- Being a hub for regional Airline Support Teams



## **Strengthening & Expansion of Rolls Royce's Service Capabilities & Support Network for Global Civil Helicopter Fleet**

Rolls Royce has been working towards strengthening & expanding its global engine MRO service network aimed at enhancing service levels & capabilities. Rolls Royce established a new 24/7 operations center for its global civil engine helicopter fleet operators aimed at enhanced service quality. The center will provide support for the Rolls Royce's M250 and RR300 engine programs and align support processes as well as operations with the company's existing defense operations center based at Indianapolis to provide seamless support across military & civil engine programs. The center tracks service issues across operational fleets for over 4,500 individual helicopter owners and about 16,000 civil aircrafts, benchmarks & shares engine data and enhances overall service responsiveness. Rolls Royce's Indianapolis based operations center also provides support for the USAF's C-130J aircraft fleet that incorporates the Rolls Royce AE 2100 engines.

## Pursuit of a Comprehensive Engine Design Strategy aimed at Gaining & Protecting Market Shares in the Global Wide-body Civil Aircraft Engine Market

Rolls Royce is working towards the development of next generation engine technologies featuring cutting edge technologies & innovations as part of strategies to protect its 50% market share in the global wide body civil engine market and to enhance its presence further in the global military & civil aircraft engines market powered by an innovation driven growth strategy consistent with the group's comprehensive annual R&D outlay of around GBP 1 billion. The group is working on two new ultra-modern engine designs from scratch up as the successor of existing Trent engine family with the plan to achieve operational status by 2025. The engine designs include:

**Advance:** The Advance engine design offers a 20% enhancement in fuel burn rate as well as a commensurate reduction in CO2 emissions as compared to the first generation of Trent engines. Rolls Royce plans to achieve operational status for the Advance engine design through 2020.

**UltraFan:** The second engine design from the Rolls Royce stable, christened, UltraFan is going to be a geared design featuring the usage of a variable pitch fan system which will transpire into a 25% enhancement in fuel efficiency as well as CO2 emissions as compared to first generation Trent engines. The design is essentially aimed at developing a high thrust engine with a high bypass ratio while being significantly fuel efficient at the same time. Rolls Royce plans to achieve operational status for UltraFan by 2025.

Other technological innovations to be featured by the new engine designs include:

- A completely new, built from scratch **Engine Core Architecture** delivering significantly enhanced fuel burn efficiency & reduced emissions.
- Usage of **Carbon/Titanium Fan Blades** with a composite casing leading to significant weight reduction to the tune of 1,500 lb per aircraft which will further lead towards achieving an enhanced weight to thrust ratio and enhanced fuel efficiency.
- Incorporation of **Advanced Ceramic Matrix Composites** with a significantly higher thermal resistance and the capability to operate optimally & effectively in high turbine temperature conditions. Rolls Royce's acquisition of Hyper-Therm HTC in 2013, a company that specialises in ceramics will enhance the group's capabilities pertaining to ceramic matrix materials significantly.

Rolls Royce announced in July 2015 that its has partnered with Industria de Turbo Propulsores, S.A. (ITP) Of Spain (a 46.9%:53.1% JV between Rolls Royce & Sener dating back to 1989) to pursue & support a €43 million research programme which will test Intermediate Pressure(IP) turbine technologies to be incorporated into the UltraFan engine program. The IP turbine program is partially funded (€ 23.5 million) by the EU Clean Sky 2 initiative which will run through 2023 with remainder funding coming from ITP. Rolls Royce has also strengthened its long-term strategic partnership with Sener in the ITP JV for assured delivery of turbines to Rolls Royce for its Trent engine programs.

**Trent 7000 Engine Program as the Exclusive PowerPlant for A330neo:** The Trent 7000 marks an expansion of Rolls Royce's Trent engine family with the Trent 7000 slated to replace the Trent 700 engine program on the A330neo aircraft which is scheduled to enter service in Q4 2017. Rolls Royce has combined the engine architecture of Trent 1000 TEN, its operating experience with the Trent 700 & latest engine technology from the Trent XWB into the Trent 7000 program with the first Trent 7000 engine scheduled to undergo first set of tests in late 2015. The specs of the Trent 7000 include a thrust output range of 68-72,000 lb with a 10% projected improvements in fuel efficiency over the Trent 700 which it replaces with twice the bypass ratio & 50% lower noise level as compared to the Trent 700 which it replaces. Rolls Royce has once again brought its proven industry partner team; comprising of ITP, Kawasaki Heavy Industries & Mitsubishi Heavy Industries Aero Engines, Ltd.; for the Trent 7000 program which has played a key role in the development of its Trent engine family over the past decades.

**Open Rotor Engine Concept:** Rolls Royce has also been working on and has developed as well as tested the Open Rotor Engine Concept which could be further matured based on emerging market requirements and developments.

## Expansion of Presence across Military Aircraft Programs Globally

Rolls Royce plans to & is working towards expanding its presence across Military Aircraft programs globally as a strategy to derive further growth and to partially offset the ongoing, significant pressure on defense spending across most traditional markets. Key military aircraft programs being targeted, looked at and being harnessed by the company include:

- **Hawk Advanced Jet Training System for the U.S. Air Force's T-X Trainer Aircraft Program:** Rolls Royce's participation in the Hawk Advanced Jet Training System for the pursuit of U.S. Air Force's T-X Trainer Aircraft program opens further growth avenues amid difficult defense spending scenario across most traditional markets. Rolls Royce will lead the integration & support of the Adour Mk951 engine on the Hawk AJTS as the fourth partner of the program apart from BAE Systems, Northrop Grumman & L-3 Link Simulation & Training. Rolls Royce already has about 200 Adour Mk951 engines in service with the U.S. DoD.
- **A-400M Aircraft Program** entered Service in 2013 with the French Army powered by the TP400-D6 engine producing 11,000 shaft horsepower made by Europrop International, a consortium comprising of Rolls Royce, MTU, Snecma & ITP. The A400 program, with a significant order book of 174 aircrafts as of December 2013, is aimed at meeting the demand of a versatile tactical airlifter of 7 key European nations & Malaysia as the export customer. Europrop International plans to ramp up engine production to 2 engines per week at its Munich based production facility with the focus shifting from development to production & support as Airbus plans to speed up deliveries.
- **Airbus A330 MRTT Air Tanker Program:** Growing International Customer Base for the Airbus A330 MRTT Air Tanker program is likely to benefit Rolls Royce with its Trent 700 engine's growing defense applications. The A330 MRTT (Multi Role Tanker Transport) aircraft program, derived from the A330 commercial airliner, has been selected by over 6 nations comprising of Australia, Saudi Arabia, the UAE, the U.K., Qatar & Singapore with a total order book of 28 aircrafts besides 17 aircrafts in service with the first four nations. Airbus is also in final negotiation stages with India for 6 aircrafts. Further, U.K. too has been focusing on growing its tanker fleet which is likely to provide further growth impetus for the Trent 700 engine.
- **C-130J Super Hercules Aircraft Program:** Growing Volumes for the C-130J Super Hercules program are driving demand for the Rolls Royce AE 2100 engine. Lockheed Martin signed a long-term agreement with Rolls Royce in February 2014 for 600 AE 2100 engines worth \$1 billion through 2025. The agreement will service U.S. Government & International contract requirements between 2014 & 2018. Rolls Royce has so far delivered over 1500 AE 2100 engines produced in Indianapolis for the C-130J aircraft program. Rolls Royce delivered the 2000<sup>th</sup> AE 2100 engine in September 2015 to Lockheed Martin for the C-130J Super Hercules program.
- **Scott's Bell 47 Program:** Rolls Royce has partnered with Bell for the Scott's – Bell 47 light utility helicopter program. Rolls Royce secured a multi-engine order with Scott's – Bell 47 for the RR300 engine. The Scott's – Bell 47GT-6 helicopter program has orders for over 38 helicopters and is scheduled to enter service in 2016 with plans to produce over 30 aircraft annually by 2017.



## Expansion of Rolls Royce's Industrial Presence, Footprint & R&D Capabilities in the Asia-Pacific Region

Rolls Royce has been in the process of **expanding its industrial presence as well as footprint in the Asia-Pacific region** spearheaded by its Singapore based hub. The company established its primary industrial facility for the APAC region based at Singapore in 2012 that produces about 50 Trent 900 engines annually for the Airbus A380 aircraft program. Rolls Royce is working towards broadening & expanding its engine production line-up being produced at the facility. The facility commenced production of the Trent 1000 engine that powers the Boeing 787 Dreamliner aircraft program from mid-2014. The start of production of Trent 1000 in Singapore enables Rolls Royce to meet growing demand for the Trent 1000 engine program that has seen a significant order intake of-late with orders worth \$11.5 billion received by the company in 2013 alone. Rolls Royce plans to reach the installed cumulative engine production capacity of 250 engines annually by early 2016 with the addition of Trent 1000 to production line from mid 2014. The facility rolled out its first Trent 1000 engine in January 2015 which powers the 20 Boeing 787 aircrafts to be operated by Singapore based long-haul budget carrier, Scoot.

Rolls Royce also has a fan blade manufacturing capability at the facility which produces about 2100 wide-chord, hollow Titanium blades annually. The facility started production of fan blades for the Trent XWB engine program from early 2014. The facility plans to reach its total installed annual production capacity of 6000 blades through 2015. Rolls Royce plans to expand its overall production capacity at the facility as part of phase 2 of expansion to reach annual production capacity of 7600 blades in 2016.

Rolls Royce is also **focusing significantly on expanding its R&D capabilities as well as infrastructure** in the region. The company has a regional training center as well as an advanced technology center at the facility. Rolls Royce plans to further establish an advanced remanufacturing & technology center in Singapore in collaboration with IHI & Siemens which opened in Q3 2014. The center focuses on technologies for repair & restoration, surface modification & product verification. Rolls Royce also plans to establish a corporate lab at the Nanyang Technological University(NTU) with support from National Research Foundation at an estimated investment of \$60 million. The center would focus on pursuing R&D in the areas of electric power & control systems, manufacturing & repair technologies & computational engineering.

**Rolls Royce also announced plans in April 2015 to expand its existing engineering capabilities in India with plans to establish a new aerospace engineering facility based in Bangalore.** The facility will house around 500 engineers by the end of 2017 and will support regional customers with aerospace engineering solutions , development of new tools & technologies leveraging existing capabilities besides supporting the company's regional supplier base. The move is part of Rolls Royce's strategy of establishing presence across key growth markets while maintaining cost competitiveness. The latest move will enable Rolls Royce to enhance its strategic positioning with its existing customer base along-with competitiveness while pursuing potential growth opportunities aggressively. Rolls Royce currently operates two engineering service centers in Bangalore besides a production facility to manufacture components for the Trent civil aircraft engine family under IAMPL, a 50:50 JV with Hindustan Aeronautics Limited. The production facility commenced operations in 2013 & has an employee base of 100.

## Strengthening of R&D Capabilities in-line with Mid to Long-Term Technology Development Plan

Rolls Royce plans to significantly strengthen its R&D capabilities in-line with its mid to long-term technology development program. The company has been making significant capital investments towards establishing infrastructure to sustain its technology development plans spearheaded by the development of next-generation engine programs with completely new engine architectures, namely, Advance & Ultrafan, featuring several technological enhancements. As part of the same, Rolls Royce plans to establish a new test-bed for power gearboxes to be based at Dahlewitz in Germany at an estimated investment of €65 million of which about 20% will be funded by the German & the Brandenburg government. The power gearboxes will be used primarily for the Ultrafan engine program that will be a geared engine design featuring a variable pitch fan system that will deliver up to 25% reduction in fuel burn and emissions as compared to the first generation Trent engines. The Dahlewitz facility has been an integral part of the company's R&D efforts and has implemented development packages for Rolls Royce's Trent engine family over the past two decades. Rolls Royce is now working towards establishing a new test-bed at the Dahlewitz site for the Trent XWB & the latest, upcoming Trent engine platforms. The company broke ground at the site for the facility in March 2014 with the same becoming operational from late 2014. The moves are in-line with Rolls Royce's overall R&D strategy & program powered by a sustained, significant annual R&D outlay which stood at £1.1 billion for 2013.

## Plans to Focus on Service Contracts for Military Programs & Grow Long-Term Service Contracts for Civil Engine Programs

Rolls Royce plans to & has been working towards **growing service contracts over its military engine programs** as a means to partially offset significant pressure on defense spending for new hardware & equipment. The company has received a number of new, significant engine support programs for military programs worth about \$224 Million in 2015 as compared against contracts worth \$400 million in 2014 originating from the U.S. market with the company's growing focus on services. Key recent service support contracts received by Rolls Royce for military programs include:

- A \$59.5 million contract from the USMC awarded in June 2015 to provide integrated logistics support for its fleet of KC-130J aircraft , in addition, to supporting aircrafts operated by Kuwait with provisions under the FMS route. In 2014, Rolls Royce received a \$182.7 million contract from the USAF for expansion of existing MissionCare support contract for the C-130J aircraft fleet that incorporate Rolls Royce AE 2100 engines. The expansion included increased orders for spares & spare parts and deployment of Field Service Representatives at 2 new USAF bases.
- Rolls Royce also received a \$40 million contract for providing engineering support & carrying out repairs on the T56 engine under Component Improvement Program for the C-130J & P3 aircraft programs. The company also received a \$31 million 5 year full MissionCare contract from Norway under the FMS route to provide logistics & sustainment support, spare parts & field services.
- A \$93.5 Million contract from the U.S. Navy to provide intermediate, depot & logistics support for Navy's T-45 trainer aircraft fleet based at Naval Air Stations at Texas & Mississippi. The T-45 aircrafts are powered by the Rolls Royce Adour engine. Rolls Royce received contracts worth \$100 million from the U.S. DoD in 2014 to provide Contractor Logistics Support(CLS) to provide support for the F405 (Adour) engines for the US Navy's T-45 training aircraft.
- Rolls Royce received a \$39 million contract in April 2014 to support the installed AE 1107C turboshaft engines over the USMC & USAF's V-22 Osprey aircrafts under Rolls Royce's MissionCare contract which will include repairs & support services.
- Support contract for the USMC's KC-130J refuelling tankers worth \$50 million received in March 2014.
- Support contract worth \$57.1 million for AE 1107C engines for the USMC & USAF's MV-22 & V-22 Osprey aircrafts respectively received in December 2013.

Rolls Royce has also been working towards **growing long-term service contracts for its global civil engine programs**, with focus on the Trent 700 engine program, as an assured, visible, ancillary revenue stream. As part of its efforts, the company has been able to sign a number of new long-term service contracts of-late.

- The recent agreement signed in late October 2015 with China's HNA Group worth \$2.4 billion for Trent 700 & XWB engines over 44 Airbus aircrafts on order, in addition, to TotalCare service support has come as a shot in the arm for Rolls Royce. The HNA group operates Hainan Airlines, Tianjin Airlines and Capital Airlines and the agreement includes Trent 700 engines & Total Care service support for 20 Airbus A330s and TotalCare service support for Trent XWB & Trent 700 engines for 15 Airbus A350-900S & 9 A330s.
- Other key TotalCare Engine Service/Flex contracts signed by Rolls Royce include:-
  - Rolls Royce signed its largest ever order with Emirates in April 2015 worth \$9.2 billion to provide Trent 900 engines for 50 Airbus A380 aircrafts scheduled to enter service in 2016 along-with TotalCare service support.
  - A contract signed with Saudi Arabian airlines in July 2015 worth \$1.3 billion for long term TotalCare engine service support for 20 Airbus A330 aircrafts powered by Trent 700 engine
  - Rolls Royce received a \$1.1 billion contract from Air Europa for Trent 1000 engines & TotalCare long-term service support in early 2015.
  - A \$580 million TotalCare long term engine service contract worth \$580 million signed with Vietnam Airlines in July 2015 for Trent XWB engines for its fleet of 14 Airbus A350XWB aircrafts
  - A TotalCare Flex service agreement with Cathay Pacific signed in September 2015 to support Trent 800 engines for its fleet of 17 Boeing 777 aircrafts.
  - Agreement with AerCap for Trent 500 engines over 3 AerCap A340s operated by South African Airways
- Rolls Royce also expanded its existing scope of Fuel Management Partnership with Air Mauritius with the signing up of a Efficiency Service Solution in early October 2015 powered by its VisiumFUEL software, data & analytics & consultancy. The company also signed a similar 5 year efficiency service solution contract with Singapore Airlines in mid-June 2015 for its fleet of 137 aircrafts.
- Rolls Royce has also been able to make inroads into the regional aircraft market with the TotalCare Flex solution, derived from TotalCare service solution with specific focus on economic management of mature engines. The company signed an agreement with BMI regionbal in early October 2015. The agreement covers 17 Embraer 145 aircrafts powered by the AE 3007 engines.

## Rolls Royce Plans to Re-Enter the Narrow Body Engine Segment

Rolls Royce plans to re-enter the narrow body aircraft engine segment; which it exited in 2012 with the sale of its 32.5% share in International Aero Engines (IAE) consortium, which makes the ubiquitous V2500 engine program for the Airbus A-320 aircraft, to Pratt & Whitney for \$1.5 billion besides variable payments for 15 years based on aircraft flight hours. The move comes in wake of Rolls Royce's continuing difficulties in the wide body segment with Boeing's decision to go with GE's 9X engine as the sole power plant for its 777X program, uncertainty over the future of A350-800 program with Airbus moving & upgrading customers to the A350-900 & 1000 programs besides continuing budgetary pressures impacting acquisition of new transport aircrafts. The narrow body segment of the civil aircraft market is the largest, in terms of volumes, and is split amongst IAE, CFM & Pratt & Whitney across key aircraft programs. Rolls Royce's strategy, with respect to plans to re-enter the narrow body segment, is likely to be based around technology; as indicated by the company's decision to develop next generation Advance & geared core Ultra-fan engine designs; and is likely to materialize over medium term only as it has already missed out on the second generation upgrades of key aircraft programs from both key industry air-framers, namely, Airbus's A-320neo & Boeing's 737 MAX which are split amongst CFM & P&W. The regional segment, too, witnessing a surge of new aircraft programs from existing as well as new industry players led by Bombardier's C Series, Embraer's E2 Jets, Sukhoi's SuperJet 100, Mitsubishi's MRJ, COMAC's C919 & Irkut's MS-21 has been split amongst P&W & CFM with their latest GTF & LEAP engine programs. Thus, the next generation upgrades from airframers for the narrow body segment is unlikely to be initiated in the present decade and without a path-breaking technology transpiring into optimized operating economics which could surpass P&W's PW1000G & CFM's LEAP program; it is going to be a herculean task for Rolls Royce to stage an effective comeback to the segment.

## Rolls Royce Plans 5-Year Industrial Modernization Plan for its Indianapolis based Manufacturing Operations

**Rolls Royce plans to transform & modernize its Indianapolis based existing manufacturing operations**, in addition, to conducting technology research. The company announced an investment of \$600 million towards the pursuit of modernization plan announced in early October 2015. Rolls Royce's 5 year upgrade & modernization plan will include the incorporation of latest production systems, machinery, manufacturing methods & assembly operations, in addition, to consolidation of operations & significant reduction of utility costs across the Tibbs Avenue & Raymond Street based facilities. The move & investment by Rolls Royce is aimed at enhancing the competitiveness of its U.S. based manufacturing operations. Rolls Royce's presence in Indianapolis dates back to 1915 and it currently has an employee base of over 4,000 in Indianapolis where Rolls Royce operates an advanced aerospace technology research & design lab and manufactures engines for a number of U.S. defense programs & commercial helicopters, which include:-

- F-35B Lightning II
- C-130J Super Hercules
- V-22 Osprey
- RQ-4C Global Hawk & Triton UAVs
- Naval Ship to Shore Connector Program

**Rolls Royce earlier in March 2015 also opened an advanced LiftWorks repair facility in Plainfield, Indianapolis** which will carry out component level repairs for the F-35 Lightning II fighter jet propulsion system known as LiftSystem. The company also has a LiftSystem production facility based in Plainfield itself adjacent to the new LiftWorks facility which will enable significant cost savings by leveraging significant synergies of operations across the two facilities. The latest LiftWorks facility marks a \$10 million investment from Rolls Royce & the USMC.



## Focus on Continual Improvements aimed at Performance Enhancement over existing Engine Programs

Rolls Royce has been focusing on continual improvement & performance enhancement over its existing engine programs aimed at extending performance & capabilities further besides opening avenues for potential engine retrofitment over existing military aircraft programs.

- Rolls Royce undertook a self-funded initiative to pursue proactive, continual modification & improvement program to enhance & upgrade the functioning of its AE 1107C engines installed over the USMC's MV-22 Osprey aircraft, which included, a 17% increase in engine power, increased reliability, improved hot & high performance & extension of overall time on wing.
- Rolls Royce's development of a higher thrust version of its Trent XWB engine that powers the Airbus A350-1000 aircraft program further highlights the company's focus on extending performance & capabilities of its engine programs. Rolls Royce has developed a 97,000 lb thrust version of its Trent XWB engine, termed, Trent XWB-97. The XWB-97 produces 13,000 lb of additional thrust as against its predecessor, the Trent XWB, which produces 84,000 lb of thrust. The enhancement of thrust in the Trent XWB-97 originates from a larger, scaled up engine core, an improved high temperature turbine technology and a higher flow fan. The development of the Trent XWB-97 will enable the Airbus A350-1000 to increase overall aircraft payload weight & maximum take-off weight. Rolls Royce began assembly of the Trent XWB-97 in early April 2014 and the engine undertook its first test flight over an A380 flying test bed in early November 2015. The first commercial flight of the XWB-97 is expected towards the end of 2016 and entry into service scheduled for 2017.
- Rolls Royce has also developed an engine enhancement for its T56 engine, termed as the series 3.5 package, translating into a 13% enhancement in overall fuel efficiency. The company completed assembly & ground testing of its first enhanced T56 engine over a Hurricane Hunter WP-3D aircraft owned by the National Oceanic Atmospheric Administration (NOAA) in February 2014 which demonstrated fuel savings to the tune of more than 12%. The NOAA Hurricane Hunter completed its first test flight with the T56 engine incorporating series 3.5 package in mid-June 2015. Additionally, based on aircraft operating data generated by the NOAA aircraft, Rolls Royce has reported way reduced engine temperatures that extends parts life, leads to superior high altitude performance, faster engine starts & improved reliability by around 22%. Key improvements included under Series 3.5 enhancement package include compressor inlet housing, wheel knife seals, compressor blades & new turbine blades as well as vanes. The NOAA has been the first to receive series 3.5 enhancement package over its WP-3D aircrafts which began in early 2015. Additionally, ground & flight tests conducted over the C-130 aircraft program with the series 3.5 enhancement package demonstrated a 7.9% reduction in fuel consumption besides a significant extension in overall parts life and a 22% enhanced reliability as the turbine's operational temperature is significantly lower. A USAF study concluded that the insertion of series 3.5 enhancement package to the service's existing transport aircraft fleet would generate substantial savings to the tune of \$2 billion besides extending the operational lifespan of the fleet by decades. Rolls Royce & the USAF are in discussions for incorporation of the series 3.5 enhancement package over existing, in-service C-130 aircraft fleet with the Congress giving green flag to the same with an initial funding of \$15.7 million. The enhancement package is likely to be introduced over the existing C-130 fleet of the USAF, Air Force Reserve & Air National Guard with deliveries scheduled to commence in 2016. The U.S. Air Force awarded a \$36 million contract to Rolls Royce in mid-October 2015 for insertion of series 3.5 package into its fleet of C-130 aircrafts powered by the T-56 engine. The deliveries of the Series 3.5 kit to the USAF are scheduled to commence from December 2015 to the USAF & U.S. Air National Guard. Rolls Royce has already received the requisite USAF qualification & met regulatory requirements for the Series 3.5 enhancement package for the C 130. Additionally, the company has also received FAA certification for the commercial version of the T56 engine. The 3.5 series flight test & certification for the P3C Orion aircraft program is scheduled to be completed in late 2014. Additionally, international operators of the C-130 & P3 aircraft programs, too, are mulling plans for incorporation of the series 3.5 engine enhancement over their existing aircraft fleet. With over 400 legacy C 130 aircrafts in active service globally & around 70 L-100 Commercial Hercules variants; of which 200 are operated by the USAF alone; it is going to be the start of an extended engine enhancement process for Rolls Royce over medium term.

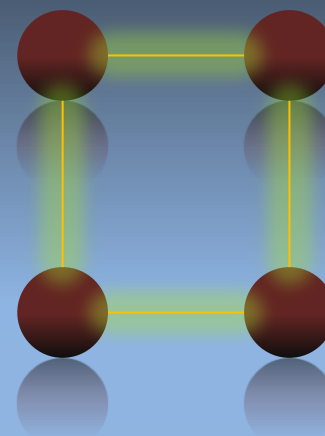
## Focus on Business Aviation with the Introduction of New Pearl Engine Family by Rolls Royce

Rolls Royce has blown the trumpet for a new war in the business aviation segment with the introduction of its new Pearl engine family for business jets. The Pearl engine family has been custom built and will be the exclusive power plant for Bombardier's latest Global 5500 and Global 6500 business jet programs. With the introduction of new engines for business jets, Rolls Royce has upped the ante on Pratt & Whitney which has garnered significant market share in the business aviation segment as of late with its Geared Turbofan technology based PW812 engine family.

The launch of new Pearl engine family is likely to also further consolidate Rolls Royce strong presence in the business aviation segment with a strong, in-service fleet of over 3,000 business jets powered by Rolls Royce engines (especially the iconic BR700) and a strong 42% market share, as outlined by Rolls Royce itself.

The new Pearl engine features advanced technologies derived from Rolls Royce's Advance2 technology demonstrator programs with incorporation of proven technologies from the BR700 as well. The Pearl 15 has been certified to deliver up to 15,125lb of thrust. The Pearl 15 provides 9% more thrust than the BR700 at the time of take off while providing a 7% improvement in specific fuel consumption to operators and being 2 decibels quieter than the BR700 and features a new Engine Health Monitoring System. The Pearl engine has been developed by Rolls Royce's Dahlewitz based facility and received EASA certification in February 2018 and is planned to enter service at the end of 2019 post the current flight tests underway at Bombardier's Flight Test Centre in Wichita, Kansas.

# Business Outlook for 2019



## ■ Part 1.3

## Outlook

Rolls Royce's near term strategy focus remains laser sharp on improving operating efficiency, performance & competitiveness, especially, for the embattled civil aerospace business which continues to jostle with the unanticipated, in-service issues with faster deterioration of parts on the Trent 1000 engine. Consequently, the civil aerospace segment has booked an operating loss of GBP 162 million for 2018 in an otherwise high growth market trajectory. The same has been highlighted by the continued pursuit of active restructuring initiatives by the group through portfolio restructuring and simplification of the overall corporate structure. The stronger operating performance, with the group aiming for GBP 1 billion in free cash flow by 2020, is being aimed at forming the backbone of the investment & technology led growth platform over medium term. The other pivot of the strategy for Rolls Royce remains as continued investments in technologies for the future for the long term with key technology focus areas being:-

- Advanced Materials for development & utilization in next generation engine platforms, namely, Advance & Ultrafan
- Hybrid & Electric Technologies for next generation propulsion technologies for emerging areas like Urban Aerial Mobility
- Artificial Intelligence for potential applications in aviation MRO
- Digitalization of processes for enhanced efficiency & quality and an imperative for long term growth platforms as highlighted by the development of the R<sup>2</sup> Data Labs & creation of digital ecosystem through active collaboration with Academia & Start-Ups

The Advance & Ultrafan are likely to enhance competitive positioning for the long term, however, their development horizon will ensure that Rolls Royce will not be able to participate in the emerging growth opportunity, in the form of Boeing's upcoming Middle of the Market aircraft program or the NMA, which will be Boeing's first scratch-up program since the 787 in 2003. Further, announcement of the winding of the A380 program by 2021, as announced by Airbus in early 2019, means Rolls Royce will be delivering its final set of the Trent 900s over the next 2 years. Lastly, the risks & potential impact of a no-deal BREXIT continue to create further, near term uncertainty for Rolls Royce with significant planning required to mitigate its effect.

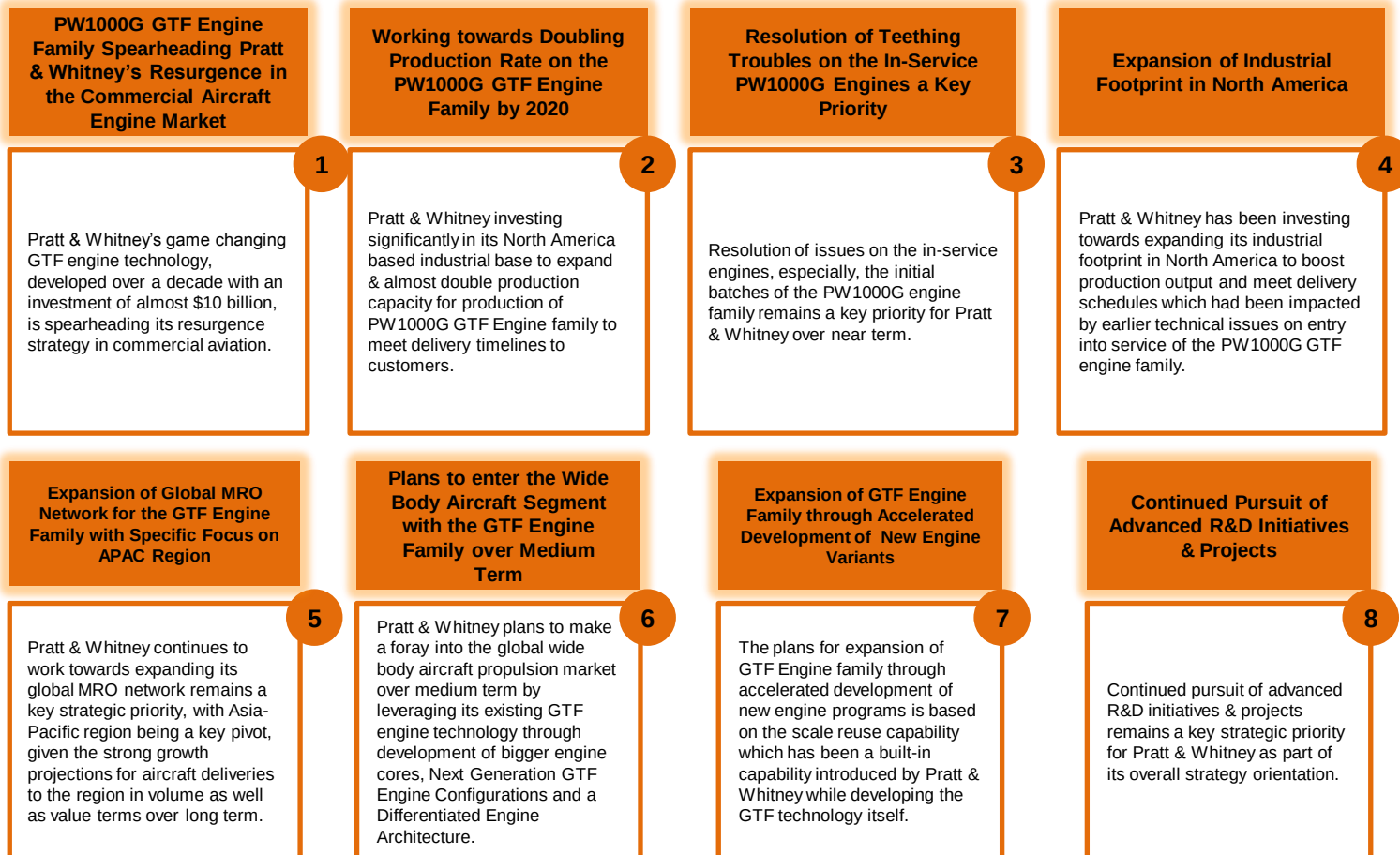
# Strategy Focus



## Part 2.1



# Pratt & Whitney – Overall Strategy Focus

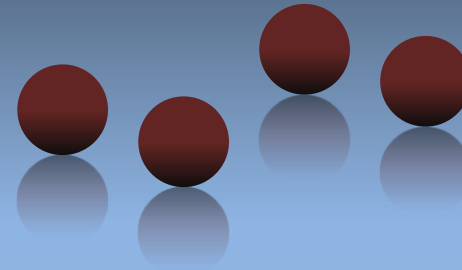




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# Key Strategies & Plans



## ■ Part 2.2



## The PW1000G Engine Family Spearheads P&W's Resurgence Strategy for the Commercial Aviation Segment with its Geared Turbofan Technology - 1

Pratt & Whitney plans to & is poised to extend its market presence and share in the narrow body aircraft segment across commercial & regional aviation sectors further with the introduction & selection of its PurePower PW1000G engine program & its variants; powered by the Geared Turbofan Engine technology; over a number of upcoming aircraft programs in the commercial & regional aviation segments. The development & introduction of the GTF engine spearheads Pratt & Whitney's strategy for resurgence of its civil aerospace business, especially, in the narrow body segment over medium term with the company having built up a sizeable order backlog of more than 8,000 engines, as of August 2017, across a number of key aircraft programs. Key aircraft programs to be powered by and which have chosen to integrate Pratt & Whitney's PW1000G power plant include:

- PW1500G for the Bombardier's ambitious C-Series with the aircraft program in active service now.
- PW1200G over the Mitsubishi Regional Jet (MRJ). Pratt & Whitney PW1200G engine will be the exclusive power-plant on over 32 MRJs under an agreement signed between P&W, Japan Airlines(JAL) & Mitsubishi Aircraft Corporation in February 2015. The first MRJ powered by the PW1200G made its test flight on November 11, 2015.
- PW1100G-JM for the Airbus A320Neo Aircraft Program which entered service in early 2016.
- PW1400G-JM for Irkut's MC-21 Regional Aircraft Program:- Pratt & Whitney delivered the first PW1400G-JM engine to Irkut Corporation in late August 2015 for the MC-21 aircraft program. The FAA certification for the PW1400G-JM engine program was received in May 2016.
- PW1700G & PW1900G for the Embraer's E2 Jets. The PW1900G started its flight test program with the first successful flight in early November 2015. Pratt & Whitney delivered the first two PW1900G production engines to Embraer for the E190-E2 aircraft program in January 2018.
- Pratt & Whitney has been able to make further inroads in the business aviation market with the selection of its PW812D engine for the Dassault Aviation's Falcon 6X business jet program in February 2018. The Falcon 6X succeeds the 5X which was cancelled by Dassault due to delays and development problems over the Safran's Silvercrest engine which had been selected by Dassault to power the 5X.

A number of these upcoming aircraft programs scheduled to enter service in the later half of the current decade. The commencement of flight testing phase of PW1900G engine for Embraer's E2 Jets in early November 2015 marks the initiation of flight testing phase of PW's PurePower engine family's fourth airframe customer.

The PW1000G engine with a development horizon of over 20 years significantly extends P&W's technological as well as competitive edge featuring:

- Fan-drive Gear System
- Enhanced Aerodynamic Efficiency
- Usage of Light-Weight Materials
- Significant Technological Advances in high-pressure spool, low-pressure turbine, combustor & engine controls



## The PW1000G Engine Family Spearheads P&W's Resurgence Strategy for the Commercial Aviation Segment with its Geared Turbofan Technology - 2

### Teething Troubles and Glitches Impacting Delivery Schedules, Order Intake and Creating Issues for Operators

The PW1000G promises to deliver significant enhancement in overall fuel efficiency & operating costs for the airliners besides achieving significant reduction in environmental emissions and noise levels. However, the program has been impacted significantly by delivery delays and teething troubles with periodic reports of glitches and engine failures which have impacted delivery schedules across OEMs, especially, for Airbus and Bombardier and created operational issues for aircraft operators, especially, with one leading Indian Airlines, Indigo (which operates one of the largest fleet of A320neos) reporting regular instances of mid-air engine failures with the airlines withdrawing some aircrafts from service while also receiving warnings from regulatory bodies over the same. The same had impacted the order intake for the PW1000G GTF engines as against the LEAP engine family from CFM's stable in 2017 by a ratio of almost 10:1. However, Pratt & Whitney reported that over the 1 year period from July 2017 to July 2018 it received orders for over 2,000 new GTF engines as well as commitments, including, options & spares from 22 customers. Pratt & Whitney had almost 200 GTF engine powered aircrafts in service globally as of July 2018 with total engine flight hours pegged at 1 million over the A320neo, A220 & E190-E2 aircraft programs.



## **P&W Poised to Extend Technological Capabilities & Overall Competitive Advantage with the Development of Sixth Generation Military Engine for the USAF**

Pratt & Whitney's R&D on the development of the sixth generation military engine, termed, Variable Cycle Adaptive Engine; in collaboration with the U.S. Air Force Research Laboratory (AFRL) under the USAF's Adaptive Engine Technology Development (AETD) program is likely to extend P&W's technological capabilities and overall competitive advantage, especially, in the military engine segment. The development of the 6<sup>th</sup> generation engine by P&W will leverage the company's significant experience in designing & developing its 5<sup>th</sup> generation engines, namely, F119 & F135 which power the F-22 Raptor & F-35 Lightning II aircraft programs of the USAF. The development of the next generation engine intends to include & test adaptive bypass flow which will incorporate a third air flow stream in addition to the two usually associated with currently operational 5<sup>th</sup> generation engines. The addition of the third air stream is likely to:

- Enable enhanced Fuel Efficiency with the Adaptive Engine Technology Development (AETD) program aiming to achieve fuel efficiency improvements to the tune of 25%-33% as compared against the currently operational 5<sup>th</sup> generation engines with the enabling of modulation of the engine operating conditions & bypass ratio for optimization of engine performance for a range of flight conditions and mission requirements.
- Achieving enhanced Thermal Management of Air System leading to significant increase in capability towards heat dissipation capacity at the air vehicle level
- Significant Reduction in Heat Reduction for Fighter Aircrafts

The development of the sixth generation engine is aimed at achieving enhanced efficiency in:

- Long-Range Persistence Scenarios
- Performing High Thrust Combat Manoeuvres
- Transonic & Supersonic Flight Conditions

Pratt & Whitney tested the adaptive bypass flow over its adaptive supersonic fighter engine fan rig based on a full-scale F-135 development fan in August & September 2013. P&W conducted demonstration testing of the advanced high pressure ratio core in late 2015 followed by full engine testing of the three-stream adaptive fan, compatible augmentor and exhaust system in 2016. Pratt & Whitney announced in September 2017 that it successfully tested the adaptive 3 stream fan in an engine with an F135 core under the AETD program. Pratt & Whitney had also announced plans to conduct additional adaptive engine testing on a new high efficiency engine core developed under the AETD program in late 2018. In addition to the development of a demand-modulated engine architecture, Pratt & Whitney is also maturing an adaptive technology suite which includes control systems, power systems and thermal management systems.

**Pratt & Whitney received a \$437 million contract modification from the U.S. Air Force Life Cycle Management Center (AFLCMC) in September 2018 for next generation adaptive propulsion risk reduction for potential air superiority applications.** Pratt & Whitney plans to leverage its proven fifth generation propulsion technology, the learning obtained from the design & testing activities under the Adaptive Engine Technology Development (AETD) program towards maturation & refining of technologies under the Adaptive Engine Transition Program (AETP).





## Expansion of P&W's North American Industrial Footprint - 1

Pratt & Whitney has been working towards expanding, modernizing & revitalizing its North America based industrial footprint with significant developments & growth in its commercial aviation business led by the PW1000G engine family for which the company has an order backlog of more than 8,000 engines besides the F-35 JSF program in the military segment for which P&W is the sole engine manufacturer. Pratt & Whitney has signed up long-term supply agreements worth \$10 billion for parts & materials with over 90 suppliers of-late with plans for production ramp-up to meet delivery schedules across commercial & military segments. Key strategic initiatives being made by P&W towards the same include:-

**Establishment of Pratt & Whitney's New Connecticut based Headquarter besides Expansion of Manufacturing and R&D Capabilities:** Pratt & Whitney plans to invest over \$500 million through 2018 to establish its new global headquarters based at Connecticut. The new 425,000 square foot headquarters will be based at P&W's existing East Hartford based corporate campus located near Rentschler Field and will also house P&W's planned global engineering centre of excellence. The company held a groundbreaking ceremony at the site in July 2015. the construction at the site commenced in late 2015 and the facility was scheduled to be completed in late 2017. Besides, P&W headquarters, UTC plans to invest to the tune of \$4 billion towards research & capital expenditures across its aerospace businesses in in Connecticut. The investments will also include enhancement of company's manufacturing operations across the East Hartford and Middletown based facilities. The construction work commenced in 2014 and is likely to be completed through 2018. Other key investments being made by UTC include:

- Creation of UTC Aerospace Systems (UTAS) customer training centre based at Windsor Locks which will include new & refurbished space.
- Establishment of new research labs & infrastructure at United Technologies Research Centre (UTRC).
- Investments in new research projects & capital investments at Pratt & Whitney, Sikorsky, UTAS & UTRC.

**Expansion of Manufacturing Facility in North Berwick:** P&W plans to expand its North Berwick based manufacturing facility in view of the \$1 billion contract from the Department of Defense for 38 engines for the F-35 Lightning II aircraft program. The facility expansion would include addition of 19,000 square foot of floor space for shipping, receiving & packaging of engine modules. The plant expansion is also being undertaken in view of P&W's ongoing negotiations with the U.S. DoD for contracts for additional engines for the F-35 aircraft program.

**Opening of New Engineering and Technology Centre on Pratt & Whitney's East Hartford Campus** with the opening ceremony taking place in November 2017. The new 425,000 square feet facility will serve as P&W's latest, global engineering hub for the design of Pratt & Whitney's PW1000G GTF engine family and the F135 engine program which powers the F-35 program.



## Expansion of P&W's North American Industrial Footprint - 2

**New Fan Blade Manufacturing Facility at Lansing, Michigan Opened in September 2017:** The facility, based at Pratt & Whitney's AutoAir plant in Lansing, Michigan will produce fan blades for the Geared Turbofan Engines. The expansion of the AutoAir plant is in line with Pratt & Whitney's strategy of ramping up production rate to meet delivery schedules on the 8,000+ units strong order backlog for the GTF engines. The new 93,000 square feet facility supplements an already existing fan blade production line at AutoAir and is part of Pratt & Whitney's over \$1.2 billion investment in cutting edge manufacturing technologies across its global industrial footprint. The facility has witnessed a 10 fold ramp up in production of fan blades over the past 20 months. The new facility will play a key role in Pratt & Whitney's target of delivering over 350 to 400 GTF engines annually.

**Modernization & Upgrade of Longueuil, Canada based Manufacturing Facility by P&W Canada:** Pratt & Whitney Canada is modernizing , upgrading & revitalizing its Longueuil based manufacturing facility with plans to add 3 new production lines incorporating cutting-edge technologies & equipment at an investment of \$275 million through 2018 which include plans to establish a World-Class Centre of Excellence for Intelligent Manufacturing at an investment of \$80 million. The new production lines; featuring a high degree of automation, utilization of closed loop process controls & high precision machining technologies; will produce components for Pratt & Whitney's latest PurePower engine range with the production having commenced in 2015.

**Expansion of Town of Wallkill facilities:** P&W plans to expand its Town of Wallkill based jet engine component manufacturing facility with a \$140 million investment. The expansion will also include employment of 100 new personnel.

**Expansion of Columbus, Georgia based Engine Overhaul Facility for Providing MRO Services on GTF Engine Family:** Pratt & Whitney plans an investment of \$65 million at the Columbus facility which will be the North American maintenance hub for the GTF engine family. The facility will provide MRO support to PW1100G-JM and PW1500G engines. The facility will carry out disassembly, inspection, assembly & testing for the GTF engine family going forward. Additionally, Pratt & Whitney plans to establish a new test facility adjacent to the existing test cell for which the construction work commenced in 2016. The new Engine Test Cell was unveiled by Pratt & Whitney in April 2018 and will enable it to double the facility's engine test capacity.



## Poised to Leverage Continued Momentum in the Helicopter Segment to Drive Further Business Growth - 1

Pratt & Whitney is well poised to leverage the continued growth momentum in the global helicopter & rotorcraft industry, especially, in the civil segment. P&W's helicopter engine business achieved a number of significant milestones across key programs over the recent years which are likely to be key to drive further business growth over near to medium term.

- P&W's PW210A which received certification in 2014 for Leonardo's AW169 helicopter program
- P&W's PT6C-67E engine for the Airbus Helicopter's H175 program too which entered service in 2014
- Earlier in December 2013, Pratt & Whitney's PT6T-9 engine entered service with the Bell's 412 EPI program

Further, growth projections for the civil helicopter segment project stable demand growth over medium term. The demand for civil helicopters is likely to be led by North America amongst traditional markets and Latin America amongst Emerging Markets. Pratt & Whitney; which has been instrumental in the development of the medium twin helicopter segment over the years has significant opportunities to make further inroads & extend its presence in the segment.

### Pratt & Whitney Expanding Service Network in Latin America

Pratt & Whitney has been focusing significantly on expanding its after-sales service network in the Latin America region as of late as part of its sustainment support strategy for civil helicopter operators. **Pratt & Whitney appointed its third Designated Maintenance Facility (DMF) in Brazil in August 2018 at LABACE 2018** with the appointment of Carapicuíba, São Paulo based Helipark Manutenção Aeronáutica Ltda. Helipark will provide line maintenance support as well as mobile repair team (MRT) field services to the Latin America based operators for the following helicopter engines:-

- **T6B-37A, PW206C, PW207D, PW207D1, PW207D2 and PW210A**

**Pratt & Whitney had, earlier in 2017, appointed RICO Taxi Aereo Ltda. and ABA Manutenção de Aeronaves Ltda. as Designated Maintenance Facilities (DMF), as part of its sustainment support network expansion, to support PT6A operators in Brazil as well as local agricultural customers.** Pratt & Whitney's sustainment support network in the region includes:- P&WC Do Brasil maintenance facility, the Covington Aircraft Engines Inc.'s PT6A Satellite, a P&WC parts distribution centre located in Sorocaba, seven Field Support Representatives (FSRs), and specialized Mobile Repair Teams.



## Poised to Leverage Continued Momentum in the Helicopter Segment to Drive Further Business Growth - 2

Pratt & Whitney unveiled 6 New Services for Helicopter Engine Operators at Heli Expo 2018 as Part of its Service Business Transformation. The services launched included:-

- Expansion of Certified Pre-Owned Engine Program for Helicopters with the broadening of program to cover PT6C-67C, PW206C and PW207C helicopter engines, in addition, to the existing PT6A Certified Pre-Owned (CPO) engine program
- Small Fleet Pay-per-Hour (PpH) Solution for Small Fleet Operators
- ESP Platinum Program with environmental repair at overhaul for operators of PW206/207 engines
- Enhanced Line Replaceable Unit (LRU) Availability Program
- Tailored Customer Services for Defence and Security Mission-Readiness
- Mission-Ready Services for Heads of State and other VIP Transport Helicopter Fleets



## Strengthening & Expansion of Global MRO & Engine Support Network with Specific Focus on the Asia-Pacific Region - 1

Pratt & Whitney has been working towards strengthening & expanding its global MRO & support network with specific focus on the Asia-Pacific region, in-sync with the trend of growing commercial airplane deliveries to the Asia-Pacific region by Airbus & Boeing amongst commercial aircraft producers and almost all regional aircraft manufacturers. The long-term industry outlook for the commercial aviation industry projects the Asia-Pacific region to account for the largest (about 34%) share of the global air traffic through 2032. Further, the APAC region is projected to account for about 36% of new 100 seat aircraft deliveries through 2032. Pratt & Whitney consequently is working towards expanding its MRO & support network and establishing its industrial footprint in the region to capitalize on emerging growth opportunities effectively and to enhance the overall quality of aftermarket support.

**Establishment of P&W's Regional Hub & a New MRO & Engineering Facility based at Singapore:** P&W opened its latest MRO & Engineering facility, Pratt & Whitney Component Solutions Pte Ltd (PWCS), based at Singapore's Seletar Aerospace Park in February 2014 which performs component level repair for the PW4000 engine program. The facility (including MRO & Manufacturing facilities) has been established at an investment of \$110 million and will also be the company's regional office/hub for the Asia-Pacific region and will house Global Services Engineering – Asia which will provide specialized aftermarket repair design for P&W's businesses based in the region. Pratt & Whitney has also established a 180,000 square foot manufacturing facility at the site (P&W NGPF Manufacturing Company Singapore Pte. Ltd. or PWMS); its first in Singapore; which produces hybrid metallic fan blade parts and high pressure turbine disks for the P&W's PurePower engine program. The facility commenced operations in February 2016. Additionally, Pratt & Whitney has further strengthened its collaboration relationship with Hanwha Techwin, a key risk revenue sharing partner for P&W in the region since June 2015. P&W and Hanwha Techwin signed a key agreement in September 2016 as per which Hanwha acquired a 30% equity stake in P&W NGPF Manufacturing Company Singapore Pte. Ltd. (PWMS). Additionally, the two partners plan to further expand their risk & revenue sharing collaboration which increases Hanwha's share to 2% & 3% respectively on Pratt & Whitney's 24,000 & 30,000 pound thrust GTF programs respectively. Hanwha also has 1% share on the 17,000 pound thrust GTF engine program.

Pratt & Whitney has announced in early February 2018 in a company press release that its Singapore based engine center, Pratt & Whitney Eagle Services Asia (ESA), a joint venture with SIA Engineering Company (SIAEC), is expanding capabilities and will provide Engine overhaul services for the GP7200 and PW4000 engines. The move effectively brings GP7200 overhaul capability to Asia with Singapore becoming the only facility to offer engine overhaul & repair capabilities in Asia. **Pratt & Whitney plans to further bring engine overhaul capabilities for its PW1100G-JM Geared Turbofan Engine to the Singapore facility by early 2019. Pratt & Whitney announced that the facility inducted its first PW1100G-M engine for overhaul in early January 2019.** The Singapore based engine facility already is the global center of excellence for GP7200 low pressure compressor (LPC) module overhauls and the latest decision will further expand the capabilities to include disassembly, assembly and testing of the LPC module, which includes MRO for the five-stage low-pressure compressor, fan hub, drum, blades and stator assembly.



## Strengthening & Expansion of Global MRO & Engine Support Network with Specific Focus on the Asia-Pacific Region - 2

ESA has invested around \$85 million in the Singapore facility towards modernization & facility upgrade, including, new tooling, machinery & employee training etc. to add MRO capability for the GTF engine family. The addition of the GTF & GP7200 engine MRO capabilities, in addition to the existing PW4000, is going to enhance the strategic significance of the Singapore facility for Pratt & Whitney going forward. **In 2019, a new ground-based flow system for engines is planned to be introduced, in a phased manner, at the facility which will enable improved visibility of operations through the GTF engine overhaul line aimed at enhanced productivity while ensuring space optimization. The flow lines will have a capacity to be able to accommodate 6 GTF engines at a given time.** The GTF engine MRO network now covers 3 continents with engine centres operated by P&W, MTU Aero Engines, Japanese Aero Engines Corporation (JAEC), Lufthansa Technik and Delta TechOps.

**Setting Up of an Engine Training Facility in India:** Pratt & Whitney announced plans in 2014 to establish its third global engine training centre based at Hyderabad airport, India aimed at training engineers & technicians on P&W's existing & upcoming engine models. PW has its other two customer training facilities based in East Hartford, CT and Beijing respectively. The Hyderabad facility became operational in September 2015 and has been set up in view of the projected doubling of installed commercial aircraft base in India incorporating Pratt & Whitney & IAE Engines (V2500) from 300 to 600 over near to medium term with increasing number of aircrafts being ordered by Indian Airlines with P&W's GTF engine technology. The centre plays a key role in PW's global customer support strategy and will provide training to around 2,000 trainees annually for GTF & V2500 engines initially with a potential for doubling of that capacity.

**Renewal of Beijing Customer Training Agreement with CAS:** PW also renewed its existing agreement with China Aviation Supplies (CAS) Holding Company in September 2015 for the Beijing Customer Training Centre for a period of 15 years. The move holds strategic significance for P&W given its strong market presence & market share in the Chinese narrow body aircraft market with a fleet of over 1,200 in-service aircrafts in China powered by Pratt & Whitney and V2500 engines with the number projected to grow rapidly over near to medium term with a growing order book for the GTF engine program. The Beijing based centre has been operational since 2005 and has played a key role in the region's aviation value chain by having provided over 300,000 hours of training on Pratt & Whitney and V2500 engines to over 14,000 students based across China, Russia, Singapore, Thailand & Korea. The centre has upgraded its services with the addition of a training GTF engine at the facility in November 2018 and the utilization of advanced digital tools for training, including, virtual reality.

**Shanghai Engine Centre to provide MRO Support to the V2500 Engine Program:** Pratt & Whitney along-with its JV partner, China Eastern Airlines Corporation Limited (CEA), signed an agreement in February 2016 as per which the Shanghai Engine Center, a JV between Pratt & Whitney and China Eastern Airlines Corporation Limited; will provide MRO support to the V2500 engine program from 2017 onwards. **The Shanghai Engine Center has also received Civil Aviation Administration of China (CAAC) certification in November 2018 to add MRO capability for the V2500 engine program to its aftermarket portfolio.**

**Strategic Agreement with Lufthansa Technik for provision of Mobile Engine Services:** The agreement between Pratt & Whitney and Lufthansa Technik was signed at the Farnborough Air show in July 2016 and covers the PW1100G-JM, PW1500G, PW1900G and V2500 engine families respectively.





## **Strengthening & Expansion of Global MRO & Engine Support Network with Specific Focus on the Asia-Pacific Region - 3**

Pratt & Whitney also signed a key, comprehensive service and support agreement with Hawker Pacific in July 2017 to provide maintenance, repair, engineering and customer support services to P&WC engine customers as well as mobile repair technician services across the Middle East and Asia-Pacific regions. The agreement is broad based and is applicable on:-

1. Most Pratt & Whitney Engine models
2. Diagnostics and prognostics products across the following segments:
  - Business aviation
  - General aviation
  - Offshore
  - Utility
  - Regional aviation
  - Defense and security



## Steady Expansion of Global F135 Engine MRO Network with Upcoming Centres in Turkey & The Netherlands

Pratt & Whitney plans to steadily expand its global MRO network for the F135 engine with upcoming engine centres across key markets & regions.

**PW earlier in 2014 announced plans to establish an engine centre based in Turkey** for its flagship military engine, i.e., the F135 developed and being produced for the F-35 JSF aircraft program. Pratt & Whitney and Turkish officials have signed a letter of intent for long-term cooperation has agreement towards the same. The agreement includes Final Assembly of the F135 engine in Turkey besides Maintenance, Repair & Overhaul(MRO), periodic checking & inspection and upgrade capabilities for the engine. The facility would support Turkey's fleet of F-35 Lightning II JSF with orders for the first 2 F-35s placed by Turkey in early May 2014. The facility would also be the hub for region's operational F-35 JSF aircraft fleet and would provide repair, inspection & MRO services. The agreement has already received the approval from the U.S. Government.

**PW has recently signed an agreement with the Dutch Ministry of Defense in September 2015 to establish a maintenance facility** for the F135 engine. The planned facility will provide MRO&U (Maintenance, Repair, Overhaul & Upgrade) capabilities for the F135 engine program. The establishment of an MRO&U hub based in the Netherlands is likely to play a key & crucial role for the F-35 program in the region given the strategically significant upcoming Finnish, Danish & Belgian fighter jet acquisition programs wherein the F-35 program is a key contender. The initial contract covers the initiation of planning & management activities required for the setting up of the facility & initiation of MRO&U activities scheduled to become operational in 2019. The facility, to be based at Logistics Center Woensdrecht, is planned to eventually have depot level capabilities with functional & operational spectrum of activities ranging from engine assembly, disassembly, cleaning, inspection & testing. The subsequent, follow on contracts between P&W, the Dutch MOD & the F-35 Joint Program Office are likely to follow for the establishment of depot level capabilities at the facility.

## Full Lifecycle Demonstration of the Cold & Hot Sections of the F135 Engine Program

Pratt & Whitney also completed the full lifecycle capability demonstration of the Cold sections of the F135 engine program as part of Accelerated Mission Testing (AMT) in October 2017 which follows the successful full lifecycle demonstration of the hot sections of the engine earlier in 2015. The demonstration included over 9,400 total accumulated cycles (TAC) representing over 14 years of operations and focused on the Conventional Take-Off and Landing engine variant (CTOL).



## Expansion of PW's PT6A Engine Family with the Introduction of New Engine Variants

Pratt & Whitney has expanded its PT6A turboprop engine family with the introduction of new engine variants, namely, PT6A-140A & PT6A-140AG targeted at the general aviation market which brings the total number of engine models available under the PT6A Turboprop engine family to around 70 which power over 125 aircraft programs globally. The latest engine variants feature a high degree of flexibility & ease of maintenance with a modular design & externally mounted fuel nozzles. The engines also feature an increased & one of the longest Time between Overhaul (TBO) intervals besides nil mandatory requirements for engine warm up or cool down. The latest engine variants also optimize performance especially for hot & high conditions & reduce overall operator costs with a 15% enhancement in power to 867 SHP & 5% improvement in specific fuel consumption. The PT6A-140A & PT6A-140AG also offer a competitive Total Cost of Ownership (TCO) with a 50% higher minimum component life limit than close, key competing engine models besides the possibility of extending Time between Overhauls (TBO) by up to 8,000 hours or 12 years based on operations rather than engine cycles. The expansion of the PT6A engine family is likely to enable Pratt & Whitney to further tighten its grip on the general aviation market further with a broader & even more customized product offering.

## Increased Cycle Limits for the PT6A-140 Engine Family, announced in December 2018, to Enhance Operating Economics for Operators

Pratt & Whitney has received approval from Transport Canada for offering a significant increase in cycle times on its PT6A-140 engine family in December 2018. The increased cycle times will be available on the PT6A-140, -140A and -140AG engines respectively. This increase in cycle times will lead to an extension of life of Low Cycle Fatigue (LCF) Parts, including, power and compressor turbine disks and the impeller. This cycle time increase will translate into reduced operating & maintenance expenses for the global operator base of the PT6A-140 engine family.

The cycle limits have been increased by almost 60% for the following parts:-

- Power Turbine Disk: From 12,000 cycles earlier to 16,000 cycles
- Compressor Turbine Disk: From 10,000 cycles to 16,000 cycles
- Impeller: From 19,000 cycles to 29,000 cycles



## **Establishment of Long-term Supply Sources for F135 Engine Components with the Selection of Danish Multicut A/S & Australian Levett Engineering – Pratt & Whitney Offers Growth Option 2.0 Upgrade for the F135 Engine – Received Lot 11 LRIP Contract**

**Pratt & Whitney has established & put in place a key, long-term supply source for its F135 engine components as part of its global supply chain for the F-35 JSF program. The F135 engine powers the F35 Lightning II Joint Strike Fighter and Pratt & Whitney selected Denmark based Multicut A/S, a precision component manufacturer, as the supplier of F135 engine components. The 10 year long term procurement agreement was signed between Pratt & Whitney and Multicut in early November 2015 & marks the beginning of a long-term relationship between the two companies. Denmark is amongst the nine partner nations in the F-35 program (having joined the JSF program in 2002 during the System Development & Demonstration phase) and is in the process of selecting a replacement for its ageing fleet of F-16 aircrafts with the F-35 JSF being touted as the clear favourite (for the potentially \$4.5 billion worth contract) amongst the three competing aircraft programs, namely, the F-35 JSF, F/A 18 Super Hornet & the Eurofighter Typhoon. Reports appearing in media (led by Aviation Week in a news release published on November 18, 2015) indicate that a decision is likely to be made in favour of the F-35 in December 2015 given Denmark's earlier participation in the F-35 JSF program as one of the partner nations and traditional close ties between Danish & the U.S. Air Forces which has been evident with the Danish Air Force's contribution to the F-35 flight test program at Edwards Air Force Base. The Danish Air Force contributed an F-16 aircraft, test pilot & maintenance personnel for the test program. The latest selection of Multicut by Pratt & Whitney, therefore, is a clear indicator towards the outcome of the Danish fighter competition.**

**Pratt & Whitney had earlier in mid-2015 also signed a Long-Term Agreement for procurement of F135 engine components with Australia based Levett Engineering. The 5 year LTA is valued potentially at \$4 million. Levett has been an existing supplier on the F-35 program and has been supplying second & third stage vane transfer tubes to Pratt & Whitney for the F135 engine since 2009. The latest contract includes the supply of transfer tubes and engine main bearing housings & supports.**

### **Pratt & Whitney Offers Growth Option 2.0 Upgrade for the F135 Engine, under its Future Adaptive Spiral Technology approach, as per the F-35 Program's Continuous Capability Development and Delivery (C2D2) Strategy**

Pratt & Whitney offered the Growth Option 2.0 upgrade for the F135 engine, under its Future Adaptive Spiral Technology approach enabling timely insertion & integration of next generation propulsion technologies into current and future programs. The F135 engine powers the F-35 Lightning II aircraft and the upgrade was offered in June 2018 which offers increased Power and Thermal management system (PTMS) capacity, thereby, enabling the F-35 to utilize an enhanced spectrum of offensive and defensive weapon system technologies.

**Pratt & Whitney also received the Multi-Year Lot 11 LRIP Contract for F135 Engines in May 2018 worth \$2 billion. The contract covers procurement of 135 engines, program management, engineering support, production support, spare modules and parts. P&W has delivered almost 375 F135 engines so far (till Lot10 deliveries).**



## Expansion of the PW1000 GTF Engine Family with the Introduction of a Higher Thrust Version for the Airbus A321neo & A321LR Aircraft Programs - 1

Pratt & Whitney has developed a higher thrust variant of its PurePower Geared Turbofan Engine aimed at the A321neo & its long range variant, the A321LR, aircraft programs. The latest entrant to the GTF engine family, the PW1135G-JM engine, will provide a thrust of 35,000 lb as against 33,000 lb provided by the PW1100G-JM engine, which is in the process of certification for the A320 neo aircraft program which was completed in December 2016, thus, making the PW1135G-JM the most powerful of the GTF engines to be introduced by Pratt & Whitney so far. The increased thrust emanates from the additional thrust margin left out by P&W while developing the GTF engine family and thus has not necessitated any structural or design alterations. The expansion of thrust by 2,000 lb will enable the airlines to cover additional distance while operating from airports located at hot & high altitude locations while higher payloads & carrying additional passengers. Additionally, the introduction of the PW1135G-JM engine opens further avenues for the airlines with the opening of new, potential routes; including extended range operations (ETOPS), with the A321LR featuring Additional Centre Tanks (ACT); for them without impacting overall operating cost base & emissions levels. Pratt & Whitney has already been & is likely to further continue with the pursuit of its strategy of developing & introducing engine variants leveraging scale reuse capability which has been built in by PW while developing the GTF based PurePower engine family at the outset. **The strategy is aimed at further expansion of the GTF engine family over near to medium term through accelerated development of new engine programs.** The A321LR has been a key success for Airbus which gives a 25%-30% lower operating costs on routes of upto 4,100 nmi than a traditional Boeing 757 as the wide body for this kind of a route is uneconomical. The A321LR with an increased MTOW of 97 tonnes carries 206 passengers. The A321LR conducted its maiden flight in January 2018 and received EASA as well as FAA certification in October 2018 with deliveries commencing in November 2018. The A321LR has been a leader in its segment so far having accumulated a market share of over 80% in the middle of the market segment. The A321neo has accumulated orders for over 1,300 aircrafts.

PW's GTF technology based PurePower engine family has already leveraged the scalability & reusability significantly for rapid development of engine variants for a number of aircraft programs, which include, Bombardier's C-Series (now A220), Embraer's E2 Jets, Mitsubishi's MRJ Regional Jet & Irkut's MS-21. Pratt & Whitney had over 80 customers, with both announced & unannounced firm orders as well as options, for the PW1000G Engine family as of December 2016. Further, the development of GTF technology based PW800 engine, which received certification in February 2015; is aimed at expanding Pratt & Whitney's presence in the business jet market going forward. The PW800 engine claims a 10% improvement in fuel efficiency over existing generation engines and the engine has been selected by Gulfstream to power its upcoming G500 & G600 business jet aircrafts. The Airbus 321neo powered by the PW1135G-JM engine completed its maiden flight in March 2016.



### **Expansion of the PW1000 GTF Engine Family with the Introduction of a Higher Thrust Version for the Airbus A321neo & A321LR Aircraft Programs - 2**

Pratt & Whitney has received a major boost in order intake with Delta Airlines selecting the PW1135G-JM engine to power over 100 A321neo aircrafts it ordered in December 2017 with the order including a 20 year EngineWise services agreement. The aircraft deliveries are scheduled to commence from Q1 2020. Earlier in 2016, Delta had provided a significant boost to the GTF engine family and the C-Series with the placement of the largest order for the C-Series aircrafts which included firm order for 75 C-Series aircrafts powered by GTF engines in April 2016.





## **Plans for Development of Next Generation GTF Engine Configurations and a Differentiated Engine Architecture Targeted at the Wide Body Aircraft Segment**

Pratt & Whitney plans for making continued, further technological enhancements to its first generation Geared Turbofan technology going forward with plans to develop next generation GTF engine configurations to achieve an overall fuel efficiency improvement in the 20%-25% range by 2025 to effectively meet the open rotor technology being developed by CFM as its flagship, future technology for the narrow body segment. The next generation GTF engine will feature enhanced & optimized bypass & pressure ratios, advanced aerodynamic features, incorporation of material science innovations & cooling techniques & methodologies. Pratt & Whitney plans to increase the bypass ratio by 25% from 12:1 in the present generation GTF engine to 15:1 in the next generation configuration while it also plans to increase the turbine pressure ratio by 20% from 50:1 to almost 60:1 which will drive significant enhancements in performance & efficiencies. PW also plans to change the gear ratio in the next generation GTF engine configurations from 3 to 4 which will optimize the fan speed of the current larger fan further for optimized propulsive efficiencies transpiring into further enhanced fuel burn rate & efficiency. Additionally, Pratt & Whitney is also working towards developing a high performance engine core with a higher thermal efficiency which will translate into same engine thrust level with a higher fuel efficiency. Pratt has also been working towards material science innovations and has developed advanced materials, similar in properties to the CMCs being utilized by arch-rival CFM, with a current Technology Readiness Level 5. Further, PW plans to develop a differentiated engine architecture, leveraging the significant learning curve it has already covered over the development of the Geared Turbofan technology based PurePower engine family, targeted at the wide body aircraft segment in the 70,000-1,00,000 lb thrust range by the middle of next decade as part of its overall strategy of expansion of its commercial segment business.

### **Pursuit of Further Advancement on the GTF Technology in Collaboration with the FAA**

Pratt & Whitney also tested the next-generation Geared Turbofan (GTF) engine propulsor technology with the completion of around 175 hours of ground testing in October 2017. The technology development initiative is being pursued under FAA's Continuous Lower Energy, Emissions and Noise (CLEEN) Program aimed at development of a range of environment friendly technologies. The new demonstrator is a second generation, ultra-high bypass fan design based on an existing development engine from a certified Geared TurboFan product. The initiative is aimed at further advancement of compressor & turbine technology in collaboration with the FAA.



## **Continued Improvements over the GP7200 Engine's Performance, Reliability & Durability. Expansion of Global MRO Network with Plans for Opening of 2 New MRO Centers.**

Engine Alliance (EA); Pratt & Whitney's 50:50 owned JV with GE, for the GP7200 engine program which powers the Airbus A380 aircraft has continued with its series of improvements over the GP7200 aiming to deliver steady improvements in performance, reliability & durability for its airlines customers. EA announced plans to develop an upgraded High Pressure Turbine for the GP7200 program at the Farnborough Airshow held in 2014 with the upgraded HPT aimed at delivering performance retention by up to 0.5% with EGT margin retention by up to 10 degree Celsius. EA has, since then, delivered over 50 upgraded engines, in addition, to introducing a retrofit kit for the global airlines operators. Going forward, EA plans to introduce a new engine control software for the GP7200 program, likely to be available in late 2015, which is aimed at delivering a 1% enhancement in overall fuel burn rate during climb by lowering metal temperature of the HPT by up to 40 degree Celsius. At present, over 92 Airbus A380 aircrafts powered by the GP7200 engines are in active service across 5 key airlines globally led by the Emirates. The continued introduction of enhancements over the GP7200 program is likely to enable Engine Alliance to capture a larger market share over the A380 program going forward vis a vis the Rolls Royce Trent 900. Additionally, it could play a role in the possible introduction of choice of engines by Airbus for the A380neo program over near term, if it materializes.

### **Expansion of Global MRO Network**

Engine Alliance also plans to expand the global MRO network for the GP7200 engine program, as announced by the company at Farnborough Air show in July 2016, with the establishment of two new MRO facilities. The two new engine overhaul centres are to be established at Dubai & Singapore respectively. The new MRO facilities would be created & managed under:-

- Emirates Engine Maintenance Centre (EEMC)
- Pratt & Whitney Eagle Services Asia Pte Ltd (ESA)

The move by Engine Alliance is aimed at supporting the growing global fleet of in-service GP7200 engines and also targets the scheduled first MRO visits for the GP7200 operators to the existing service facilities. The Singapore facility, which is already operational, is the first overhaul center for GP7200 engine program in Asia. Additionally, Engine Alliance is also adding capacity at its existing MRO facilities based at GE Wales and Air France Industries to support the GP7200 engine program effectively.

**Engine Alliance also established a key Centre of Excellence for the overhauls of GP7200's low pressure compressor (LPC) in February 2016 for the Asia-Pacific region.** The COE, managed by & under Pratt & Whitney's Eagle Services Asia Pte Ltd ("ESA"), is based at Singapore.



## Transformation of Global Logistics Function at Pratt & Whitney to Support Rapid Engine Production Rate Ramp Up

Pratt & Whitney has recently transformed its global logistics & supply chain operations aimed at meeting the rapid ramp-up of engine production required to meet the growing order book of aircrafts across leading industry OEMs. The company has invested significantly over the recent years towards modernizing its production methods, processes & technologies besides expanding & transforming its overall assembly operations, operating model and business strategy with the company moving from a centralized Connecticut based assembly operations to a multiple global sites based model. The transformation of global logistics & supply chain operations is consequently aimed at creating standardized logistics processes across the company's key global sites. Pratt & Whitney, as part of the same, opened its latest Northeast Logistics Centre (NELC) based in Londonderry, New Hampshire in September 2015, in collaboration with its global logistics partner, UPS. The 600,000 square foot facility will serve as the company's global distribution hub which will play a crucial role in channelling the distribution between Connecticut, Suppliers & the company's other manufacturing facilities & engine centres located globally.



## **P&W Working towards Doubling Engine Production Capacity by 2020**

Pratt & Whitney has been working towards doubling its overall engine production capacity & levels by the end of decade to meet the required rate of production to deliver on the burgeoning order backlog, especially, for the GTF engine programs. As part of the same, the company has signed over 200 long-term procurement agreements with its global supplier base worth over \$18 billion. The LTAs cover Pratt & Whitney's large engine programs, which include, PW1000G engine family, F135 engine for the F35 program & the PW800 engine for business jets. Pratt & Whitney is also transitioning to a globally diversified industrial base as against the centralized production model which the company has followed traditionally. Pratt & Whitney is also investing to the tune of \$1 billion globally towards expanding the production capacity for its F135 & PW1100G engine programs. The company has also undertaken a transformation of its Middletown, Connecticut based production facility with the modernization of production systems, technologies & infrastructure aimed at expansion of production capacity along-with enhanced operating efficiency. The modernization has included the incorporation of new assembly lines, test cells & cutting-edge technologies. The facility will produce the PurePower PW1100G-JM engine and the F135 engine respectively & features the latest automated horizontal engine assembly lines for the PW1100G-JM engine. The horizontal assembly line enables vertical lift & features rotation capability, thereby, increasing production efficiency with an ergonomically advanced design. The Middletown facility assembled its first set of GTF engines on the horizontal assembly line in April 2015. Pratt & Whitney has also installed the horizontal assembly line at its Mirabel Aerospace Centre, Quebec, Canada.

## **Ramp up of Production Capacity at P&W's Singapore based Production Facility in 2018**

Pratt & Whitney has announced (in February 2018) a significant increase in production capacity at its Singapore based production facility which produces fan blades and high pressure turbine disks for the PW1000G Geared Turbofan Engine family. The announcement is in-line with the company's plans to ramp up production of its PW1000G GTF engine family in 2018. The facility is operated by P&W NGPF Manufacturing Company Singapore Pte Ltd, a joint venture between Pratt & Whitney and Hanwha Techwin Co. Ltd.; and commenced production operations in 2016 and is likely to achieve full production capacity in early 2019.



## Establishment & Expansion of MRO & Support Network for Pratt & Whitney's GTF Engine Family - 1

Pratt & Whitney has been in the process of formalizing & putting the MRO network in place for its GTF engine programs, in collaboration with IAE LLC. P&W entered into definite agreements with its OEM collaborators, namely, Japanese Aero Engines Corporation (JAEC) and MTU Aero Engines AG (MTU) under the IAE umbrella in mid-2015 to formalize a MRO network for the PW1100G-JM engine. Key, initial MRO locations that have been finalized by the alliance partners so far include:-

- Christchurch, New Zealand
- Hanover, Germany based facility operated by MTU
- Mizuho, Japan based facility operated by JAEC
- Singapore as the Asian MRO Hub for the PW1100G-JM with engine service likely to commence from 2019 onwards

**Pratt & Whitney's Columbus, Ohio based Engine Center has played a pivotal role** in the establishment of global MRO network for the GTF engine family. The Columbus based facility has been the nerve center for the GTF's MRO and has truly turned into a MRO Powerhouse with respect to infrastructure, tooling and workforce. The investments in the facility over the recent years have included the addition of capabilities to carry out disassembly, inspection, assembly and testing, establishment of a new test facility adjacent to the existing test cell and an upcoming 200,000-square-foot facility to overhaul GTF engines, which is scheduled to be operational by mid-2018.

**Pratt & Whitney is planning to expand the MRO network further in future** beyond its partner network to include airline centers & independent MRO facilities with the projected growth in engine delivery volumes & their utilization level. Pratt & Whitney plans to have a global network of over 10 MRO facilities in place by 2020 when the first set of GTF engines are likely to require their first overhauls. **Pratt & Whitney also signed a key, strategic agreement with Lufthansa Technik in July 2016** as per which Lufthansa Technik becomes a part of P&W's global MRO network for the PW1000G GTF engine family. Lufthansa Technik will be providing full scale MRO services for Pratt & Whitney's PW1100G-JM and PW1500G engine programs. Pratt & Whitney, thus, now has Japanese Aero Engines Corporation (JAEC), MTU Aero Engines AG (MTU) & Lufthansa Technik as its principal global MRO Service partners.

Other initiatives being undertaken by Pratt & Whitney to support the GTF engine program's EIS include:-

- Establishment of a third customer training facility based in Hyderabad, India which opened in September 2015 besides existing customer training centers located across Beijing & Connecticut.
- Creation of a specialized team of over a dozen Field Representatives focused on the GTF engine program's entry into service. The representatives will work in close tandem with customers & local support teams over a 3 to 6 month horizon.
- Creation of Regional GTF Engine Lease Pool. Pratt & Whitney created an engine lease pool of 10 GTF engines by the end of 2015 distributed across 4 regional storage locations located close to operators.
- Expansion of PW's Global Operations Center as the backbone of Customer Support aimed at providing enhanced customer support to customers & operators.



## Establishment & Expansion of MRO & Support Network for Pratt & Whitney's GTF Engine Family - 2

Pratt & Whitney continues to work towards the expansion of its Global MRO network for the GTF engine family with the addition of multiple new, third party MRO service providers globally in 2018. This is the first time Pratt & Whitney has added third party suppliers to its network.

As part of the first phase of network expansion conducted in **April 2018, Pratt & Whitney announced the addition of 5 MRO providers to its global network**, which included:-

- Turbine Controls
- StandardAero
- TWIN MRO
- ACMT
- Lewis & Saunders

As part of the **second wave of expansion of its Global MRO network**, Pratt & Whitney added **5 new third party MRO service providers to its network in December 2018**.

These providers include:-

- FAG Aerospace (Schaeffler)
- American Cladding Technologies
- Tube Processing
- MDS Coating Technologies
- MB Aerospace

**Pratt & Whitney has also added Horiguchi Engineering Co., Ltd as another supplier of engine stand for its PW1100G-JM engine family**, in addition, to Dedienne Aerospace.

The same was announced by Pratt & Whitney in December 2018.





## Pratt & Whitney Leveraging Big Data to Enhance Engine Performance & Offer Increased Value to Global Operator Base

Pratt & Whitney has been pursuing a series of big data projects aimed at enhancing engine performance & offering increased value based service offerings to its global operator base with company's growing focus on services & engine performance metrics. The company undertook over 14 different big data projects in 2015 aimed at the same. Pratt & Whitney has been focusing on expanding its technical capabilities significantly with reference to predictive analytics as it focuses on enhancing engine performance & offering optimized value to customers through development of analytics based engine diagnostics & proactive maintenance solutions to optimize flight operations & reduce maintenance downtime, as outlined by Matthew Bromberg, President, Pratt & Whitney Aftermarket in a media press release in mid-June 2015. Pratt & Whitney is also collaborating with leading industry & academic partners, which include, IBM, the MIT Leaders for Global Operations and the United Technologies Research Center to pursue a range of analytics driven projects focusing on engines & services solutions. Some of the key projects being undertaken by the company include:-

- Development & testing of predictive models to monitor engine event performance aimed at development of a proactive approach for planning & assessing requirements for engine maintenance translating into performance optimization & reduction of maintenance costs. The project has so far focused on the PW4000 engine models with Pratt & Whitney seeking potential airline collaborations to fine tune processes & procure additional data to develop model performance metrics. The company is also focusing on the development of a similar predictive analytics based model for the V2500 engine program.
- Projects aimed at optimizing PW's engine leasing business are also underway
- Project to refine & optimize the process of allocation of shop visits to PW's MRO partner network.
- Analysis of shop visit pricing for development of predictive maintenance cost models & refinement of their accuracy.

**The success of the projects led to Pratt & Whitney offering its range of predictive analytics and intelligence solutions under the EngineWise services brand in 2017.** With the EngineWise branding, Pratt & Whitney is investing in a range of data acquisition and predictive analytics tools aimed at providing proactive maintenance solutions to operators which increases fleet utilization while reducing operational disruptions significantly. Pratt & Whitney has announced the availability of the new EngineWise data analytics capabilities for its V2500 engine in February 2018 with the company receiving the supplemental type certificate for the Airbus A321 aircraft program. Digital technologies based preventive maintenance services are going to be a key focus area for Pratt & Whitney in 2018 and through the end of present decade as the company works towards offering its suite of data analytics based services on other programs.



## Plans for Further Enhancement of GTF Engine's Fuel Efficiency & Reduction of Emissions with Pursuit of Advanced Technology Research Projects

Pratt & Whitney aims to further enhance the fuel efficiency & overall emissions level for its family of Pure-Power Geared Turbofan engines with pursuit of advance research projects focusing on development & maturation of advanced core engine technologies focusing on engine compressor & turbine modules. The project has been initiated by the Federal Aviation Administration (FAA) under the second phase of Continuous Lower Energy, Emissions and Noise (CLEEN II) initiative. The announcement for Pratt & Whitney's selection was made by the FAA in early September 2015. The current phase follows the first phase of the CLEEN project which was initiated by the FAA in 2010. Under the second phase, Pratt & Whitney will conduct research on advanced technologies for potential incorporation into ultra high bypass ratio engines, which include the PurePower GTF engines; aimed at enhancing overall fuel efficiency & reducing emission levels. The research will specifically focus on enhancing thermal efficiency of the engine core.

Pratt & Whitney is also a subcontractor on another research project, led by the UTC Aerospace Systems, which focuses on development of Advanced Nacelle technologies & Aerostructure's ecoIPS to reduce overall fuel burn rate & engine noise levels. Pratt & Whitney plans to use a PW1000G engine to test the demonstration unit.

The pursuit of these advanced research programs are likely to enable Pratt & Whitney to further refine & advance the capabilities of its Geared Turbofan engine technologies, thereby, further enhancing its competitive positioning & advantage over medium to long term.

Pratt & Whitney is also participating with an industry team to develop & test 3D printed turbomachinery components, including, first rotating part for development programs at Pratt & Whitney. The same was announced in July 2017 with other research partners being Norsk Titanium, the Notre Dame Turbomachinery Laboratory (NDTL) and TURBOCAM International. The industry team is currently exploring the applicability of Norsk Titanium's Rapid Plasma Deposition (RPD) material for turbomachinery applications. Pratt & Whitney & parent UTC have been focusing significantly on 3D printing as a technology with UTC recently establishing an Additive Manufacturing Center of Expertise in East Hartford, CT at an investment of \$75 million.



## **Focus on Rapid, Cost Effective Internal Military Engines Development Capability with the Establishment of GatorWorks**

Pratt & Whitney, aiming to further accelerate its internal development capability for military engines, has established its newly created, internal rapid prototyping entity focused on rapid and agile development of reliable military engines at a much lower development cost. The establishment of new entity was announced by Pratt & Whitney in June 2018. As per Pratt & Whitney, the objective of GatorWorks is “to deliver state-of-the-art engines in half the lead time and for half the cost of traditional procurement cycles”.

GatorWorks will leverage internal enterprise-wide capabilities for the following functions & activities:-

- Rapid prototyping
- Iterative design
- Procurement
- Testing of prototypes and products

GatorWorks will feature & comprise of:-

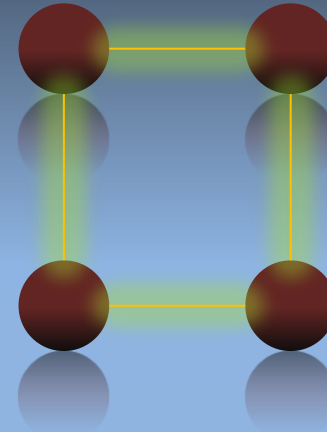
- Cross-functional teams
- Vertical Integration
- Select suppliers and partners



**United  
Technologies**



# Business Outlook for 2019



## Part 2.3



## Outlook

Pratt & Whitney, after a marked resurgence & comeback to the commercial aircraft engines market with its GTF technology has been on a roll having positioned itself on a number of key & upcoming aircraft programs, like the A320neo, A220, Embraer's E2 jets and the Chinese & Russian forays into commercial aviation. At present, Pratt & Whitney is trying to move towards the growth phase by ramping up production rates which were affected due to entry into service issues in 2017 which limited the delivery numbers for both Pratt & Whitney as well as its biggest customer, Airbus.

2018, however, has turned out to be on a different plane altogether with Pratt & Whitney producing over 1,000 large commercial & military engines in 2018, one of its highest in last over 30 years. The company has dealt with in-service issues effectively and has been able to grow its MRO business and expand its global MRO network with the doubling of number of engines overhauled across the globe in 2018 as against 2017. The well developed, mature and deep rooted military aircraft engines business remains the core growth engine and a key anchor for Pratt & Whitney with presence over a number of military aircraft programs in the U.S. market which form & are likely to remain a key element of the force structure of the U.S. Armed Forces over long term, led by the F-35 Lightning II JSF (F135 engine), F-22 Raptor (F119), F-15 Eagle & F-16 Falcon (F100), B-52 and the C-17 Globemaster III (F117) and the most recent KC-46A tanker for the USAF. Further, with a global in-service fleet of 11,000+ military engines, with an average life span of 35-40 years for military aircrafts, Pratt & Whitney is well positioned to make the most of it. Pratt & Whitney has intelligently rebuilt its commercial aircraft turbofan engines business over the past decades by leveraging its technological capabilities and the military aircraft business effectively combining it with the heft & financial resources of its parent UTC as the key global Tier 1 aerospace supplier. The signing of a key \$2 billion multi-year procurement contract with the Department of Defense in 2018 for supplying 135 F-135 engines, which powers the F-35 Joint Strike Fighter Program, gives further medium term visibility to the order book with the selection as engine provider for the upcoming B-21 Raider program further broadening growth horizons for the long term.

Pratt & Whitney's parent UTC continues to reinvent itself with the consolidation of aerospace business, under Collins Aerospace, post acquisition of Rockwell Collins being the latest stroke from the old maestro which will give it further heft in the commercial aerospace business as an even more formidable force to deal with. The company also continues to make significant investments towards future growth technologies & areas, led by:-

- Connected Products as part of focus on Digitalization which also saw expansion of data analytics capabilities with the acquisition of an analytics software company, Predikto Inc., in 2018
- Autonomy
- Electrification



## Outlook

Going forward, Pratt & Whitney needs to maintain its laser focus on disciplined execution as the core element of its strategy to further build on its rapid production ramp up on the GTF engine family with the upcoming Boeing's NMA program likely to be the much coveted growth catapult for which Pratt & Whitney will again have a face-off with the old foe, CFM International as the third industry force, Rolls Royce, has announced its non-participation.

Also, rapid expansion of the global MRO network for the GTF engine remains a key priority for Pratt & Whitney with the company targeting establishment of 8 facilities across the globe by 2020. Further, Pratt & Whitney will need to focus on developing capabilities rapidly in the Hybrid-Electric propulsion domain which remains a key imperative for the industry and a tremendous growth opportunity for future as arch-rival Safran & Rolls Royce are investing significantly towards developing the same.



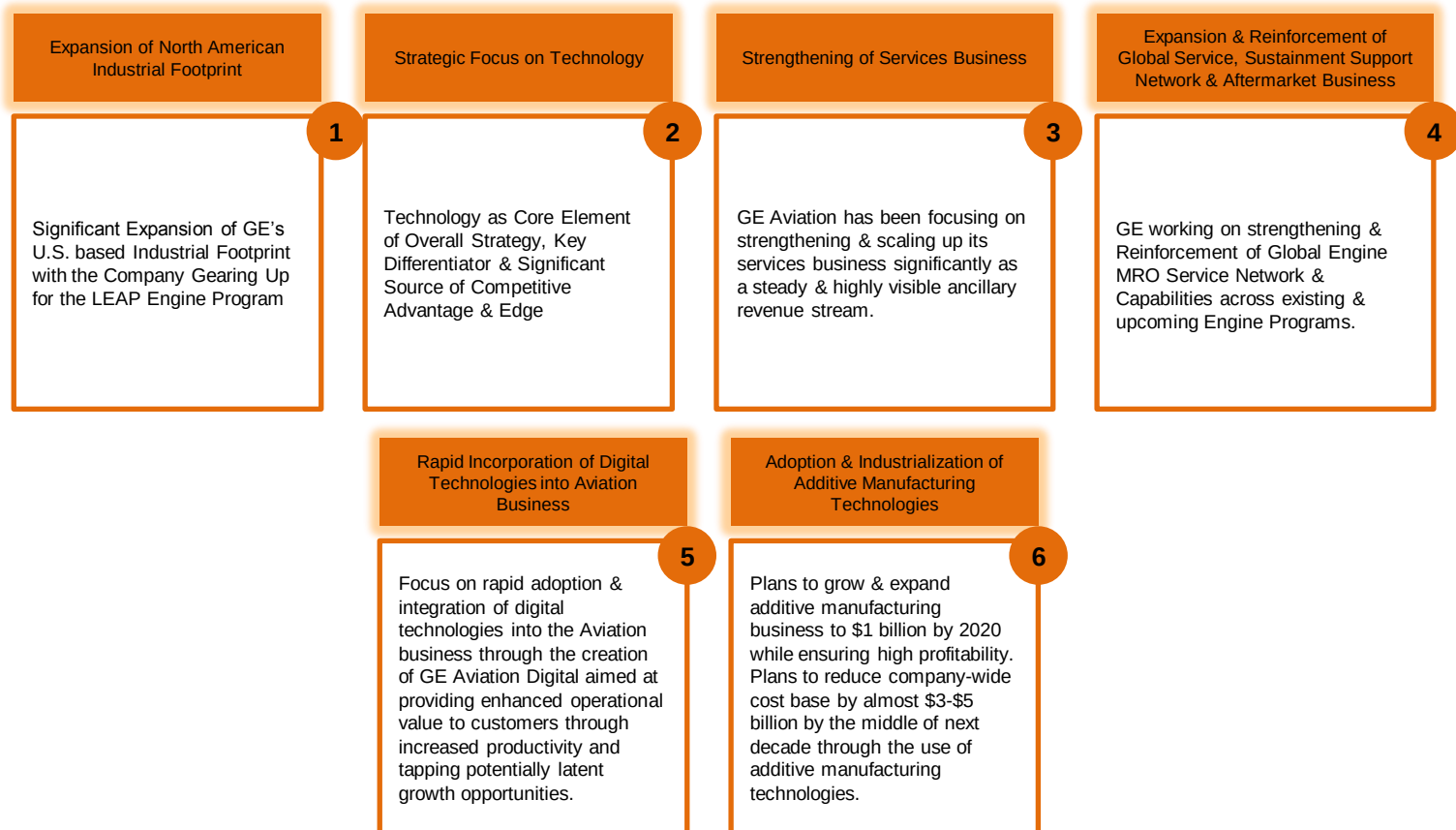
# Strategy Focus



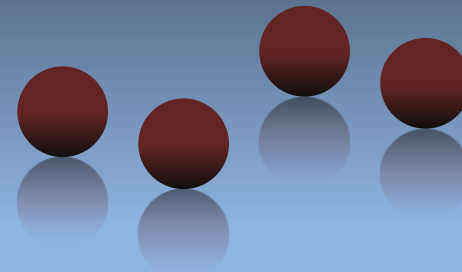
## Part 3.1



# GE Aviation – Overall Strategy Focus



# Key Strategies & Plans



## ■ Part 3.2

# Strategic Initiatives & Developments - 1



## Strategic Initiatives & Developments - 1

1.

### GE's Passport Engine Enters Service

GE's latest offering for the business aviation segment, the Passport engine, formally entered service on the Bombardier Global 7500 business jet in December 2018. This marks a major milestone for GE and is likely to give a significant boost to GE's engines portfolio for the large cabin, long range business jet segment going forward given the engine's scratch up design minimizing cabin noise while boosting performance & efficiency.

2.

### GE Expands Turkey Presence with a New R&D Technology Lab

GE has expanded its presence in Turkey with the opening of a new Additive Manufacturing Technologies R&D Lab in November 2018. The lab's technology development focus would be on next generation manufacturing technologies based on industrial additive manufacturing machines manufactured by GE Additive. GE's total R&D investments in Turkey over the years are worth \$15 million. GE has been operating the Aviation Turkey Technology Center for the last 18 years and employs around 400 engineers.

3.

### Marine Propulsion Business Gets a Boost with New Navy Contract

GE's Marine propulsion business has got a major boost with the award of new contract by the U.S. Navy in November 2018 for 10 Arleigh Burke class destroyers to be powered by 40 of GE's LM2500 marine gas turbines. GE maintains a very strong presence in the U.S. naval ships propulsion systems market with almost 97% of the U.S. Navy's existing commissioned ship fleet powered by GE's gas turbines.

4.

### GE Wins the U.S. Army's ITEP

GE has won the U.S. Army's strategically significant, Improved Turbine Engine Program (ITEP), which entails the re-engining & retrofit on the Sikorsky's UH-60 Black Hawk & Boeing's AH-64 Apache fleet with a new turboshaft engine. The ITEP is also likely to be considered for the FAR leg of the Future Vertical Lift (FVL) program going forward. GE is offering its XT901-GE-900 engine for the ITEP.

## Strategic Initiatives & Developments - 2

5.

### GE to Develop First Civil Supersonic Engine in over 55 Years

GE is working towards the development of the world's first supersonic civil engine class in over 55 years, named Affinity, for Aerion Corporation's upcoming AS2supersonic business jet slated for flight testing in 2023. The Affinity would be a medium bypass ratio engine based on the CFM56's core featuring a twin shaft, twin fan turbofan design while featuring a FADEC.

6.

### Upcoming Test Flight of the GE9X Engine in 2019

The world's largest commercial turbofan engine, the GE's 9X which will power the Boeing's latest 777X aircraft, is scheduled to start with a comprehensive flight test program starting February 2019. The engine has been an ambitious project for GE with years of development effort likely to culminate in planned FAA certification in 2019 followed by entry into service.

7.

### Expansion of GE's MRO Presence in Malaysia

GE is further expanding its MRO footprint in the APAC region with the planned expansion of its Subang, Malaysia based engines MRO facility with an announced investment of \$80 million in September 2018. The expansion is aimed at providing MRO support for the CFM's LEAP engines family in the region.

8.

### GE Divests Middle River Aircraft Systems Business

GE announced in September 2018 that it is divesting its Middle River Aircraft Systems business with the sale to ST Engineering for \$0.6 billion. The deal is likely to be closed in early 2019 post regulatory approvals & clearance. Middle River Aircraft Systems produces engine nacelle systems and thrust reversers and other complex, specialized aerostructures. MRAS was acquired by GE in 1997.

## Significant Expansion of GE's U.S. based Industrial Footprint with the Company Gearing Up for the LEAP Engine Program - 1

GE is working towards expanding its U.S. based industrial footprint with the company gearing up for its latest LEAP engine program. As part of its growth plans and to sustain anticipated growth, GE plans a capital expenditure of \$3.5 billion towards production plant & equipment across its global sites with a majority of the investments to be based in the U.S. with jet engine deliveries from GE & CFM international poised to grow from 2,442 in 2013 to 2,850 in 2016.

- As part of the expansion plans, GE broke ground on a new jet engine manufacturing facility, its seventh facility in the U.S., based at Lafayette, Indiana in late 2014 at an estimated investment of \$100 million. The new 225,000 square foot facility will produce the new LEAP engine family for GE's JV with Snecma, CFM International that has a rapidly growing order backlog. The facility will assemble the engine's hot section as well as conduct the engine's final assembly. GE commenced workforce hiring for the facility in 2015 with plans to increase overall workforce at the site to 200 by 2020 with the LEAP engine program slated to enter service in 2016 which will follow a significant ramp up of production through 2019. The overall cumulative global installed jet engine base of GE & CFM is projected to grow from 34,000 in 2013 to 41,000 engines by 2020. Additionally, GE announced plans in January 2015 for further expansion of the Lafayette facility at an additional investment of \$15 million. The investment is aimed at building MRO Capabilities for the LEAP engine family at the facility through addition of new machinery, equipment & tooling. GE also plans to further expand overall headcount at the site to 230 by 2020. The expanded 300,000 square foot Lafayette facility is scheduled to become operational in January 2016 and will play a key role in the LEAP engine's entire life cycle ranging from assembly to MRO support.
- GE has also expanded its West Jefferson based machining facility to expand overall production capacity to support the rapidly growing order backlog for the LEAP engine program. The company has invested to the tune of \$65 million towards expansion of the facility by another 80,000 square foot. GE opened the expanded facility in July 2015 and plans to increase the headcount at the facility through 2017 with addition of 105 new positions. The latest facility expansion is part of GE's larger plan of expanding and modernizing its North Carolina based industrial base with planned investments of \$195 million through 2017.
- GE has announced plans to expand its Hooksett, NH based facility by over 55,000 square foot at an investment of \$50 million aimed at expansion of overall production capacity to support production rate increase in-line with a growing order backlog, especially, for commercial engines led by the LEAP engine program. The Hooksett facility produces rotating engine parts for GE's commercial & military jet engines. The company held a groundbreaking ceremony at the site in May 2015.

## Significant Expansion of GE's U.S. based Industrial Footprint with the Company Gearing Up for the LEAP Engine Program - 2

- GE Aviation has also established a new 170,000 square foot production facility for producing advanced composites based at Asheville, NC at an estimated investment of \$125 million. The facility commenced production in October 2014 and will produce Ceramic Matrix Composite (CMC) based engine components, specifically, the high pressure turbine shroud for the LEAP engine program with each engine to incorporate 18 such turbine shrouds. The company plans to ramp up workforce further to 340 by 2019. The Asheville facility has achieved a key milestone with the delivery of 25,000<sup>th</sup> CMC component from the facility in August 2018.
- GE plans to assemble its Passport engine at the Strother, Kansas based manufacturing facility. The company plans to invest over \$7 million towards machinery & equipment at the Strother facility to assemble, maintain, repair & overhaul the Passport engine which will power Bombardier's under-development Global 7000 & 8000 business jet programs. GE plans to install equipment & tool and train employees in 2015 which will be followed by rolling out of first Passport engine in 2016. The facility is scheduled to reach full production capacity in 2017. The Passport engine made its first test flight in December 2014.
- GE Aviation is also in the process of setting up a new test site for the GE9X engine program at its Evendale based headquarters. The facility will test the combustor for the 9X engine incorporating the 3<sup>rd</sup> generation TAPS(Twin Annular Pre-Swirl) design.
- GE Aviation plans to invest to the tune of \$27 million at its Newark, Delaware based facility through 2019 which focuses on maturation of CMCs(Ceramic Matrix Composites) prior to full-scale production.
- Earlier in 2012, GE Aviation formed a JV with Parker Aerospace, christened Advanced Atomization Technologies, LLC based at Clyde, New York for production of advanced fuel nozzles for GE's commercial jet engine programs. Also, GE Aviation acquired Morris Technologies in 2012 to strengthen its industrial base for additive manufacturing for jet engines. GE Aviation has been producing the fuel nozzle tip for the 9X engine program based on Morris additive manufacturing technology.
- GE has also invested to the tune of \$350 million over the last 2 years towards revamping & expanding its Southern Ohio based operations, specifically, its Cincinnati, Dayton, and Peebles based facilities.
- GE's latest Limestone County based production facility is scheduled to commence operations from the fall of 2018. The latest GE facility will be the only U.S. based facility to mass produce Silicon Carbide based ceramic fiber used in the production of CMC tapes used in the production of aircraft engines. The facility has been established at an estimated investment of \$200 million and will start producing CMC tape from Q4 2018 with the production of Silicon Carbide based fibre likely to start from April 2018.



## Technology as Core Element of Overall Strategy, Key Differentiator & Significant Source of Competitive Advantage & Edge

Depth of technological capabilities has been the core strength, key differentiator & a significant source of competitive advantage for GE Aviation over the years with the company investing significant resources towards development of a range of path-breaking R&D technologies over the past decades, especially, pertaining to material science innovations that are proving to be a definitive edge over its LEAP & 9X engine programs. Key technological developments include:

**Ceramic Matrix Composites(CMC):** CMC is GE Aviation's revolutionary material science innovation that aims to replace the conventional Nickel super alloys over the upcoming LEAP & Passport engine programs. The material, essentially, Silicon Carbide made through a proprietary resin transfer molding process delivers significant advantages in terms of weight, enhanced capability to withstand very high temperatures and enhanced durability & fuel efficiency. GE Aviation plans to initially use CMCs in the Passport engine's exhaust mixer, centerbody and core cowls with plans to further increase the scope of usage over medium term.

**Advanced Combustor Technologies featuring Twin Annular Pre-Mixing Swirler (TAPS):** TAPS has been developed by GE under FAA's CLEEN program aimed at funding the development of advanced technologies which will reduce CO2 emissions, enhance fuel efficiency & reduce noise levels. TAPS II makes its commercial debut on the CFM LEAP engine scheduled for EIS in 2016. GE also has been working on TAPS III since 2011 which will be incorporated into the 777X program. TAPS III is GE's third generation combustion system that pre-mixes air & fuel prior to combustion for leaner burn & lower emissions than conventional combustion systems. TAPS III also features utilization of additive manufacturing for fuel nozzle tips besides a new combustor dome design & utilization of CMCs for inner & outer liners aimed at increasing durability and for ensuring enhanced lean burn combustion.

**TitaniumAluminide (TiAl) Alloys** for Fan Blades translating into reduction of fan weight to the tune of 50%.

**Oxide Oxide(OxOx):** GE has been pursuing R&D on CMCs & OxOx since the 1980s and introduced them in 2011 over its F414 engine exhaust seals on the F/A 18 Super Hornets for enhanced engine durability. Passport will be the first non-military engine to receive OxOx CMCs.

**SuperFinish:** is a proprietary surface coating developed by GE Aviation for application on high pressure compressor blades & blisks that renders fan blades almost 4 times smoother than traditional blades enabling the air to pass more smoothly over them as compared to traditional blades & helps blades retain their efficiency & performance by creating a protective layer translating into lower fuel consumption and enhanced fuel efficiency. Passport integrated propulsion system will be the first non-military engine to feature Super Finish and along-with CMCs & OxOx, the passport engine will deliver an 8% enhancement in overall fuel efficiency as compared to traditional, comparable engines.

## Strengthening & Reinforcement of Global Engine MRO Service Network & Capabilities across Engine Programs - 1

GE Aviation has been strengthening its global MRO service network across engine programs to enhance quality of service levels. The company has taken a number of strategic measures of-late towards the same.

- **Strengthening of Service Capabilities in Europe:** GE Aviation has strengthened its relationship with GA Telesis Engine Services(GATES), a Finland based full-service engine MRO service partner, by signing a 3 year agreement in April 2014 to provide replacement engine parts for the CF6-80C2 engine program. The agreement is likely to further strengthen GATES MRO capabilities & service quality. GATES currently provides engine MRO service for the CF6-80C2 & CFM56-5B/5C engine programs with an annual overhaul capacity of 200 engines.
- Further **strengthening of presence & capabilities in the Asia-Pacific region with the expansion of GE's Propeller MRO facility** based at Brisbane, Australia which provides MRO services for the region's installed regional turboprop aircraft fleet comprising of the Bombardier Q400 & Q400NextGen, SAAB340 & Fokker 50 respectively. The investments in the facility are targeted at enhancing productivity which include focus on reduction of lead time, inventory levels& freight costs. The facility has strategic significance given the fact that it is the only certified facility in the APAC region to provide MRO services for Bombardier's Q400 Turboprop aircraft program and in view of the significant growth of the Q400 aircraft program's installed aircraft base as well as new order intake for Bombardier from the region with several key orders emanating from China & Russia besides plans by Bombardier to locally produce Q400NextGen aircraft in Russia.

**Plans for Establishment of MRO Facility for LEAP-1C Engine in China** which powers COMAC's C919 aircraft as the sole powerplant. The The Aero Engine Maintenance Training Center (AEMTC) based in Guanghan City since the last two decades will provide MRO & training services for the LEAP-1C engine program.

- **Establishment of Engine MRO Service network for the AW189 Helicopter Program** with the signing of first engine service agreement between GE Aviation & the Lynn, MA based Bristow Group in Feb 2014 for the CT7-2E1 turboshaft engine under GE's Maintenance Cost Per Hour (MCPH) program which powers the latest twin-engine AW189 helicopter from the AgustaWestland stable. The Bristow Group has been providing engine MRO services for the CT7 turboshaft engine for over a decade, including, the CT7-8A engine that powers Sikorsky's S-92 helicopter program. Extending the agreement to the AW189 program further strengthens GE Aviation's service network. GE Aviation has delivered over 30 CT7-2E1 turboshaft engines for the AW189 program since the engine's certification in mid-2013. Further, with a substantial order backlog for the AW189, increasing utilization & growing customer interest from the offshore oil & gas exploration sector for the program; GE's focus has been to establish & expand the service network for the program with similar arrangements/agreements likely to be made across other key markets over near term. GE has also been focusing on helicopter operators for long-term engine service agreements. GE signed two engine service agreements with Era Group in September 2015 for the CT7-2E1 & CT7-8A turboshaft engines that power the AW189 & S-92 helicopters respectively. The agreement will cover the AW189 & S-92 helicopter fleets operated by Era Group, one of the world's largest helicopter operators.

## Strengthening & Reinforcement of Global Engine MRO Service Network & Capabilities across Engine Programs - 2

- GE Aviation has been working towards **strengthening of its long-term OnPoint solution offerings for global business jet operators**. GE Aviation expanded its operator base to include Boeing Business Jets powered by GE & CFM56 engines in Q4 2013 under its OnPoint service agreement for engine MRO services. GE Aviation further plans to develop a new & enhanced service program for the upcoming Passport integrated propulsion system being developed for Bombardier's Global Express 7000 & 8000 long-range jets based on significant prognostic capabilities of the Passport engine emanating from the incorporation of Full Authority Digital Engine Control System(FADEC).
- Significant strengthening of **GE Aviation's engine MRO services network for Business Aviation** in the Asia-Pacific region with the appointment of ExecuJet's global MRO facilities as authorized service centres for the CF34 engine program. The agreement was announced in mid-November 2015. The agreement adds ExecuJet's MRO facilities based in China, Australia(3 facilities based in Sydney, Perth & Melbourne), Malaysia, South Africa(Cape Town & Johannesburg) and Nigeria respectively to the GE's global MRO network for the CF34 engine program. The latest ExecuJet facilities will carry out line maintenance inspections, routine installed engine maintenance and repair & overhaul services in addition to providing GE Aviation's OnPoint service solution and the engine warranty support. The expansion of the global MRO network will enable GE to provide enhanced sustainment support for its global in-service base of 6,000 CF34 engines powering the Bombardier Challenger aircraft across 200 operators globally. GE Aviation had earlier in April 2015 appointed Bombardier Singapore Service Centre as its authorized service facility for the CF34-3 engines. In October 2014, GE had appointed Lufthansa Technik AERO Alzey GmbH (LTAA) as the authorized service facility for the CF34 engine program. GE had also appointed Hong Kong based Metrojet and Jinan based STAECO as the authorized service centres for the CF34-3 engine in Q4 2013. Signing up of new service facilities further extends GE Aviation's global service network that comprises of over 50 authorized service centres and significantly strengthens engine MRO service capabilities for the CF34, H80 & M601 engine programs in the strategically significant Asia-Pacific region which is likely to register a significant growth with respect to business aviation over near to medium term. GE Aviation also signed a maintenance support agreement with Lufthansa Technik AERO Alzey GmbH (LTAA) in mid-November 2015 as per which LTAA will provide mobile maintenance support team for the CF34 engine powering the Bombardier Challenger & Embraer Lineage aircraft programs across Russia, Europe, the Middle-East & Africa. GE's stepping up of efforts seem to be showing results with GE Aviation ranked as number 1 in the 2015 Turbine Powerplant Product Support Survey conducted by the Pro Pilot magazine in October 2015.
- **GE Aviation appointed Switzerland based RUAG Aviation as its Authorized Service Centre for the CF34-10E engines in May 2016** as part of efforts to expand global MRO network for its iconic engine program for regional aviation segment.
- GE Aviation appointed Indianapolis, Indiana based Comlux as an authorized service centre for the CF34-3 engine program in Q3 2013 that will provide line maintenance inspections, engine maintenance, OnPoint service solution & engine warranty support. **GE Aviation also introduced the new Prognostic Health Management Plus solution for its CF34-3 engines** which power Bombardier's Challenger 600 aircraft family. The solution will provide enhanced engine prognostics in addition to C-FOQA Centerline in real-time.

## Establishment of Next Generation Global MRO Network for the GENx & GE9X Engine Programs

GE has been working towards expanding & reinforcing its global MRO network as part of its plans to establish a next generation support network for its flagship nx engine series aimed at sustaining & augmenting continued growth momentum for the program with a rapidly growing order backlog. GE's nx engine program entered service in 2011 and powers Boeing's 787 Dreamliner & the 747-8 aircraft programs. Additionally, the GENx engine program; after receiving production output increase in 2012 & 2013; is scheduled to undergo a further ramp up of production rate from 2014 onwards with the production output slated to grow by almost 50% to reach 300 engines annually. Key measures being taken up by GE to expand & reinforce its global MRO network for the nx program include:

- **Mainland Europe:** GE Aviation signed an agreement with Air France-KLM for the Maintenance, Repair & Overhaul of GENx-1B engine program under GE branding while GE will provide technical support & assistance on scoping of overhaul work and licensing of component level repair, material support as well as conducting technical training programs for personnel.
- **Polish Site for XEOS JV with Lufthansa Technik:** GE Aviation has signed a key agreement with Lufthansa Technik to establish a new MRO Facility based in Poland for GENx-2B and GE9X engine programs respectively. The selection of the site in Poland was announced by the two companies in December 2016 following the agreement to form the JV earlier in mid-2015. The two companies plan to invest over \$267 million in the new JV, christened XEOS, for setting up of the 350,000 square feet facility which is scheduled to commence operations from September 2018 with the MRO capability for the GE9X engine to be available from 2021. The facility will initially focus on providing MRO support to a global fleet of around 500 in-service GENx-2B engines.
- **Asia-Pacific:** GE Aviation formed a JV with its Taiwan based long-term service partner, Evergreen Aviation Technologies Corp. (EGAT), christened, GE Evergreen Engine Services to provide MRO service for the GE's nx engine series. GE Aviation plans to provide tooling & personnel training in late 2014 which will enable the facility to achieve limited MRO capability in early 2015 with full overhaul capability likely to be achieved by 2019 when the nx engine program achieves full scale production.

GE Aviation's existing MRO network for the nx engine program includes:

**Europe:** GE Aviation's overhaul facility based at Prestwick, Scotland and Lufthansa Technik for providing MRO services for the GENx-2B that powers the Boeing 747-8.

**Latin America:** GE Aviation's Petropolis, Brazil based MRO facility

**Middle East:** The Abu Dhabi Aircraft Technologies (ADAT), based in UAE is GE's MRO partner in the Middle East region. ADAT received FAA approval in late 2013 for the GENx-1B and -2B engine configurations.

**Asia-Pacific:** Air India for providing MRO services for GENx-1B.

## **GE Plans to Strengthen Presence in the General & Business Aviation Segments with the Development of New Turboprop Engine Family incorporating Cutting-Edge Military & Commercial Engine Technologies**

GE plans to expand its presence in the general & business aviation segments with the development of its new turboprop engine family. The development of the new turboprop engine family targeted at the business & general aviation segments is aimed at expanding GE's market presence in the segment, with specific focus on the 850-1600 SHP power output range; & to take on Pratt & Whitney's dominant position led by the company's iconic PT6 engine family. GE has been researching the turboprop aircraft market for almost past half decade and the company plans to leverage & integrate some of its available technological capabilities & proprietary technologies developed for the military & commercial aviation segments into the latest turboprop engine family with GE aiming to give a fillip to its market presence in the business & general aviation segment. The engine has also been selected by Textron Aviation for its latest single engine turboprop aircraft as announced by the two companies at the NBAA 2015. The latest engine is the first engine model in the GE's planned latest engine family and features a power output of 1300 SHP besides a 20% enhancement in fuel efficiency along-with a 10% increase in power output driven by a 16:1 overall pressure ratio. The engine is based on the modular architecture of T700/CT7 turbo shaft and features an all-Titanium 3D aero compressor design, usage of additive manufacturing for structural components & an integrated electronic propulsion control. The engine also features an extended 30% additional time between maintenance overhauls with GE's incorporation of cutting-edge design & production technologies. GE plans to conduct the detailed design review for the new engine program in 2017 which will be followed by the full engine tests likely in 2018 with GE's new Turboprop Centre of Excellence (TCE), to be established in Czech Republic, Europe at an investment of \$400 million (as announced in September 2015) to spearhead the development, testing & production of the new engine. The facility is also conducting detailed research on a proposed 5,000 SHP turboprop engine which GE is targeting at the regional aircraft market. The engine program will be derived from the GE38-1B turboshaft engine; which is GE's largest turboshaft engine being developed for Sikorsky's CH-53K military helicopter program, and will leverage the facility's (TCE) turboprop technological expertise and GE's technological prowess in commercial & military engine technologies. The GE38-1B engine powered the maiden flight of a CH-53K prototype successfully in late October 2015 paving the way for engine qualification.

## **Broadening of Engine Fuel Sources for Engine Testing with the 10 year Procurement Agreement for Synthetic Biofuel**

GE Aviation is exploring sourcing of alternate fuel sources for jet engine testing aimed at broadening of fuel sources & extending capabilities in the alternate fuel domain. The company plans to use cellulosic synthetic biofuel as the alternate fuel source for engine testing as part of the new engine development process. GE Aviation has signed a 10 year agreement with the Washington D.C. based D'Arcinoff Group (DG) for the purchase of cellulosic synthetic biofuel with a minimum procurement level of 500,000 gallons annually besides options to purchase upto 10 million gallons of fuel annually. The synthetic biofuel would be used at GE Aviation's primary jet engine testing facility based in Peebles, Ohio in combination with traditional jet fuel. GE Aviation consumes almost 10 million gallons of jet fuel annually across its engine testing facilities and the supply of synthetic biofuel scheduled to begin from 2016 will:

- Enhance the company's capabilities & experience in the alternate biofuels domain
- Significantly reduce environmental impact by reducing emissions as part of the engine testing process
- Give a significant fillip to the biofuel industry

GE Aviation has been exploring & experimenting with biofuels since 2007 in collaboration with industry OEMs, airline partners & government entities with the testing having ranged from the F414 jet engine over an F/A-18 Super Hornet fighter jet to the CF6 commercial engine over the Boeing 767 commercial airplane. The D'Arcinoff Group (DG) will produce the cellulosic synthetic biofuel at its upcoming D'Arcinoff Group Energy Program facility based at Hudspeth County, Texas which is scheduled to become operational from 2016.

## Focus on Strengthening & Scaling Up Services Business with Expansion of Service Offerings - 1

GE Aviation has been focusing on strengthening & scaling up its services business significantly as a steady & highly visible ancillary revenue stream. The same is also evident with almost 80% of the company's order backlog as of December 31, 2013 comprising of long-term service contracts. Further, the services component of GE Aviation's order backlog registered a 22% growth in 2013 besides a 37% growth in services segment revenues for the company for 2013 translating into \$700 million with the company focusing on services business and a growing global installed base of engines made by GE & CFM international respectively. Additionally, first of the 27,000 in-service CFM56 engines reached the overhaul stage in 2013 as the average age of the operational CFM56 engine base is pegged at 13 years indicating towards significant anticipated potential growth in the revenue stream likely to originate from the MRO services required by these engines over medium term transpiring into a significant growth avenue for both GE & Snecma.

As part of its growing services focus, the company has signed a series of large ticket long-term engine maintenance contracts of-late which include:-

**A \$16 billion contract with Emirates at the Dubai Air Show 2015 for OnPoint solution agreement covering provision of MRO services for over 300 GE9X engines** which will power the 150 Boeing 777X aircrafts to be procured by Emirates over the next 12 years. The \$76 billion order for 150 777X aircrafts was placed by Emirates in 2013 at the Dubai Air Show & included 300 GE9X engines worth \$16 billion. Emirates also signed another OnPoint solution agreement worth \$36 million for providing maintenance & inventory support for avionics, electrical power & mechanical systems for the Emirates in-service fleet of over 100 Boeing 777 aircrafts & for over 44 777-300ER aircrafts which are on order.

**Extension of TrueChoice Flight Hour Agreement by Emirates at the Farnborough Air Show 2016 for its 54 GE90-115B engines** powering its fleet of Boeing 777-300ER aircrafts. The agreement is valued at over \$400 million.

Signing of an MoU for expansion of TrueChoice Flight Hour Agreement by Volga-Dnepr Group worth \$1.4 billion **at the Farnborough Air Show 2016**.

**Long term service agreement between CFM International & LATAM** for provision of continued support for operations & maintenance of LATAM's 35 CFM56-5B engines.

**Strengthening of Support capabilities in the Middle-East** with the opening of GE's Middle East Aviation Technology Center in early November 2015 based at the Dubai Airport Free Zone. The center will leverage cutting edge technology, which includes, data analytics, aviation domain experience & software capabilities for enhanced productivity, performance & efficiency. The centre will provide end-to-end customer solutions to regional customers ranging from engines, services to software. The facility will:-

- Provide enhanced level of service support for GE's engines & systems
- Be the Regional & Customer support hub
- Enable provision of data-driven customized services & solutions powered by GE's Predix software platform

**GE has also signed an MOU with Dubai's Museum of Future** at the Dubai Air Show 2015 for accelerating innovation & advanced technology across the globe.



## Focus on Strengthening & Scaling Up Services Business with Expansion of Service Offerings - 2

GE, additionally, has been focusing on further developing & expanding its range of services across respective segments:

**Organization of Service Offerings with the Introduction of TrueChoice Product Suite in April 2016:** The new engine maintenance offerings from GE Aviation, Services includes 4 distinct product families with overall focus on providing a broad range of options and customization across the entire engine lifecycle to the global commercial aviation industry. The four product families include:-

- **TrueChoice Flight Hour** with focus on optimization of overall cost of ownership over the engine lifecycle with flexibility in risk transfer & payment options.
- **TrueChoice Overhaul** with focus on time and material overhauls through customized worksopes focusing on workshop visit, economic priorities & ownership horizon.
- **TrueChoice Material** with focus on New & Used OEM Parts, advanced repairs & technology upgrades.
- **TrueChoice Transitions** with focus on offering broad range of options for changing ownership horizons.

**Flight Efficiency Services(FES):** GE Aviation's Flight Efficiency Services entail analysis & utilization of real-time data analytics aimed at optimization of an aircraft fleet's flight performance transpiring into enhanced operating efficiencies for the airline. The FES components include:-

- **Fuel Efficiency Enhancement:** Analysis & enhancement of aircraft's fuel efficiency by up to 2% through real-time data analysis, constant monitoring & benchmarking.
- **Flight Analytics:** Analysis of flight data across the entire flight envelope and benchmarking it against historical flight data & utilization of software & simulation models to generate scope for efficiency enhancement or Required Navigation Performance(RNP).
- **Navigation services** which include Performance based Navigation(PBN), optimum flight path design & consultation reduce track miles flown
- **Fleet synchronization** service that provides optimum traffic flow and sequence management to reduce hold times while landing along with increased predictability.

GE has expanded its services solutions offerings for Southwest Airlines with the inclusion of Flight Efficiency Analytics service in September 2015 covering Southwest's fleet of over 700 Boeing 737 aircrafts. The service will include Fuel Management Solution & will provide enhanced operational efficiency. GE already provides Flight Operational Quality Assurance (eFOQA) analytics and crew recovery solutions to Southwest Airlines. GE also signed up Qantas Group for the FES service in September 2015 which includes provision of Flight Efficiency & Flight Operations Quality Assurance (eFOQA) analytics for Qantas & Jetstar.

GE Aviation signed up EVA Airways & Garuda Indonesia as latest airlines opting for its Flight Efficiency Services in Q4 2013. Brazil's GOL & Green Skies Airlines and Alitalia have been other key global airlines that have opted for GE Aviation's FES while Emirates is conducting 18 month trials for the Fleet synchronization & flow management system at Dubai International Airport. With strong growth projections for the airlines industry over medium to long-term emanating from long-term, sustainable drivers and a high oil price environment; GE Aviation's Flight Efficiency Services offering is poised for significant growth with GE's substantial focus on services apart from equipment.

## Focus on Strengthening & Scaling Up Services Business with Expansion of Service Offerings - 3

GE, additionally, has been focusing on further developing & expanding its range of services across respective segments:

**TRUEngine LLP:** GE Aviation introduced its TRUEngine LLP(Life-limited parts) program in September 2013, with the extension of its existing TRUEngine program, aimed at providing potential customers an easy & reliable means to identify & evaluate life-limited parts based on their operating history in OEM configuration. The service covers most critical engine components & other rotating components that include disks, spools and shafts. The program has been rolled out by GE for the CFM56, CFM34, CFM6, GE 90 & GENx engine programs.

**Enhancement of the OnPoint Solution Offering for Business Aviation Operators & Development of an Enhanced Service Program for the Passport Integrated Propulsion System based on its significant, sensor based Prognostic Capabilities.**

**Development & introduction of 4 new software applications powered by GE's Predix which were showcased at the Farnborough Air Show 2016, which include:-**

**Flight Phase Analyzer** that optimizes flight operations of airline customers based on analysis of engine operating data which is uploaded on a continuous basis and generation of key insights.

**Water Wash App** provides visibility into the significant time on wing gains which could be achieved through effective water wash. The app also provides tools to optimize the procedure for obtaining maximum fuel efficiency gains.

**ClearView Invoicing App:** Provides new functionality & visibility into invoicing details to customers which aims to significantly simplify & enhance overall customer experience with the TrueChoice Flight Hour maintenance program .

**Engine Health Validator App:** Evaluates engine condition and enables reduction in time & costs in the transfer of engine assets between lessees.

## Pursuit of Advanced Technology Research Programs to Further Extend Competitive Edge - 1

GE Aviation has been pursuing a series of advanced technology research programs aimed at enhancing its competitive advantage & edge with respect to propulsion technologies. GE has been working, especially, with the U.S. Army over 6 cooperative agreements for developing technologies for the U.S. Army's next generation turboshaft engine design. Key programs include:-

**Future Affordable Turbine Engine (FATE) program:** The FATE program was awarded to GE by the U.S. Army in 2011 to develop & demonstrate advanced technologies for next generation turboshaft engines in the 5,000-10,000 SHP class with potential applications, which include, upgrades to existing rotorcrafts and to propel future rotorcrafts under the Future Vertical Lift (FVL) program. The technologies may also be incorporated into existing engine programs which include the T700 & GE38 besides their potential inclusion into latest engine programs like the GE3000 which is going to be offered for the U.S. Army's Improved Turbine Engine Program (ITEP) to provide next generation medium lift propulsion. GE plans to use additive manufacturing for producing structural components for the FATE program. The FATE engine program aims to meet a number of performance parameters as compared to current generation engines, which include:-

- Reduction in fuel consumption by over 35%
- Achieving a 80% improvement in power to weight ratio
- Achieving a 20% improvement in engine design life
- 45% reduction in production & maintenance costs

GE completed component tests for the FATE program in October 2015, which included, testing of compressor, combustor & turbine. These tests were preceded by inlet particle separator testing and will be followed by full FATE engine demonstrations. The inlet particle separator is aimed at enhancing the engine's capability to separate dust & sand while ingesting air. GE has recently started using an advanced & improved sand ingestion facility to determine & test engine's sand separation efficiency. GE is also collaborating with the U.S. Army to design & test an improved inlet particle separator for the GE3000 engine which would be capable of being retrofitted on existing helicopter fleet as well.

## Pursuit of Advanced Technology Research Programs to Further Extend Competitive Edge - 2

**Advanced Affordable Turbine Engine Program (AATE):** The AATE program aims to validate technologies required to meet & achieve performance goals outlined for the U.S. Army's Improved Turbine Engine Program (ITEP). As part of the same, GE has completed engine testing on its second GE3000 engine. GE plans to use additive manufacturing for producing structural components for the AATE program.

**Advanced Variable Speed Power Turbine (AVSPOT):** The AVSPOT program is a collaboration between GE, U.S. Army & NASA which aims to conduct research for development of turbines capable of meeting the range & life requirements of current & next generation rotorcrafts. The program aims to develop turbine engines which can allow the rotor to be slowed for high speed rotorcraft performance improvements.

**Autonomous Sustainment Technologies for Rotorcraft Operations (ASTRO):** The ASTRO program has been initiated by the U.S. Army's Applied Technology Directorate (AATD) and aims to develop & demonstrate a range of technologies for designing & maintaining current & future vertical lift fleets.

**Capabilities Based Operations & Sustainment Technologies – Aviation (Cost-A):** The program is aimed to demonstrate a broad spectrum of integrated, high value diagnostics, prognostics & system health management technologies aimed at reducing scheduled & preventive inspections while increasing safety.

**Rotorcraft Advanced Engine Integrated Control System (RAEICS):** The RAEICS program is intended at developing innovative, high performance systems for the FVL platforms under a 24 month project. The program was awarded to GE Aviation by the U.S. Army to carry out conceptual design & trade analysis.

**GE has also forged a cross functional partnership with the European industry & academia for the development of technologies and crew interface standards for the next generation of passenger aircrafts under the Joint Undertaking for the Clean Sky 2 program.** The same was announced at the Farnborough Air Show 2016. As per the agreement, GE will work towards integrating 5 key technologies under project REACTOR aimed at addressing the pilot workload & stress within the cockpit of the future under new Human-Machine interaction interface.

## **GE Banking on Improved Engine Turbine Program (IETP) to Spearhead Medium to Long Term Growth**

GE Aviation has been banking heavily on the U.S. Army's Improved Engine Turbine Program (IETP) to spearhead and drive growth over medium to long term given the strategic nature and the size, scale as well as scope of the program. The IETP program aims to replace the existing T700 engines on around 3,000 UH-60 Black Hawks and AH-64 Apaches in-service with the U.S. Army over medium term with a new, more powerful & fuel efficient engine to be developed under the IETP program. GE faces a stiff head on battle with Advanced Turbine Engine Co. (ATEC); a JV between Pratt & Whitney and Honeywell; over the IETP program with both the teams having been involved in the development of their respective competing engine technologies & concepts since many years. The DoD awarded preliminary design review phase contracts to both GE and ATEC worth \$102 million and \$154 million respectively in August 2016 with work likely to be completed in August 2018. The Army will then test and choose amongst the two competing engine designs for the engineering & manufacturing development phase (EMD phase) of the program. The Army anticipates a total development cost of \$720 million for the IETP program.

The performance specifications for the engines to be developed under the IETP program have been stringent with an almost 25% improvement in fuel consumption at a 50% increased power output of 3,000 SHP (as against 2,000 SHP for the T700 engines used in Black Hawks and Apaches) to partially offset the weight increases over the years, a 65% improvement in thrust to weight ratio and a 20% improvement in engine design life.

GE has offered its GE3000 turboshaft engine for the preliminary design phase of the program featuring a number of cutting edge technologies, including, a simpler single spool design architecture reducing the maintenance footprint & costs by up to 35% while improving reliability, usage of cutting-edge manufacturing technologies like additive manufacturing and 3D Aerodynamics and incorporation of proprietary material sciences advances like Carbon Matrix Composites (CMCs) while also incorporating technologies from the FATE program. Pratt & Whitney and Honeywell, on the contrary, have pitched (under ATEC) their HPW3000 turboshaft engine featuring a two spool architecture while claiming potential annual savings to the tune of \$1 billion for the Army in fuel and maintenance costs respectively. It is going to be a high stakes battle for GE which aims to protect & keep its existing market positioning over the T700 program with the GE3000 engine and after having lost two key military aviation programs to Pratt & Whitney recently, including, the F135 engine for the Lockheed Martin's F-35 program and the upcoming B-21 Raider LRS-B program awarded to Northrop Grumman Corporation. The IETP battle, however, has significant long term implications for both the competing teams given that the engines developed for the IETP & FATE programs are likely to be utilized for the JMR-FVL program going forward.

## Expansion & Industrialization of Additive Manufacturing & CMC Manufacturing Capabilities - 1

GE Aviation has been working actively towards expanding & industrializing its additive manufacturing and CMC manufacturing capabilities as well as footprint with continued series of investments across North America and Europe as part of the group's underlying strategy to develop a global network of additive manufacturing capability. GE has invested over \$1.5 billion towards manufacturing & additive technologies at its Global Research Centre, has developed additive applications across 6 GE businesses and has been pioneering services applications across the company. GE has also taken a number of key strategic initiatives as part of the same, which include:-

**Acquisition of a 75% stake in Germany based Concept Laser GmbH**, a leading global supplier of additive manufacturing equipment, worth \$599 million in October 2016 with the agreement enabling GE to take full ownership of Concept Laser in a specific number of years. The step marks a key move in GE's strategy & commitment for additive manufacturing as part of its long term plan. Concept Laser is a key player in the global metal additive manufacturing space with around 200 employees, manufacturing operations in the U.S., China and a global network of around 35 distributors. The move strengthens GE's capabilities in the additive manufacturing machinery space significantly besides further expanding GE's additive manufacturing footprint in Europe which already has over 12,000 employees, including, 400 GE engineers working across Italy, Poland, Czech Republic and Germany on the development of GE's latest Advanced Turboprop Engine (ATP) for Textron Aviation's latest single engine turboprop aircraft, the Cessna Denali. The ATP engine will have almost 35% of the components made from additive manufacturing, which is far more than for any production engine in the aviation history ever. GE engineers have been able to reduce the number of parts from 855 traditionally to a mere 12 complex parts produced through additive manufacturing representing a mere 35% of the total engine architecture.

**Expansion of Additive Manufacturing Capacity in North America** with GE Aviation's Auburn, Alabama plant using additive manufacturing technologies extensively to produce aviation components, including fuel injector nozzles for the LEAP engines first of which entered service smoothly over the Airbus A320neo program in July 2016 and have been performing well. These are the first sophisticated aviation components to be produced on a mass scale using additive manufacturing technology. GE's Auburn plant has around 28 additive printing machines, in addition, to the Cincinnati, Ohio based Additive Technology Centre which also houses a significant additive manufacturing capacity. Collectively, GE Aviation plans to produce over 4,000 fuel injector nozzles for LEAP engines in 2016 leveraging the Cincinnati & Auburn facilities with the production output scheduled to go up to 12,000 in 2017 and 40,000 by 2020. Additionally, the GE3000 and FATE engine programs, which represent the future of GE's rotorcraft engines business, too, use additive manufacturing technologies as well as CMCs, in addition, to their usage in military engines & engines being developed for general aviation segment and as these programs take shape GE Aviation would need additional production capacity for additive & CMC manufacturing capacity.

## Expansion & Industrialization of Additive Manufacturing & CMC Manufacturing Capabilities - 2

### Establishment of New Facilities in North America aimed at expansion of CMC production capacity significantly

**GE Aviation broke ground on the construction of two new adjacent production facilities based at Huntsville, Alabama in June 2016** to mass produce Silicon Carbide, a key ingredient used in the production of Carbon Matrix Composite (CMCs) components. The move is in-line with GE Aviation's strategy to expand the utilization of CMCs across its range of jet engines across segments and land based gas turbines used in the production of electricity with GE putting a fully integrated supply chain in place for the mass production of CMCs to support the same. GE Aviation is investing to the tune of \$200 million towards the establishment of these two production facilities in Alabama which are scheduled to become operational during the first half of 2018.

The first of the two production plant will produce silicon carbide (SiC) ceramic fiber which will be the first such facility for SiC ceramic fiber production in the .S. while the second facility will use this ceramic fiber to produce unidirectional CMC tape used for fabrication of CMC components. GE is also establishing an additional production facility for SiC ceramic fiber in Japan through NGS Advanced Fibers Co. owned by Nippon Carbon (50%), GE(25%) and Safran (25%). NGS is currently the only producer of large scale SiC ceramic fibers in the world and the upcoming Huntsville facility will complement NGS.

**GE Aviation also opened its latest CMC manufacturing facility based at Asheville, North Carolina in 2014** which fabricates CMC shrouds for the LEAP engine's high pressure turbine section.

GE is anticipating a ten fold increase in its demand for CMCs over the next 10 years given their increasing utilization over a broad range of GE Aviation's upcoming engine programs, which include:-

- LEAP Engine Program which replaces the iconic CFM56 engine program
- GE9X engine which will power the Boeing's upcoming 777X program
- GE3000 Engine for the IETP Program which aims to replace the legendary T700 engine
- U.S. Army's FATE Program



## Expansion & Industrialization of Additive Manufacturing & CMC Manufacturing Capabilities - 3

### Acquisition of Key Additive Manufacturing Equipment Suppliers by GE in September 2016

GE Aviation announced acquisition of two key suppliers of additive manufacturing equipment in early September 2016 worth \$1.4 billion. The two suppliers, include, Sweden based Arcam AB and Germany based SLM Solutions Group AG respectively. The acquisitions by GE are aimed at further expanding its presence & capabilities in the strategically significant digital industrial space and are part of GE's underlying plan to expand its additive manufacturing business to \$1 billion by 2020 with high profitability while reducing costs across the organization to the tune of \$3-\$5 billion by the middle of next decade. GE has already invested to the tune of \$1.5 billion in manufacturing & additive technologies since 2010 which has enabled GE to:-

- Develop additive applications across six businesses
- Create new group-wide service applications
- Expand technological capabilities significant with GE having filed over 346 patents in the powder metals space alone so far

The acquisitions will further enable GE to:-

- Expand new product design envelope which will enable substantial reduction in product cost
- Development of new productive models for services cost & delivery
- Develop leadership in design & manufacturing of highly valued parts.
- Further enhance its competitive positioning with reduced development costs & increased profitability
- Bolster overall business by integrating well with GE Store while leveraging GE's core strengths in materials, software & product design

## Focus on Rapid Adoption & Integration of Digital Technologies into Aviation Business – GE Takes the Digital Leap

### Creation of New Digital Business by GE in March 2016

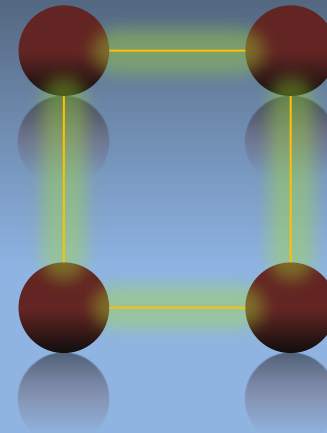
GE Aviation announced the creation of a new digital organization in March 2016 incorporating & integrating the range of digital technological capabilities across Aviation under one business entity. The move by GE is aimed at providing enhanced value to customers by leveraging the increased utilization of digital technologies & analytics based data science for creation of revolutionary & transformative service offerings & solutions.

GE Aviation Digital has been established as a separate division under GE Aviation which supports customer operations through unique & transformative services as well as solutions based on

- Data sciences
- Deep domain expertise
- Software capabilities

These services & solutions based on digital backbone & data sciences aim to deliver enhanced value to customers through increased productivity and tapping latent growth opportunities. Development of cloud based operating system providing data analytics solutions for industrial internet in the form of Predix and connecting the continuous stream of data generated by the in-service fleet of aircraft systems & engines to it through Teledyne's wireless technology under GE's strategic Digital Alliance partnership agreement with Teledyne controls in November 2016 is going to take things to the next level by enabling much more effective engine maintenance operations. The defining factor will be the increased & continuous data monitoring through Teledyne's GroundLink technology, which utilizes higher bandwidth and cellular wireless networks for data transfer, which is likely to translate into much higher effectiveness of the predictive maintenance and health monitoring systems than current generation solutions which are based on limited data sets & streams.

# Business Outlook for 2019



## ■ Part 3.3

## Outlook

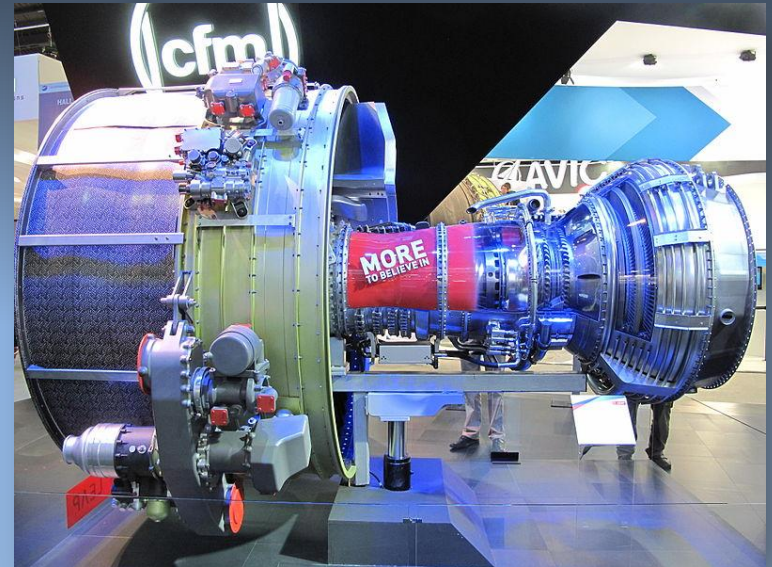
GE Aviation has a lot of action lined up for itself in 2019 starting with the ambitious and world's largest commercial turbofan engine, the 9X, slated to power the flight test program of Boeing's 777X starting in February 2019 culminating in years of efforts translating into actual flight test hours. The 9X subsequently is scheduled to receive FAA certification in 2019 itself which will be followed by Entry into Service (EIS) somewhere in early 2020 with the deliveries of the 777X aircraft to customers by Boeing. Given GE's immaculate track record of trouble free EIS of its engines, the 9X should prove to be no exception either.

The long term outlook for the global commercial air travel remains robust with fundamentals firmly in place as the current aviation supercycle; the longest in the industry's history so far starting in 2004; continues unabated driving demand for new aircrafts as part of fleet expansions as well as replacement. The crude oil prices have softened again, amid projections of slower global economic growth owing to impact of trade wars; after marking an abrupt spike since 2014 towards the second half 2018, thereby, boosting profitability of airlines globally which touched almost \$37 billion in 2018 and likely to be sustained over near term as well. The backdrop augurs well for GE Aviation as the leading propulsion systems manufacturer commanding dominant market shares in the narrow body as well as wide body aircraft segments of the global commercial aircraft market.

The Passport's entry into service over Bombardier's Global 7500 program augurs well for GE going forward with the behemoth looking to capture a larger share of the large cabin, ultra long range business jet market over medium term with strong competition from the Pratt & Whitney's PW800 and a looming threat from Rolls Royce's upcoming Pearl engine family for business jets announced in May 2018 at the EBACE. The first member of the family, Pearl 15 has already captured and powers the Bombardier's under development Global 5500 & 6500 aircraft programs. In the narrow body space, ramping up production of LEAP engines rapidly to the levels of CFM56 by 2020 remains a key priority for CFM International. CFM delivered 739 LEAP engines during the first 9 months of 2018 and it is on track to deliver 1100 to 1200 LEAP engines for the year 2018. In the rotorcraft space, the outcome of the U.S. Army's Improved Turbine Engine Program (ITEP), likely to be decided in 2019, is going to be crucial for GE, given the strategic nature and scope of the program which aims to replace & re-engine the existing fleets of UH-60 & AH-64 Apache helicopters estimated at around 7,000 to 8,000 helicopters. CFM56's burgeoning service numbers as they reach their mid life of operating cycle continues to churn MRO revenue streams for GE.

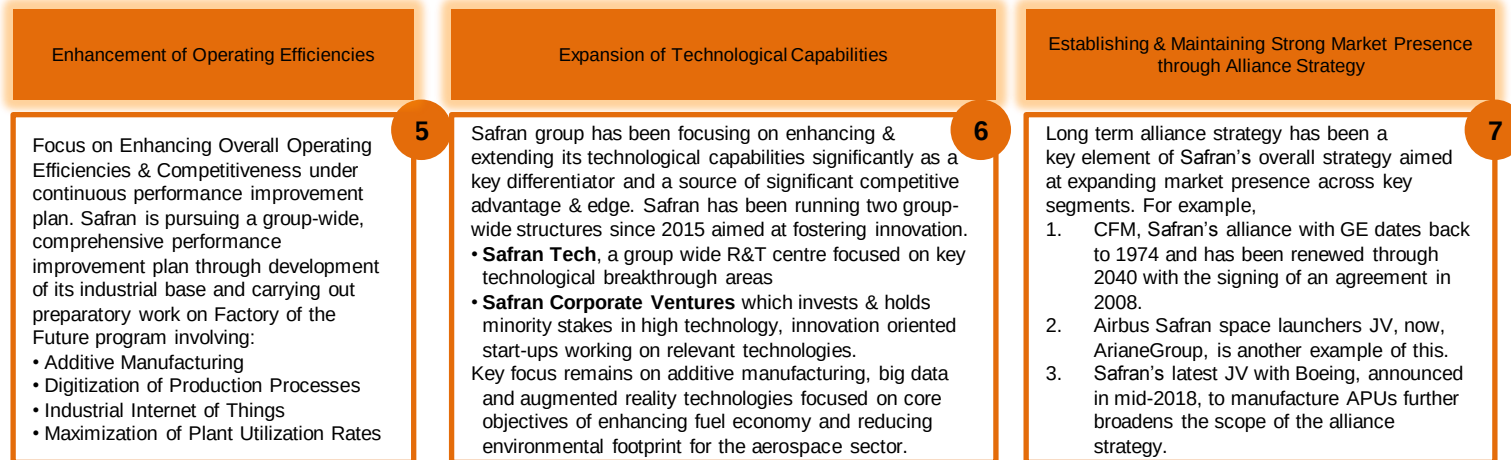
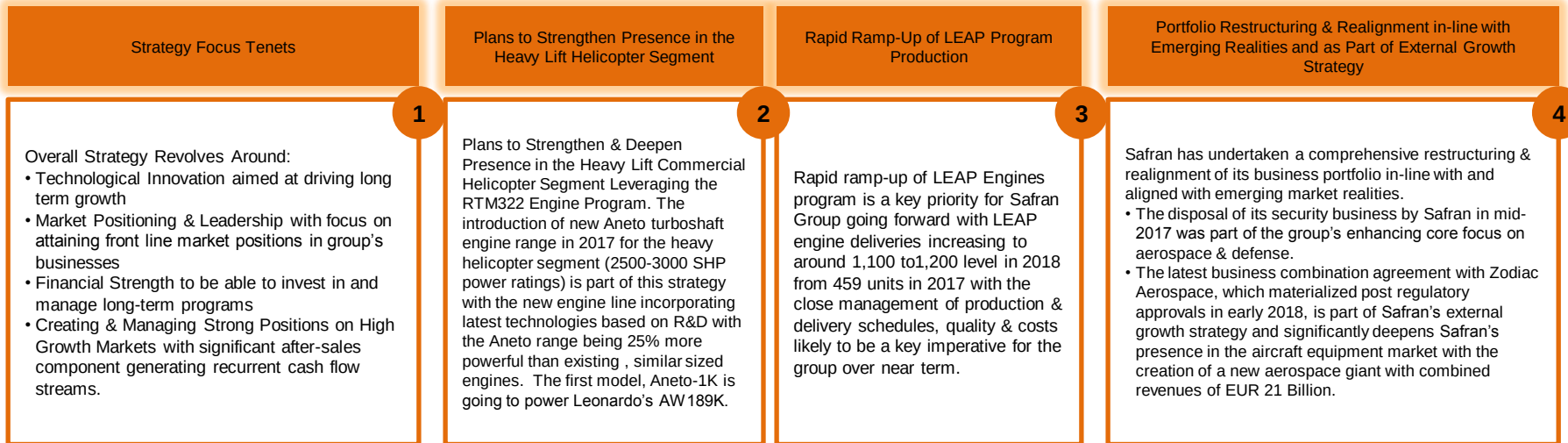
Overall, with a strong aviation supercycle in full bloom, huge order backlog for services, multiple new aircraft programs in the offing, especially, the upcoming ITEP & the Affinity supersonic business jet engine program and rapidly evolving digitalization & additive manufacturing prowess as well as scale of operations across GE's industrial base driving huge productivity & efficiency gains. However, GE as an organisation has a major task ahead of it to reinvent itself by putting its house in order with profitability pressures, leadership crisis and mounting pressure from investors for restructuring of the legacy conglomerate organizational structure.

# Strategy Focus

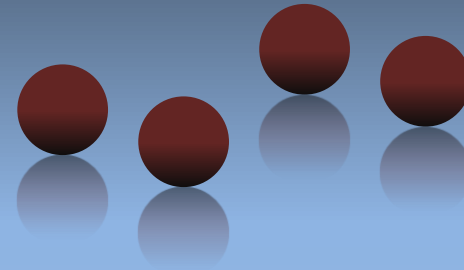


## ■ Part 4.1

# Safran SA – Overall Strategy Focus



# Key Strategies & Plans



## ■ Part 4.2



## Overall Strategy Revolves Around Technological Innovation, Market Positioning & Leadership and Financial Strength

Safran's overall strategy revolves around the following four key cornerstones:

- Technological Innovation aimed at driving long term growth with focus on emerging technologies landscape marked by additive manufacturing, augmented reality, big data riding on the backbone of digitization of production processes and industrial facilities.
- Market Positioning & Leadership with focus on attaining front line market positions across group's businesses from a long term growth perspective. The undisputed market leadership of the CFM56 engine program in the medium thrust engines segment proved the rationale behind the strategy.
- Developing Financial Strength to be able to invest in and manage long-term programs effectively
- Creating & Managing Strong Positions on High Growth Markets with significant after-sales component generating recurrent cash flow streams

Creating & sustaining strong positions on long-term markets, across both traditional & emerging markets, has been the group's preferred strategy focusing on growth and profitability emanating from sustained demand levels.

## Plans to Strengthen & Deepen Presence in the Heavy Lift Commercial Helicopter Segment Leveraging the RTM322 Engine Program

Safran Helicopter Engines has significantly reinforced its product development capabilities, product portfolio & has fine tuned its overall strategy for the heavy lift helicopter segment, focusing on the 3000 shp turbine engine range, with the acquisition of Rolls Royce's 50% stake in the RTM322 engine helicopter engine program for €293 million, which was being pursued jointly by both the companies under a JV.

With the acquisition, Safran Helicopter Engines opened further market avenues & plans to make further inroads into the commercial helicopter market targeting the heavy lift segment as a strategy to partially offset ongoing spate of spending cuts in the military helicopter segment. Safran Helicopter Engines plans to & is working towards developing & introducing a commercial variant of the RTM22 which will power commercial heavy lift helicopters in the 2500-3000 hp range competing directly against GE's T700 & the CT7 which powers the AgustaWestland's AW189, Sikorsky's S-92 & the Bell's upcoming 525 Relentless program.

The acquisition also brought along a significant global installed base of the RTM322 engine which powers 3 key European helicopter programs, namely,

- The Tri-Engine AW101 Merlin
- Dual Engine NH90 made by NH Industries
- WAH-64D Apache Longbow helicopter made by GKN Westland under License from Boeing

The significant installed engine base is likely to ensure significant, steady revenue stream originating from MRO activities over medium to long-term.

Safran Helicopter Engines also unveiled a new helicopter engine family for the heavy lift segment in 2017. The new engine family, named, Aneto is focused on the heavy lift segment in the 2500 to 3000 SHP power rating. The introduction of new Aneto turboshaft engine range in 2017 for the heavy helicopter segment is part of this growth strategy. The new engine line incorporates latest technologies based on R&D with the Aneto range being 25% more powerful than existing , similar sized engines. The first model, Aneto-1K is going to power Leonardo's AW189K (as an engine option) as its launch customer. The Aneto is being targeted at the following market segments-

- Offshore Oil & Gas Transport
- Military Transport
- Search & Rescue
- Firefighting

Aneto-1K made its maiden flight in March 2017 and is scheduled to enter service in 2018.

## LEAP Engines to Spearhead CFM's Renewal of Core Civil Aircraft Engine Portfolio in over Four Decades

Safran has been pacing rapidly towards renewal of its core civil aircraft engine portfolio, its first in over 4 decades, led by the development of the latest LEAP(Leading Edge Aviation Propulsion) engine program introduced in 2008 as the successor to its flagship, best-selling CFM56 engine, which commands a 80% share in the short to medium haul aircraft market. The LEAP engine program, being developed in collaboration with GE, will offer a 15% reduction in overall fuel consumption with a commensurate reduction in CO2 emissions besides achieving significant reduction in Nitrogen Oxide emissions & noise levels. The program features following significant technological advances in material science, aerodynamics & lean combustion technologies: increased usage of composite materials made from Resin Transfer Molding(RTM) process, usage of Titanium Aluminide (TiAl) alloy for turbine fan blades which will translate into a 50% reduction in the weight of the blade, usage of Ceramic Matrix Composites(CMC) for turbine material, incorporation of a blisk fan in the compressor, incorporation of a second-generation Twin Annular Pre Swirl (TAPS II) combustor and a high bypass ratio of around 10-11:1. The LEAP program has so far been selected for the following key aircraft programs:

- COMAC's C919 Regional Jet under the LEAP-1C configuration.
- Airbus's A320Neo Program under LEAP-1A configuration as the second engine option where it competes directly against P&W's PW1000G engine.
- Boeing 737 MAX Program under LEAP-1B configuration as the aircraft's exclusive power plant.

The LEAP aircraft engines entered service in 2016 and the group plans to further extend its strong market positioning led by the CFM56 engine in the short to medium haul aircraft segment with the upcoming LEAP range. Additionally, the purchase contracts being signed by airframe manufacturers for the LEAP engine have a 7-10 year service contract & spare parts provisioning & availability ensuring enhanced visibility of secondary revenue streams.

**CFM's order book for the LEAP engine program has been growing at a phenomenal rate.** Safran booked engine orders worth over EUR 13.12 Billion LEAP engines in 2017 with orders for 2,870 new LEAP engines & 474 CFM56 engines with Safran's total order backlog standing at 13,728 engines as of December 31, 2017 and valued at EUR 44.67 Billion. CFM registered new orders for 858 LEAP and CFM56 engines worth \$15.6 billion at the Farnborough Airshow 2018. Safran has been working hard towards transitioning from CFM56 to LEAP engine family across its global industrial base as the production of LEAP ramps up. Safran plans to deliver around 1100 to 1200 LEAP engines for 2018 as against 459 in 2017. Thus, close management of production & delivery schedules, quality & costs are likely to be a key imperative for the group over near term as the LEAP program takes the leap after 2 years of its entry into service having recorded over 600,000 flight hours & 290,000 flight cycles over an in-service global fleet of 210 aircrafts (as of February 05, 2018) operated by 33 operators. **Safran delivered its 1,000<sup>th</sup> Leap engine and the in-service LEAP powered fleet crossed the 1.5 million flight hours landmark in July 2018. CFM also delivered its first LEAP engine powered A321neo aircraft in July 2018.**

## Acquisition of Zodiac Aerospace as Part of External Growth Strategy

Safran Group's business combination agreement with Zodiac Aerospace has created the world's third largest aerospace company (except OEMs) with combined revenues of EUR 21 Billion and headcount of around 90,000. Further, the move is likely to bring in a lot of synergies and scale advantages given the complementarities of the product portfolio of Safran and Zodiac. **Safran's product portfolio is focused on:-**

**engines, landing gear, wheels and brakes, nacelles, onboard power systems, actuation and avionics**

**while Zodiac's portfolio is focused on-**

**Seats, cabin interiors, power distribution, lighting, fuel, oxygen and fluid systems and safety equipment.**

The move is part of group's external growth strategy aimed at:-

- Enhanced focus on the Global Aerospace & Defense market
- Expansion of product portfolio in the Aerospace & Defense markets
- Further reinforcing market presence & positioning as the leading Tier 1 supplier of aircraft equipment to the OEMs & airlines operators
- Achieving further decorrelation of group's revenues from new aircraft production cycles through upscaling of after sales business and cabin refit activities
- The move also expands Safran's technological capabilities significantly with the Zodiac's well developed electrical systems portfolio which will enable Safran to accelerate its readiness for the upcoming era of electrical aircraft propulsion systems

Safran received regulatory approvals for the deal in late 2017 and the combined entity came into existence in February 2018. However, integrating Zodiac's operations into its own and realizing synergies while executing the operational turnaround will be a key & crucial challenge for Safran over near term.

## Streamlining, Preparing & Gearing Up Industrial Base for the LEAP Engine Program Production - 1

CFM has been working since 2013-14 towards establishing, streamlining & preparing industrial base for the production of the LEAP engine program which entered service in 2016. Snecma signed an agreement with its existing supply partner, Mecachrome in mid-April 2014 for the establishment of a production supply line for the Titanium Aluminide based turbine fan blades for the LEAP engine program. The Titanium Aluminide, an alloy of Titanium & Aluminium which replaces Nickel as the preferred material for low pressure turbines which is capable of handling very high temperatures of upto 750 degree celsius while reducing the weight of the blades by almost 50% transpiring into a significant 15% enhanced fuel efficiency for the LEAP engines.

Snecma in collaboration with its supplier, Mecachrome established a new manufacturing facility for the production of TiAl based turbine fan blades at an investment of EUR 60 million which follows an innovative resin transfer moulding process. The process renders the fan blades to be flexible which untwist as the turbine's rotational speed increases. The production line is based at Mecachrome's existing production facility based at Sablé-sur-Sarthe. The production line commenced volume production from 2015 with ramp up of production output taking place in 2016. **The ramp up is aimed at achieving the targeted overall production rate of one fan blade every 3 minutes slated for 2019 as part of the LEAP engine program's production target of 1,700 engines a year from 2019, the same level as the CFM56 & one of the highest in the industry;** with CFM scheduled to transition from CFM56 to LEAP.

**Safran has also set up a shared manufacturing facility based at Rochester, NH in the U.S., in collaboration with its U.S. based partner Albany to produce woven composite parts for the LEAP engine program** with another twin facility being established in Commercy, France which has been producing blades & casings for the LEAP engine program since 2014. The LEAP engine production is in the ramp-up phase with 1100-1200 engines scheduled to be delivered in 2018 up from 459 in 2017.

### Streamlining, Preparing & Gearing Up Industrial Base for the LEAP Engine Program Production - 2

**Safran also opened a new parts production plant located in Poland for the LEAP engines in July 2018. The new 8,000 square metre facility; which employs a headcount of 200 & established with estimated investment of EUR 48 Million; will produce low-pressure turbine blades for LEAP engines. The new facility is scheduled to deliver 180,000 blades in 2018 with a further ramp up likely to take the production output to 350,000 in 2019 and 800,000 by 2022.**

The LEAP engine program entered the comprehensive 3 year ground & flight test certification program with the first LEAP engine, LEAP-1A, entering the ground testing phase in September 2013. The program successfully entered service over the A320neo program in August 2016 post certifications. Currently, the LEAP program has recorded over 600,000 flight hours & 290,000 flight cycles over an in-service global fleet of 210 aircrafts (as of February 05, 2018) operated by 33 operators. Safran delivered its 1,000<sup>th</sup> Leap engine and the in-service LEAP powered fleet crossed the 1.5 million flight hours landmark in July 2018.

**The LEAP engine program is scheduled for a rapid production ramp up from 100 engines in 2016 to deliveries of 459 engines in 2017, scheduled deliveries of 1100-1200 engines in 2018, 1800 in 2019 and 2,000 in 2020. In just 4 years, the LEAP program is scheduled to reach 2,000 annual engines production level which took 20 years for the CFM56 program to reach courtesy digital backbone driving production and other industrial processes across the ecosystem.**

## Focus on Enhancing Overall Operating Efficiencies & Competitiveness - 1

Safran is pushing ahead with its continuous performance improvement plan. The company rolled out its **Safran+ initiative** aimed at **enhanced operating efficiencies & competitiveness, with focus on:**

- Upgrade of Management Processes with focus on Support Functions
- Reduction in Working Capital Requirements
- Effective Control over Production Purchases
- Optimization of Upstream Supply Chain
- Enhancing effectiveness of Development, Research & Engineering programs
- Pursuit of Interactive Innovation Programs
- Pursuit of Lean Sigma Approaches aimed at Structuring & Standardization of Processes

### **The initiatives are aimed at:**

- Enhancement in operating profit
- Reducing & hedging the currency exchange rate for EUR/USD effectively through increased extension of purchase operations to international markets
- Lowering of working capital requirements
- Diversification of group's industrial facilities to international markets

### **Key focus areas for the group include:**

- Pursuit of quality enhancement initiatives, especially, Lean-Sigma, QRQC projects, Lean engineering
- Consolidation & centralization of shared service centres & support functions. The group consolidated the support functions based in France in 2013 expanded & established shared service centres across international markets in 2014.
- Simplification of group's organization & structure
- Consolidation & centralization of procurement & purchase activities and performance monitoring with focus on enhanced competitiveness
- Pursuit of innovation & sharing of best practices through implementation of programs like cooperative enterprise networks.



## Focus on Enhancing Overall Operating Efficiencies & Competitiveness - 2

At present, Safran is pursuing a group-wide, comprehensive performance improvement plan through development of its industrial base and carrying out preparatory work on **Factory of the Future program** with **Digital Transformation** at its core. The program provides a platform for convergence of a range of digital technologies aimed at transformation of the conventional production system and the entire industrial & functional value chain across the group. Key technologies & activities being focused upon by the program include:-

- Digitization of Production Processes
- Connected machines and objects (Industrial Internet of Things)
- Data continuity for production and operations
- Real-time processing of huge masses of data
- Robotics
- Cobotics
- Human-Machine interfaces
- 3D printing (additive manufacturing)
- Artificial intelligence

Objectives of insertion of these next generation technologies, include:-

- Maximization of Plant Utilization Rates
- Shorter new product development cycles
- Much faster responsiveness
- Higher productivity
- Better quality control
- Providing rapid production ramp ups

Safran invested around EUR 704 Million in 2016 towards its industrial facilities with around EUR 483 Million spent in France alone towards readying up for the transformation of its industrial base and the entire industrial ecosystem through strategic partnerships across its entire supply chain.

## Continued Investments Towards Expansion of Global Industrial Footprint to Sustain Growth Momentum

Safran Group continues to make significant investments towards expansion of its global industrial footprint to sustain growth momentum with significant & strong long-term growth projections for the commercial aviation sector. About 62% of the group's investments in property, plant & equipment for 2013 were directed at France based operations followed by North America & Asia. Key upcoming facilities for the group include:

- Establishment of a new R&D centre for the group focusing on the development of new composite materials based at Vert-le-Petit near Paris at an estimated investment of EUR 50 million.
- A new manufacturing facility for producing composite material based fan blades & casings for LEAP engines based at Commercy in North-East France.
- Establishment of a new facility for Messier-Bugatti-Dowty for machining of titanium landing gear rods based at Bidos in South-West France.
- Revamp & modernization of France based industrial footprint aimed at scaling up production & introduction of new engine & aircraft equipment programs, namely, for the LEAP, Silvercrest, A350 & Boeing 787 programs.
- Plant renewal & modernization of existing production sites & modernization of test equipment at a number of Snecma facilities. Snecma Corbeil & Snecma Gennevilliers received machining systems upgrades which are hubs for production of aircraft parts while the test equipment at Snecma Villaroche, which is the hub for production of jet engines & Snecma Istres have been modernized.
- Expansion of Snecma's existing industrial footprint in Mexico & China with the establishment of new manufacturing facilities based at Querétaro, Mexico & Suzhou, China.

## Significant Focus on Extending Technological Capabilities

Safran group has been focusing on enhancing & extending its technological capabilities significantly as a key differentiator and a source of significant competitive advantage & edge. The group's R&D expenditure in 2017 stood at EUR 1.4 billion for continuing operations; standing at 8.5% of the sales revenues for 2017 with over 50% of the expenditure being self-financed. Additionally, to further augment R&D capabilities Safran established a dedicated research centre to extend capabilities & augment skills across a broad spectrum of technological capabilities for its group companies focusing on the following key domains, namely, materials & processes, sensors, electronics, information & communication technology & digital simulation for engineering. The new centre, based at Safran's Saclay Campus, became operational from 2014 and employed a research staff strength of 300 in 2017.

The group also set up a **specialised Safran Composites Center** that focuses on conducting **advanced research on organic matrix composites** under the group company, Herakles, which leads in domain competence & capabilities pertaining to thermostructural composites, metal alloys & high performance coatings.

### Development of Hybrid Electric Distributed Propulsion System by Safran in collaboration with Bell

Safran also conducted **ground testing of its first hybrid-electric propulsion system in July 2018** as part of the development of a distributed propulsion system. The demonstration was conducted by Safran Helicopter Engines, Safran Electrical & Power and Safran Power Units, in conjunction with Safran Tech, the Group's research & technology center. The system generated 100 kW of electrical power as part of the demonstration. Safran also signed a **joint technology development agreement towards development of hybrid-electric propulsion system with Bell with the agreement between Safran and Bell signed in July 2018. The hybrid electric propulsion system will be developed for urban mobility VTOL applications as part of Uber elevate project which is scheduled to start with flight trials in 2020. As part of the collaboration, Bell will design, develop & produce VTOL systems while Safran will provide the hybrid-electric propulsion system**

**Safran's further research focus & orientation pertaining to aircraft propulsion** under ACARE(Advisory Council for Aeronautics Research in Europe) includes:

**Objectives 2020:** Target for a 20% reduction in fuel consumption alongwith a 60% & 50% reduction in pollutant emissions & noise levels. Snecma is well poised to leverage LEAP turbofan engine featuring a number of innovative technologies to meet ACARE's 2020 targets.

**Objectives 2050:** Guidelines laid out under Flightpath 2050 aimed at achieving a 75% reduction in greenhouse gas emissions, 90% reduction in Nitrogen Oxide & 65% reduction in noise levels by 2050. Snecma plans to pursue fundamental innovations in engine architecture through breakthrough concepts, including, open rotor concept & faired engine architecture to meet the Flightpath 2050 guidelines.

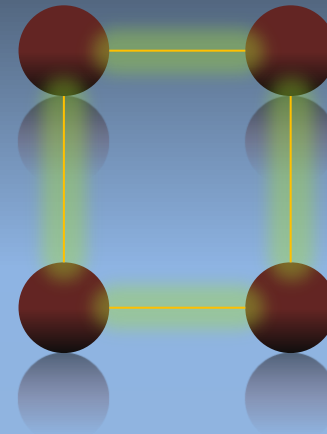
## **Plans for Key JV with Boeing for joint Production of APU's to Further Expand Safran's Presence in the Global Aircraft Equipment Market as a Tier 1 Supplier**

Safran Group is poised to take further strides in the global aircraft equipment market with the announcement of a key JV with Boeing for the production of Auxiliary Power Units (APUs) for the commercial aircraft market. As per the agreement, announced in June 2018, the two companies will design, produce and service APUs for commercial aircrafts under a newly created joint company to be owned 50:50 by Boeing and Safran respectively.

The move will deepen Boeing's presence in the aircraft equipment market under its BGS (Boeing Global Services) brand that has been giving jitters to the suppliers in the global aircraft market with world's leading global OEM straying into their well entrenched territory for over a year now. For Safran, it will provide an opportunity to the French aerospace group to further broaden its portfolio & market presence across segments as it has been producing APUs for business jets, helicopters and some military jets since 1962.

The agreement, which is likely to be signed towards the end of 2018, significantly threatens Honeywell and United Technologies as they have been key suppliers of commercial aircraft APUs for decades now holding 65% & 35% share of the global APU market, as per FlightGlobal. Honeywell is the sole supplier of APUs for the Boeing 737MAX program apart from C-Series, Comac C919 & Irkut MC-21 programs. Pratt & Whitney, on the other hand, is the sole of supplier of APUs for the Boeing 787 program. APUs produced by both Honeywell & Pratt & Whitney are offered as an option on the Airbus A320neo program as well. In the wide-body aircraft market, Honeywell supplies APUs to the A350 & 777 programs while Pratt & Whitney supplies to the A380, 787 & 747-8 programs. Thus, the move is likely to impact Pratt & Whitney the hardest given the limited production rates of the A380 & 747-8 programs over the recent years.

# Business Outlook for 2019



## Part 4.3

## Outlook

Safran, as a well diversified aerospace group, is extremely favourably positioned in the global commercial aircraft turbofan engines market with its long standing JV, CFM International, with GE Aviation dating back to past over 3 decades and commanding a dominant market position led by its iconic CFM56 engine and the latest LEAP engine family incorporating a number of proprietary technologies coming from the GE stable. CFM has been able to effectively build on the equity of its CFM56 engine, emanating from its unbeatable reliability, with a issue free entry into service of its LEAP engines family over the recent years with the sole engine supplier status on the 737MAX aircraft family further cementing its competitive positioning in the global narrow body aircraft market followed by a rapid production ramp up in a mere 2 year time span.

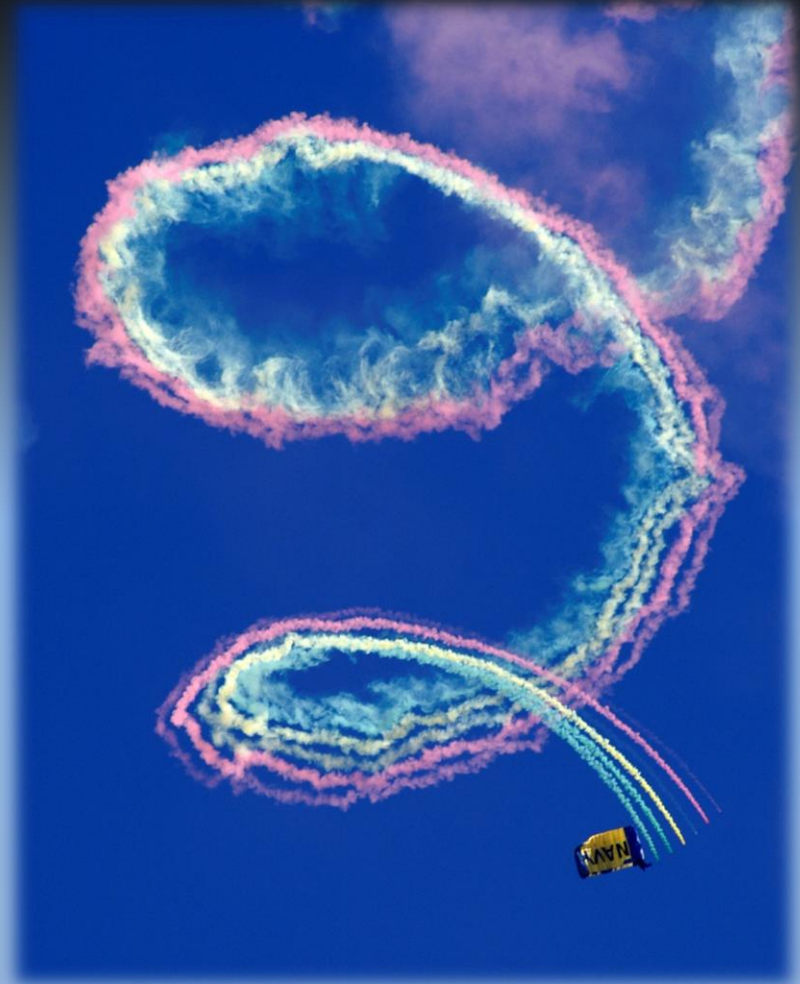
However, the LEAP engine family, too, has had its share of troubles as of late with an in-flight contained engine failure on a Southwest Airlines 737 MAX 8, originating from a carbon build up leading to blade failure, triggering caution with CFM announcing monitoring of the LEAP engine fleet globally. Also, the two crashes of the 737 MAX 8, within a span of 5 months with MCAS issues, has led to grounding of the type's fleet globally with Boeing revising the production rate of the 737 MAX series which is likely to impact the entire supplier base going forward, including, CFM which is the sole source engine supplier on the 737MAX aircraft family. However, the production ramp up on the LEAP engine program continues & remains a key strategic imperative for CFM with the company making steady progress on the same with around 1,100 engines delivered in 2018 as against 77 in 2016, the first year of production.

Further, the global, in-service CFM56 engine fleet is rapidly approaching its first mid-life overhaul which is likely to provide significant cash flow stream from the MRO activity. CFM is also looking avidly towards the Boeing's NMA program to further extend its hegemony of the commercial aircraft market. On the technologies front, the group is focusing heavily on and making rapid advancements in the Hybrid-Electric propulsion domain led by its status as the propulsion system provider for the Uber's ELEVATE program focused on the emerging Urban Aerial Mobility front, which is likely to open tremendous growth opportunities for the entire industry value chain over medium term. The earlier acquisition of Zodiac Aerospace & the latest JV agreement with Boeing in 2018 for APUs is likely to further strengthen Safran's position as the worlds third largest aerospace supplier group. On the space front, Arianespace, Airbus-Safran JV, is moving towards Ariane 6 commercial satellite launch vehicle which likely to make its debut in 2020. Going forward, integration of Zodiac Aerospace's operations into Safran group to realize synergies will be a key imperative for Safran.

Production ramp up on the LEAP family will need to be maintained with CFM targeting a output & deliveries target of 1,800 engines for 2019, in-line with its target of reaching the 2,000 engine deliveries target set for 2020. **On the technologies front, advanced materials, hybrid propulsion, additive manufacturing & digitalization remain key areas of focus for Safran with the group planning to increase its overall R&T budget to EUR 600 million in 2022.**

# Key Trends, Issues & Challenges and Risk Factors

## Part 5





# Key Trends, Insights & Growth Opportunities – 1

## Commercial Aviation to Register Rapid Growth through 2037 with Strong, Projected Airline Traffic Growth

Commercial Aviation is projected to register rapid growth over the 2018-2037 period with strong growth projections for passenger traffic growth over the forecast period which has been surpassing the global GDP growth. The global airline traffic is projected to grow at 4.7% over the 2018-2037 period as per Boeing's long term market outlook and the air traffic volume is likely to double by 2030 itself as per ICAO. The airline traffic growth is projected to be spearheaded by emerging markets led by Asia which is projected to register massive traffic growth through 2037.

Other key factors which are likely to be favorable drive passenger traffic growth through 2035 include:-

**Strong Growth in Middle Class segment globally through 2035:** The middle class segment globally is projected to grow significantly from a population size of 2.8 billion in 2015 to around 4.8 billion in 2035. The middle class segment, which accounted for around 38% of global population in 2015 is projected to account for over 55% of global population in 2035.

**Rapid Growth in Discretionary Spending across Emerging Economies:** Around 42% of the world's private consumption is likely to originate from emerging markets by 2035 as against 31% for 2015, as per Airbus projections with a rapid doubling of discretionary spending on goods & services projected across emerging markets from \$3.1 trillion in 2015 to \$6.1 trillion in 2025.

**Urbanization:** Over 60% of the world's population is projected to be staying in cities by 2035.

**Evolution & growth in the number of Global Aviation Mega-Cities** from 55 in 2015 accounting for 1 million passengers daily and 25% of global GDP to 93 mega cities in 2035 accounting for 2.5 million passengers & 35% of world GDP in 2035.

# Key Trends, Insights & Growth Opportunities – 2

## Global Commercial Aircraft Engines Market in Top Gear Propelled by the Longest Aviation Super-Cycle

The Global Commercial Aircraft engines market has been in one of its longest golden phases propelled by one of the longest aviation super-cycles so far. The same has created a huge order backlog for the industry which is likely to translate into significant top line growth potential for the aviation industry value chain over the next decade. Next generation aviation turbofan engines, featuring a high bypass ratio and extensive usage of technological innovations, ranging from material science innovations to proprietary coatings etc. from the competing engine powerhouses have just recently entered service and are likely to form a major chunk of new deliveries over medium term given the composition of order backlog across aircraft OEMs. 2018 saw the Entry into Service of three new aircraft programs, led by, Boeing's 737 MAX 9, Airbus A321 LR and Embraer's E190-E2 sporting engines from both CFM International & Pratt & Whitney respectively.

The engine manufacturers are gearing up their global industrial base for a major production ramp up to meet delivery timelines with some transitioning from production of previous generation engines to latest engine programs. CFM International has a major task lined up ahead for itself as it makes the key transition from production of CFM56 engines to LEAP engines family by the end of decade. For Pratt & Whitney, managing teething troubles over the in-service GTF engines remains the key, near term priority. For GE, 2019 is going to be an important year with the world's largest turbofan engine, GE 9X, scheduled to enter flight test phase prior to its entry into service slated for 2020. Additionally, Boeing's proposed New Mid-market Airplane (NMA) dubbed the 797; Boeing's first scratch-up commercial aircraft program since 787 in 2003; is going to be the key growth pie being looked at avidly by all engine manufacturers over near term with Boeing likely to announce the program at the Paris Air Show 2019.

The technology landscape across industry is evolving radically with the development of hybrid-electric propulsion technologies for commercial aircrafts; aimed at reducing CO2 emission levels drastically while scaling down operating costs substantially; likely to become a functional reality by the middle of the next decade with multiple industry teams across the globe laser focused on pursuing R&D on the electric propulsion technology which necessitates radical improvements in current generation battery technologies. The developments on the Urban Aerial Mobility front, however, are likely to give a further push to development & commercialization of distributed electric propulsion technologies. Further, developments on the supersonic air travel front, too, are likely to further expand growth avenues for the engine manufacturers as highlighted by the GE's decision to use CFM56 based engine core for development of propulsion system for Aerion's AS2 supersonic business jet, scheduled to enter service in the mid 2020s.

# Key Trends, Insights & Growth Opportunities – 3

## Asia-Pacific Poised to Become Key Growth Driver for Commercial Aviation Segment

The Asia-Pacific region is poised to become the key growth driver for the commercial aviation sector with strong, long-term growth drivers in the form of economic growth, rising disposable incomes, population growth & globalization. Long-term growth projections by industry OEMs indicate towards a significant role of emerging markets in the global macroeconomic dynamics & aviation industry growth.

Projections by Boeing indicate towards delivery of over 42,730 new commercial aircrafts globally worth \$6.3 trillion over the 2018-2037 period, led by the Single Aisle aircraft segment, which alone is projected to register deliveries of over 31,360 aircrafts through the forecast period accounting for over 73% of the total projected deliveries in unit terms. The growth in the commercial aviation segment is likely to be driven by significant growth in air traffic likely to be registered in the Asia-Pacific region, projected at 5.7% over 2018-2037 on the back of 3.9% GDP growth over the forecast period, which will drive growth, especially, for the single aisle passenger aircraft segment. The APAC region is projected to receive about 41% of total global aircraft deliveries through 2037 translating into 16,930 aircrafts, as per Boeing's projections. Of the 16,930 aircrafts, around 74% or 12,570 aircrafts are likely to be single aisle aircrafts followed by 3810 wide body aircrafts, likely to account for 23% of the deliveries.

The value of aircraft deliveries to the Asia-Pacific region is projected to be worth \$2.67 trillion over the 2018-2037 period. The share of APAC based aircraft fleet in the world fleet is projected to grow from 30% in 2017 (7360 aircrafts) to 37% in 2037(18,200 aircrafts).

# Key Trends, Insights & Growth Opportunities – 4.1

## Battle for Supremacy in the Commercial Aviation Propulsion Market Spearheaded by Technological Innovations – 1

The battle for leadership of the global commercial aviation propulsion market is being led by latest technological innovations as the radical Geared Turbofan technology from Pratt & Whitney takes on and challenges the undisputed market domination & market leadership of CFM, which is fighting to retain top spot with its LEAP engines family featuring significant material sciences innovation coming from the GE stable leveraging GE's significant experience in the fighter jet market. Both the technologies face the ultimate litmus test by carriers with both engine families entering service formally in early 2016 across carriers globally. Their EIS, however, has been fraught with teething troubles and operating difficulties, especially, for GTF. The situation going forward, however, is likely to evolve further with both the technologies still to go through litmus test. CFM's Leap 1A received FAA & EASA certifications in May 2016 with the deliveries of Leap 1A powered A320neo commencing to airlines in July 2016 with Turkish carrier Pegasus Airlines becoming the launch customer followed by Air Asia. CFM International had delivered over 40 Leap 1A engines to Airbus for the A320neo program by October 25, 2016. CFM still has a long way to go with the Leap 1B which powers the Boeing 737 Max scheduled to enter service in H1 2017 while the EIS for Leap 1C, which powers Comac's C919, is scheduled for 2016 year end with ground tests beginning in November 2016. Additionally, CFM has also been working towards ramping up production of the Leap engine family rapidly to the level of 2,000 engines a year by 2020 as the company transitions from CFM56 to Leap as the mainstay of its commercial aircraft engines business. CFM produced over 1,700 CFM56 engines in 2015 and plans to surpass that level for Leap by 2020.

The EIS of the Leap 1A for CFM so far has largely been free of glitches after over 4 months of active in-service duty at 10-11 cycles per day, as per Pegasus Airlines which has been utilizing the Leap 1A engine since July 2016. The PW1100G, on the contrary, has faced significant teething troubles over the A320neo with extended engine start-up times and production as well as supply chain delays which have collectively led to delays in deliveries of over 50 PW1100G engines to airline customers. Operators, however, have lauded the PW100G's frugal fuel consumption & dispatch reliability which has been over 99%. Leap 1A, however, has been able achieve double digit reduction in fuel consumption as against CFMs' claims of 15% when compared against the CFM56. CFM, however, plans to still work on introducing further reliability improvements on the Leap 1A engine going forward into 2018 aimed at extending time on wing with plans for introduction of an improved booster design with new blades.

# Key Trends, Insights & Growth Opportunities – 4.2

## Battle for Supremacy in the Commercial Aviation Propulsion Market Spearheaded by Technological Innovations – 2

How successful have the two players been in winning new customer orders, however, remains to be seen. CFM has a significant advantage at hand given the status of Leap 1B as the exclusive power plant for Boeing's 737 MAX, unlike, the A320neo on which the airlines have the option to choose from CFM's Leap 1A & Pratt & Whitney's PW1100G geared turbofan engines respectively. Technically, both the PW1100G GTF and Leap mark a significant deviation from the traditional CFM56 engine with focus on achieving high bypass ratio for enhanced fuel efficiency. Pratt & Whitney has inserted the gearbox between the low pressure turbine and the inlet fan which has enabled the engine designers to increase the size of fan blade which, in turn, has enabled them to almost double the bypass ratio from 6:1 in a typical V2500 to about 12:1 in the PW1100G. CFM, on the contrary, has stuck to the conventional engine architecture which enabled it to improve bypass ratio from 6:1 to 10:1 while inserting cutting edge technologies to improve the efficiency of the engine core to achieve the rest. CFM has added an additional stage to the high pressure compressor which has enabled it to increase the overall pressure ratio from 30:1 in CFM56 to 40:1 in Leap engine family. CFM has also added an additional second stage to the high pressure turbine while also inserting Carbon Matrix Composites (CMCs) in the shrouds of the first stage of high pressure turbine to reduce overall cooling requirements. Both the engines also feature significant usage of carbon fibre based composites for engine fan blades aimed at reducing overall engine weight.

Analysis of data extracted from FlightGlobal's Flight Fleets Analyzer database indicated that the neo aircraft family had an order book position of 4,502 firm orders from 60 airlines & lessors globally. The choice of engine had been made on around 63% of the aircrafts or 2,856 aircrafts as of March 2016. Leap 1A had won over 1,544 orders as against 1,312 for PW1100G. Leap 1A, thus, had a 54% share on neo family orders as against 46% for PW1100G as of March 2016. In terms of customers, around 30 have opted for PW1100G while 23 have gone for Leap.

The battle for engine supremacy in the narrow body segment has a long way to go before a meaningful conclusion could be drawn given the fact that Flight Global's Fleet's Analyzer Database predicts a total demand for over 9,800 Neos through 2035 which indicates a demand for over 5,300 more new aircrafts. Pratt & Whitney plans to roll out a performance improvement package for the PW1100G in 2019 aimed at achieving a further 2% reduction in fuel burn. CFM, too, is likely to follow suit with a series of upgrades for Leap engine family going forward. Overall, its going to be a very interesting and closely as well as fiercely fought battle between Pratt & Whitney, which is making a sort of comeback to the single aisle segment as a standalone engine manufacturer (post acquisition of Rolls Royce's stake in IAE earlier in 2012 which makes the V2500 for A320ceo), while CFM is working hard to protect its current market dominance with the CFM56 engine.

# Key Trends, Insights & Growth Opportunities – 5

## Asia-Pacific Region to Spearhead & Drive Military Helicopter Demand Growth through 2025

The Asia-Pacific region is poised to become the hub for military helicopters over near to medium term with orders for 2,000 new military helicopters worth \$44 billion likely to originate from the Asia-Pacific region through 2025 as key nations in the region plan to enhance force mobility and to replace ageing hardware, led by South Korea, China, India & Japan. The demand from the Asia-Pacific region is likely to be strong for light military helicopters with replacement demand being the most significant demand driver.

India plans to procure 197 Light Helicopters worth \$1.5 billion, 110 Naval Helicopters for \$2.2 billion & 22 attack helicopters worth \$900 million. India has already signed a key agreement with Russian Helicopters to locally produce over 200+ Ka-226T helicopters in India under license. All leading global OEMs, namely, Boeing, Bell, Airbus Helicopters & Sikorsky are engaged in a fierce competition to win new military helicopter contracts. OEMs who already have a strong presence in the region's aerospace & defense value chain & have a sizeable industrial footprint in the region are likely to be in a advantageous position as against their counterparts to capitalize on potential growth opportunities with the capability to fulfill industrial offset requirements. Airbus Helicopters has been able to leverage its leadership of the global civil helicopter market & deep roots in the region effectively of-late by winning a string of key military helicopter programs in South Korea. Airbus Helicopters made a key win in the South Korea's Light Civil Helicopter (LCH) & Light Armed Helicopter (LAH) Program in March 2015, which aims to develop the next generation civil rotorcraft & its military variant in the 5 ton class, based on Airbus Helicopters H155 platform, in partnership with KAI, a program worth \$3 billion which is likely to provide a significant growth avenue in the APAC region over medium term. Bell, on the contrary, has been successful in winning Japan's UH-X Multipurpose helicopter program for the Japanese Ground Self-Defense Forces (GSDF), to be worked upon in collaboration with Fuji Heavy Industries Ltd. & the Japanese Defense Ministry. The Japanese MOD's decision for selection of Fuji-Bell for the UH-X program was announced in July 2015. The UH-X program entails development of over 150 next-generation multi-purpose helicopters, worth around \$2 billion (at per unit cost of Japanese Yen ¥1.2 Billion), for the GSDF over the next 20 years beginning in FY2021. Additionally, Japan is likely to float another key AH-X program which aims to replace the country's ageing fleet of AH-1 Cobra attack helicopter fleet.

# Key Trends, Insights & Growth Opportunities – 6

## Potential Opening of Chinese Low-Altitude Airspace to be a Key Growth Avenue for OEMs

Potential opening & expansion of low-altitude airspace in China besides proposed implementation of civil aviation reforms is likely to be a window of opportunity for most global OEMs as it is likely to stimulate demand for new light business jets as well as helicopters. The Chinese authorities are working towards opening the use of aerospace, long reserved for military utilization with only 20% of airspace open to civil aviation, for civilian aircraft utilization comprising of commercial & private airplanes. The Civil Aviation Administration of China also outlined plans in November 2013 to relax & streamline requirements for obtaining a private pilot license in China aimed at bolstering the country's civil aviation sector. The Air Traffic Control Commission of China too unveiled plans to open additional low-altitude airspace over the coming years for commercial & private airplanes which is likely to further boost civil aviation in China. The opening of the Chinese low-altitude airspace has commenced in 2015 with China opening up 30% of low altitude airspace by October 2015, as per announcement made at the 4th China International General Aviation Convention held in October 2015. Airbus Helicopters, estimates that the opening of Chinese low-altitude airspace is likely to create demand for over 1,000 new helicopters over a 10 year period. Airbus Helicopters anticipates significant demand for new helicopters over near to medium term with scheduled opening of Chinese low-altitude airspace from 2015. The company expects China to become its largest market globally by 2020 generating sales of about 150 helicopters annually as against 30-40 helicopters at present. Airbus Helicopters projects a market for over 5,000 helicopters over the next 3 decades as against 330 helicopters at present driven by opening of low altitude of airspace through 2020, continued infrastructure improvement & emergence of civil private helicopter operators. With the spate of ongoing reforms, the Chinese market has the potential to overtake the United States as the world's largest market for private airplanes over the next decade, as has already happened, in the case of automotive & construction equipment industries over the past decade where China already is the world's largest market. Most key industry OEMs, led by Airbus Helicopters, Leonardo & Bell, have been at the forefront of activity with Airbus Helicopters signing up an agreement with Avicopter in March 2014 for joint production of over 1,000 EC175 helicopters. Leonardo-Finmeccanica, too, has been quite active in China having reinforced its sales & distribution network and with the establishment of a regional business headquarters in April 2014. AW has also seen a surge in its order book in China of-late having signed up a major, multi-year sales contract with the Beijing Automotive Industrial Corporation (BAIC) of China in October 2014 worth EUR 400 Million for 50 AW helicopters, including, AW139, 169 & 189s to be delivered over the 2015-2019 horizon.



# Key Trends, Insights & Growth Opportunities – 7.1

## OEMs Experimenting with Electric & Hybrid Propulsion Systems for Potential Civil Helicopter & Aircraft Applications - 1

A number of industry players are experimenting with the hybrid propulsion systems for civil helicopters aiming at reducing emissions footprint while optimizing operating economics through reduction in operating costs. A number of research programs have been underway across a number of industry players led by leading engine manufacturers Safran and Airbus towards evolution of hybrid propulsion technologies as part of their underlying strategy of pushing the performance & innovation threshold. As per Safran, the hybridization of helicopter propulsion systems will translate into a 25% enhancement in fuel efficiency through the end of present decade as against improvements to the tune of 15% likely to be achieved over conventional engine architecture by the same period, thereby, making the business case for hybrids more viable. The hybridization would involve integrating multiple power sources for different phases of the flight while optimizing the conventional engine architecture for the cruise phase with the alternate power sources providing additional power for critical phases like take off, landing, maneuvers or emergency situations. This approach in turn would lead to a significant increase in the fuel efficiency provided by the conventional turbine engine. The alternate power sources are most likely to include Auxiliary Power Units (APU) and electric motors capable of producing enough propulsive power along with evolution of the power to weight ratio of electric storage systems. Safran has been working on the hybridization of a 2000-3000 SHP class conventional engine.

Amongst other industry players, Airbus Helicopters, too, has been actively experimenting with hybrid propulsion systems over the years. The company had displayed a diesel-electric hybrid propulsion system for helicopters, termed Eco2Avia, at the ILA Berlin Airshow held in 2010. The system, developed by the former EADS Innovation Works, involves integration of electric motors with the conventional engine with the potential to reduce fuel consumption & emissions by up to 50% with the electric motors alone capable of sustaining & supporting take offs & landings, thereby, achieving significant reduction in noise levels and safety. The Opposed Piston, Opposed Cylinder (OPOC) diesel engine utilized in the prototype claims to provide a 30% improved fuel efficiency as compared with conventional turbine engines. EADS (now Airbus) also claimed that the hybrid system can accommodate a range of combustion engines. The hybrid system architecture also enables the fuselage to maintain its optimum aerodynamic profile during the cruise phase of the flight with the main rotor along with its electrical drive staying in a tilted forward position, thereby, leading to minimization of aerodynamic drag and optimization of fuel efficiency. Additionally, Airbus has been able to expand its technological capabilities in the electric and alternate propulsion systems domain significantly through the Airbus Zephyr solar powered UAV platform developed by Qinetiq and being pursued under the Airbus High Altitude Pseudo-Satellite (HAPS) program. The Zephyr holds the endurance record for an unrefueled UAV standing at around 336 hours. The challenge for manned flights, however, is to have a sufficiently high enough power to weight ratio for the drive system. The current generation propulsion system components, including, power electronics, electric motors & generators, however, will have to improve by almost 10X in future. Siemens unveiled a 50 kilogram electric motor in 2015 with a power output of 260 kilowatt capable of powering a 2 ton electric aircraft in conjunction with a conventional jet engine with further research underway over its actual in-flight performance. Siemens is also collaborating with Airbus for further scaling up of the technology aimed at developing larger electric motors capable of producing up to 1MW of power output in future.

# Key Trends, Insights & Growth Opportunities – 7.2

## OEMs Experimenting with Electric & Hybrid Propulsion Systems for Potential Civil Helicopter & Aircraft Applications - 2

### Project Zero Demonstrated Leonardo-Finmeccanica's Technological Prowess & the Capability to Harness & Leverage Group's High Degree of Vertical Integration & Sharing of Group-Wide Technological Capabilities

Project Zero, that involved the development of a pure electric tilt-rotor technology demonstrator, in collaboration with a number of Finmeccanica group companies has provided Leonardo-Finmeccanica with a test bed to actually develop & test a number of cutting-edge technologies which has enriched the company with a vast pool as well as repertoire of futuristic technologies that are likely to be leveraged and incorporated by Leonardo-Finmeccanica into the development of new products in future thereby further strengthening the technological advantage & edge from a competitive standpoint. The prototype was displayed publicly for the first time by Finmeccanica at the Paris Air Show 2013. Around 80% of the airframe structure is made of composites, including, skin, rotor blades, shroud & spokes. The tiltrotor also has the capability to be operated like an unmanned aircraft as well and also has the option of an alternate hybrid propulsion system with the usage of a light diesel engine. Additionally, the batteries that power the electric motors on the tiltrotorcraft can be charged on the ground by tilting the rotors into the wind to act as turbines.

Leonardo-Finmeccanica also announced in February 2016 that a hybrid drive system would be installed on the full scale aircraft aimed at increasing the flight endurance to 35-45 minutes from the present 10 minutes while also charging the batteries and providing additional power for maneuvering and/or hovering. Leonardo-Finmeccanica Plans to continue to use the Project Zero as a test bed to mature a range of technologies for the company's Helicopters, Tilt-rotorcrafts and Rotary Unmanned Systems as outlined by James Wang, Senior VP services & Marketing for Leonardo-Finmeccanica in an interview given at the Singapore Airshow in February 2016 as reported By Flight Global.

### Airbus, Rolls-Royce and Siemens Team Up on the Hybrid-Electric Propulsion System for Commercial Aviation

Further, Siemens, Airbus and Rolls Royce are collaborating on the development of a hybrid-electric propulsion for commercial Aviation, as reported by the media in November 2017. The three industry behemoths will collaborate under Project E-Fan X with Plans to fly a demonstrator aircraft in 2020. Under the project, Airbus will be responsible for the control architecture of the hybrid-electric propulsion system as well as batteries & its integration with flight controls, Rolls Royce will provide turboshaft engine and the 2MW generator while Siemens will produce the 2MW electric motor for the propulsion system.

Leonardo-Finmeccanica's Project Zero  
Electric Tilt-rotorcraft



# Key Trends, Insights & Growth Opportunities – 8

## **Development of Alternate Fuel Technologies aimed at Reductions in CO2 Emissions & Operating Costs for Civil & Military based Applications**

The aviation industry is working actively towards development of a range of alternate fuel technologies aimed at achieving the target of carbon neutral growth by 2020 and a further 50% reduction in CO2 emissions by 2050 as against the 2005 levels according to the International Air Transport Association (IATA) with about 40% of the expected reduction likely to be achieved through the use of alternate fuels with the rest attributable to technological developments. The development of Green Jet Fuel by Honeywell under the U.S. DARPA contract has been one of the most notable bio-fuel projects of-late with almost 800,000 gallons of fuel produced so far and about 27 demonstration flights across a diverse set of aircraft categories ranging from U.S. Military Aircrafts (F/A 18, F-15, F-22, C-17, Thunderbirds F-16, Navy MH60S Seahawk), Commercial Airliner Flights(United Airlines, Air China, Aeromexico, KLM) to Business Jets (Gulfstream G450). There have been other companies in the U.S. as well that are working towards development of Hydrotreated Renewable Jet (HRJ) fuel from algae & other second generation feedstocks. With the ASTM earlier certifying the HRJs, the industry is looking at further advancement towards commercialization. The other approach to bio-fuel production originates from the development of Alcohol-to-Jet fuels(ATJ) with its roots in alcohol-based sources which has also been referred to as the low-cost & more economical approach. The research on ATJs is being conducted by the U.S. Military & a range of companies. Additionally, ASTM, under a recent revision (June 2014, ASTM International standard D7566) certified the usage of Renewable Synthesized Iso-Paraffinic (SIP) fuel for commercial aviation usage across airlines globally. With the latest ASTM standard for jet fuel in place; the Amyris-Total alliance is gearing up to market a drop-in jet fuel incorporating a 10% blend of renewable Farnesane. Earlier in 2014, the Roundtable on Sustainable Biomaterials (RSB)'s certified Amyris's first farnesane production facility based in Brazil. Boeing & Airbus both have been carrying out research on biofuels in collaboration with industry partners across South Africa & Australia respectively. Boeing is in the process of seeking ASTM approval for utilization of green diesel as a replacement for Jet A towards the end of 2014 with the company seeking an annex to the already approved hydroprocessed esters and fatty acids (HEFA) standard which will accelerate the utilization of biofuels for commercial aviation & enable rapid commercialization of green diesel for commercial aviation usage as it is already being produced globally. The Green diesel production globally & the production of green diesel is scheduled to touch the 800 million gallons mark globally by the end of 2014 which is a mere over 1% of the global annual demand for ATF standing at over 60 billion gallons. However, green diesel could cost a mere \$3 per gallon with government incentives which along-with the fact that its is already in production globally could prove to be a significant advantage & make it score over other under development biofuels.

# Key Trends, Insights & Growth Opportunities – 9

## **Tightening Regulatory Emission Norms & Standards Necessitating Continued Investments towards New Technology Development**

The aviation industry's contribution to global CO2 emissions has been 2% and the IATA resolution intends to achieve carbon neutral growth under its Aviation Carbon-Neutral Growth(CNG) 2020 Strategy by the year 2020 and plans a further 50% reduction in net emissions by 2050 as against the 2005 levels with about 40% of the expected reduction likely to be achieved through the use of alternate fuels with the rest attributable to technological developments followed by pursuit of improvements in technology, operations and infrastructure development with deployment of single, unified global Market based Measures. Key targets outlined by the IATA resolution includes:

- Achieving annual fuel efficiency improvements of 1.5% through 2020
- Putting a Cap on the net emissions with the CNG2020
- Reducing net emissions by 50% through 2050 as compared with 2005

The aviation industry needs to invest significantly towards development of new technologies, especially, in the development of a range of sustainable alternate fuel based technologies to achieve the laid out objectives amid a difficult global macroeconomic environment, especially, for the business aviation industry.

# Key Trends, Insights & Growth Opportunities – 10.1

## Era of Supersonic Business Jets Round the Corner - 1

Gulfstream has been expanding its technological capabilities rapidly aimed at the development of a supersonic business jet over medium term to take on potential competition from the Aerion Corp, which has been working on the development of its supersonic business jet, and as the technological evolution to the next generation of business jets offering enhanced features & performance capabilities. Estimates by the Aerion Corp. indicate towards a potential demand for over 400 supersonic jets through the middle of next decade. The supersonic business jets will require a significant technological evolution prior to becoming a working reality, especially, with respect to propulsion technologies and sonic boom mitigation.

Gulfstream has been working on a range of technologies required for feasibility & viability of supersonic flights for over two decades and the company of-late has filed a series of patents related to supersonic flight related technologies which include:

**Quite Spike:** Gulfstream had tested & developed its Quite Spike device over a F-15 jet in 2007 for a NASA research project aimed at mitigating the sonic boom or the acoustic signature generated in reaching supersonic speeds. The device, essentially a retractable nose tip, reduces the acoustic effects of supersonic speeds by altering the traditional shape of the pressure wave generated by sonic boom and thus makes it indistinguishable from a regular sub-sonic aircraft flight even to people on the ground. The Quite Spike device reduces the sound generated as a result of sonic boom by almost 10,000 times as compared to the sonic boom of Concorde aircraft.

**Supersonic Nacelle Design:** Gulfstream filed the patent for a supersonic nacelle design in February 2010 aimed at reducing the drag on airframe generated as part of the sonic boom and to enhance overall propulsive efficiency to counter the drag effectively through reshaping the nacelle design. The company also filed a patent in April 2012 for **supersonic inlet design** to enhance propulsion while reducing drag & strength of sonic boom.

# Key Trends, Insights & Growth Opportunities – 10.2

## Era of Supersonic Business Jets Round the Corner - 2

Gulfstream had also conducted a series of ground tests for a supersonic design on an engine of a Gulfstream IV in 2010 and subsequently has continued with its research over the years on a range of supersonic flight technologies.

**Gulfstream filed & received a patent in July 2014** (U.S. Patent 8,789,789) for a **supersonic aircraft incorporating spike for controlling & reducing sonic boom**. In August 2014, Gulfstream was reported to have received another series of patents for :

- Systems and methods for controlling magnitude of a sonic boom.
- A propulsion system using large-scale vortex generators for flow redistribution and supersonic aircraft equipped with the propulsion system.

Gulfstream is reported to have filed two new patents since September 2016 pertaining to technologies developed for mitigation of sonic boom associated with supersonic flights. Gulfstream received a key patent on November 01, 2016 for “Sentropic compression inlet for supersonic aircrafts which shapes the compression surface of the inlet to defocus the resulting shocklets away from the cowl lip” which improves inlet and interference drag characteristics according to the awarded patent description. Gulfstream also obtained another key patent on September 20, 2016 for a computerized fuel redistribution system which aims to shift fuel loads within a wing to mitigate the magnitude of sonic boom .

Overall, Gulfstream appears to be putting the requisite pieces in the right place for a viable supersonic business jet through the end of decade despite the current regulatory prohibitions in place on supersonic flights over land by the FAA in the U.S. & by the International Civil Aviation Organization internationally on account of the sonic boom. The Aerion Corp’s supersonic aircraft project, scheduled to enter service in 2020, incorporates a low-sweep, thin wing design aimed at delivering high fuel efficiency while assuming no changes in regulatory guidelines and envisages a combination of optimum sub-sonic speeds over land and supersonic speeds over oceans for aircraft’s operations to reduce the journey time by almost half over a New-York to Paris flight to 4 hour & 15 minutes as against 7 hours & 30 minutes at present. However, regulatory approvals & changes are likely to be implemented & come into effect once requisite technologies for mitigating sonic boom effects to acceptable levels are developed, tested & matured, as has happened with all technological revolutions across industries & sectors. The FAA’s current certification process of Leonardo-Finmeccanica’s civilian rotorcraft is a prime example where certification standards were non-existent & are being developed jointly by the FAA & the Leonardo-Finmeccanica team.

# Key Trends, Insights & Growth Opportunities – 11.1

## Commercial Aircraft Engine Manufacturers Ramping Up Production Capacity to Meet Delivery Volumes - 1

Almost all the commercial aircraft engine manufacturers have been working hard towards ramping up engine production capacity significantly by the end of decade to meet existing commitments to clients and required delivery volumes.

### **Pratt & Whitney working towards doubling its engine production capacity by 2020:**

Pratt & Whitney has been working towards doubling its overall engine production capacity & levels by the end of decade to meet the required rate of production to deliver on the burgeoning order backlog, especially, for the GTF engine programs. As part of the same, the company has signed over 200 long-term procurement agreements with its global supplier base worth over \$18 billion. The LTAs cover Pratt & Whitney's large engine programs, which include, PW1000G engine family, F135 engine for the F35 program & the PW800 engine for business jets. Pratt & Whitney is also transitioning to a globally diversified industrial base as against the centralized production model which the company has followed traditionally. Pratt & Whitney is also investing to the tune of \$1 billion globally towards expanding the production capacity for its F135 & PW1100G engine programs. The company has also undertaken a transformation of its Middletown, Connecticut based production facility with the modernization of production systems, technologies & infrastructure aimed at expansion of production capacity along-with enhanced operating efficiency. The modernization has included the incorporation of new assembly lines, test cells & cutting-edge technologies. The facility will produce the PurePower PW1100G-JM engine and the F135 engine respectively & features the latest automated horizontal engine assembly lines for the PW1100G-JM engine. The horizontal assembly line enables vertical lift & features rotation capability, thereby, increasing production efficiency with an ergonomically advanced design. The Middletown facility assembled its first set of GTF engines on the horizontal assembly line in April 2015. Pratt & Whitney has also installed the horizontal assembly line at its Mirabel Aerospace Centre, Quebec, Canada.



# Key Trends, Insights & Growth Opportunities – 11.2

## Commercial Aircraft Engine Manufacturers Ramping Up Production Capacity to Meet Delivery Volumes - 2

**CFM has been working towards establishing, streamlining & preparing industrial base for the production of the LEAP engine program** which entered service in July 2016 with the Leap 1A. Snecma signed an agreement with its existing supply partner, Mecachrome in mid-April 2014 for the establishment of a production supply line for the Titanium Aluminide based turbine fan blades for the LEAP engine program. The Titanium Aluminide, an alloy of Titanium & Aluminium which replaces Nickel as the preferred material for low pressure turbines which is capable of handling very high temperatures of upto 750 degree celsius while reducing the weight of the blades by almost 50% transpiring into a significant 15% enhanced fuel efficiency for the LEAP engines. **Snecma in collaboration with its supplier, Mecachrome plan to establish a new manufacturing facility for the production of TiAl based turbine fan blades at an investment of EUR 60 million which follows an innovative resin transfer moulding process. The process renders the fan blades to be flexible which untwist as the turbine's rotational speed increases. The production line will be based at Mecachrome's existing production facility based at Sablé-sur-Sarthe. The production line commenced volume production in 2015 with scheduled ramp up of production output scheduled for 2016 aimed at achieving the targeted overall production rate of one fan blade every 3 minutes slated for 2019 as part of the LEAP engine program's production target of 1,700 engines a year from 2019, the same level as the CFM56 & one of the highest in the industry; with CFM scheduled to transition from CFM56 to LEAP. Safran is also setting up a shared manufacturing facility based at Rochester, NH in the U.S., in collaboration with its U.S. based partner Albany to produce woven composite parts for the LEAP engine program with another twin facility being established in Commercy France which began production of blades & casings for the LEAP engine program in 2014.**

**GE is working towards expanding its U.S. based industrial footprint** with the company gearing up for its latest LEAP engine program. As part of its growth plans and to sustain anticipated growth, GE plans a capital expenditure of \$3.5 billion towards production plant & equipment across its global sites with a majority of the investments to be based in the U.S. As part of the expansion plans, GE broke ground on a new jet engine manufacturing facility, its seventh facility in the U.S., based at Lafayette, Indiana in late 2014 at an estimated investment of \$100 million. The new 225,000 square feet facility will produce the new LEAP engine family for GE's JV with Snecma, CFM International that has a rapidly growing order backlog. The facility will assemble the engine's hot section as well as conduct the engine's final assembly.

# Key Trends, Insights & Growth Opportunities – 11.3

## Commercial Aircraft Engine Manufacturers Ramping Up Production Capacity to Meet Delivery Volumes - 3

- GE has also expanded its West Jefferson based machining facility to expand overall production capacity to support the rapidly growing order backlog for the LEAP engine program. The company has invested to the tune of \$65 million towards expansion of the facility by another 80,000 square foot. GE opened the expanded facility in July 2015. The latest facility expansion is part of GE's larger plan of expanding and modernizing its North Carolina based industrial base with planned investments of \$195 million through 2017.
- GE has announced plans to expand its Hooksett, NH based facility by over 55,000 square foot at an investment of \$50 million aimed at expansion of overall production capacity to support production rate increase in-line with a growing order backlog, especially, for commercial engines led by the LEAP engine program. The Hooksett facility produces rotating engine parts for GE's commercial & military jet engines. The company held a groundbreaking ceremony at the site in May 2015.

# Key Trends & Growth Opportunities – 12

## Future of Industry to be Shaped by Technology

The future of the global commercial aviation industry is going to be shaped by technology. Key technologies include:

- **Advanced Aerodynamic Designs:** Development of advanced & optimized aerodynamic designs delivering enhanced aerodynamic efficiency while reducing overall drag on the airframe structures.
- **Material Science Innovations for Airframes & Engine Architectures:** Further commercialization & greater utilization of organic composites in the production of airframe structures delivering enhanced thrust to weight ratio besides enhanced corrosion & fatigue resistance thereby extending the airframe's operational life span. Significant usage of Ceramic Matrix Composites (CMCs) by GE for hot sections in its latest aircraft engines as against Nickel Aluminum alloys earlier has enhanced performance significantly while reducing weight and improving fuel efficiency in double digits. Pratt & Whitney's Geared Turbofan engine technology is another example of the new engine technology architecture which is driving the industry's future.
- **Enhanced Fuel Efficiency of Aircraft Engines** for reduced Environmental Impact: Enhancement of fuel efficiency for aircraft engines driven by technological innovations featuring material science innovations including the usage of Carbon Matrix Composites(CMC), specialized coatings like Super Finish & Oxide Oxide(OxOx) delivering enhanced durability, performance & fuel efficiency besides significantly enhanced maintenance intervals & built-in diagnostic capabilities.
- **Development of a range of Alternate, Renewable Fuels** to reduce environmental impact of the aviation industry.
- **Safety Enhancing Technologies** like the incorporation of Synthetic Vision Systems(SVS), based on 3D technology; which were developed originally for military aircrafts; on business jets providing enhanced visibility for night operations as well as during adverse weather conditions.
- **3D Printing Technology which is revolutionizing the Manufacturing Footprint the World Over, especially, for New Product Development**
- **Internet of Things driving the Industrial Revolution 4.0 and Digital Technologies** enabling the OEMs to offer a suite of data analytics based predictive maintenance services and solutions apart from optimization of aircraft operations.
- **Development of Turbo Electric and Hybrid propulsion technologies** by the industry for commercial aircraft operations which is likely to transform the industry during next decade.

# Key Issues, Challenges & Risk Factors

## ■ Section 6

# Key Issues, Challenges & Risk Factors - 1

## Global Macroeconomic Environment Remains Uncertain

The macroeconomic environment across traditional markets remains uncertain with the increasing environmental uncertainty & the resurgence of the wave of protectionism and onset of trade wars likely to impact world's economic growth significantly. Other regional issues, like the Government shutdown in the U.S., BREXIT in Europe and the change in the traditional, rule based world order. The World GDP is projected to grow over near term, however, it is likely to be limited, marginal and range bound. The IMF, in its latest world GDP growth projections released in January 2019, has made its second & further downward revisions in the world growth for 2019 & 2020 pegged at 3.5% & 3.6% respectively, which are 0.2% & 0.1% below the projections made in October 2018. IMF has further warned that global economic expansion is losing momentum with increased trade tariffs between China and the U.S., weakness for German auto manufacturers due to new fuel emission standards, growing risks of a 'no-deal' BREXIT, slowing down of Chinese economic growth, softening of domestic demand in Italy emanating from sovereign and financial risks, weakening sentiment in global financial markets & the contraction in Turkey. As per IMF, the Advanced countries are projected to grow by 2% in 2019 and 1.7% in 2020 while emerging economies are also projected to witness economic slowdown going forward with IMF projecting 4.5% economic growth for emerging economies in 2019 and 4.9% for 2020 following 4.6% economic growth in 2018. Chinese economy registered a growth rate of 6.6% for 2018, its slowest since the 1990 while India is projected to be the fastest growing economy in the world, at 7.5% in 2019 and 7.75% in 2020 after growing at 7.3% in 2018, and overtake China while benefitting from low crude oil prices and limited monetary tightening. China is projected to grow at 6.2% in 2019 & 2020 due to required financial tightening and the impact of trade tariffs.

BRIC markets, too, grapple with their unique set of challenges & issues spearheaded by China, which is projected to slow down in 2019 and 2020 with the ongoing trade war with the U.S. which are likely to impact economic growth significantly leading to significant deceleration of growth momentum which is likely to persist over near term. Brazilian economy continues to be on the path to stabilization following the political crisis. India, on the contrary, has been pacing up well with respect to economic reforms, deregulation & stronger policy focus on growth being on priority with GDP growth estimated at just over 7% for FY 2019-20 with increased focus & allocation towards infrastructure development likely over near term. Russian economy, too, continues to be impacted by the plummeting of oil prices and western sanctions over the Ukraine crisis, which have impacted economic growth severely. The focus, stance & timing of fiscal & monetary policy interventions in the U.S., thus, are going to be crucial for the U.S. & the global economy over near term with the FED likely to gradually raise interest rates further in 2019 and 2020.

# Key Issues, Challenges & Risk Factors - 2

## **Tightening Regulatory Emission Norms & Standards Necessitating Continued Investments towards New Technology Development**

The aviation industry's contribution to global CO2 emissions has been 2% and the IATA resolution intends to achieve carbon neutral growth under its Aviation Carbon-Neutral Growth(CNG) 2020 Strategy by the year 2020 and plans a further 50% reduction in net emissions by 2050 as against the 2005 levels with about 40% of the expected reduction likely to be achieved through the use of alternate fuels with the rest attributable to technological developments followed by pursuit of improvements in technology, operations and infrastructure development with deployment of single, unified global Market based Measures. Key targets outlined by the IATA resolution includes:

- Achieving annual fuel efficiency improvements of 1.5% through 2020
- Putting a Cap on the net emissions with the CNG2020
- Reducing net emissions by 50% through 2050 as compared with 2005

The global aviation industry needs to invest significantly towards development of new technologies, especially, in the development of a range of sustainable alternate fuel based technologies to achieve the laid out objectives amid a difficult global macroeconomic environment.

Additionally, the Paris Agreement signed in December 2015 led to an accord amongst countries to limit the increase in global temperatures to within 2 degree Celsius threshold above pre-industrial levels. Civil Aviation, however, was excluded from the last year's agreement with the implementation of measures to limit CO2 emissions from international flights likely to be voluntary for the first 5 years for certain countries as ICAO's 191 member nations have not been able to come to an agreement with respect to the defining metrics. The outcome of the path going forward is going to be crucial for the aviation industry which already has started making its way ahead with the development of hybrid and electric propulsion systems for aircrafts.

# Key Issues, Challenges & Risk Factors - 3

## Volatility in Global Crude Oil Prices & Uncertainty over Supply Emanating from the Middle East & Venezuela

There has been a significant drop & sustained weakness in the global crude oil prices over the recent years (2014-2017) emanating from continued swings in the geo-political dynamics, differential pace of economic recovery across traditional markets and uncertainty in supply arising from ongoing political uncertainty in the Middle East region. The increase in maintenance of production output by OPEC block has led to an almost 50% drop in oil prices since late 2014 which has persisted over the years with OPEC regulating production levels to take on Shale oil & Russian exports. The global oil price dynamics are complex with crude oil prices remaining volatile since August 2018 after settling down in late 2018 indicating signs of supply side impact with the prices on the rise from the start of 2019. The fluctuations in global crude oil prices have an impact on fleet utilization level & profitability across operators.

The global crude oil price dynamics include:

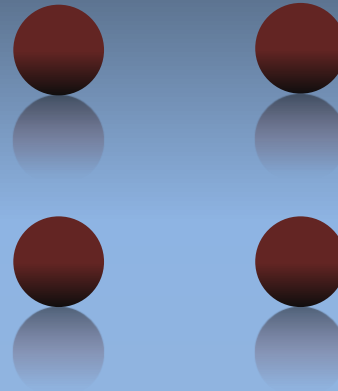
- Oil production level likely to be maintained by OPEC
- Oil exports by Iran, Libya & Russia
- Ongoing political instability, conflict & war operations in the Middle East
- Oil demand originating from South Korea & China
- U.S. economic growth & policy measures by the administration concerning oil export policy

**U.S. Oil Export Policy:** The U.S. policy stance with respect to Shale oil exports is likely to be a factor driving global crude oil prices in future as U.S. transforms itself from being a key oil importer to achieving self-sufficiency in terms of energy with the discovery of Shale oil reserves in just over a decade and is poised to become the world's largest oil producer. There has been a restriction on the U.S. crude oil's exports to international markets as per a regulation dating back to the 1970s.

However, with the U.S. poised to become one the world's largest oil producers, there has been a growing public opinion & debate towards lifting that regulation supported by multiple voice from the industry and the regulatory agencies. A new state by state analysis has shown that 18 U.S. states could gain over 5,000 jobs each in 2020 from exports of U.S. crude oil as access to foreign customers will create economic opportunities across the country. The U.S. will have to walk a very thin line, in terms, of managing domestic consumer interests as well as managing its strategic external interests in the global energy market dynamics & geo-political equation. However, given the current Trump Administration's stance & track record with respect to legacy acts, agreements and treaties it seems very much likely & highly possible.



# Force Field Analysis



## ■ Part 7

# Force Field Analysis – Global Commercial Aircraft Market

## Driving Forces

Development of Latest Engines Programs by Industry Players Featuring Next Generation Technologies Catalyzing Fleet Replacement Activity

Strong

Medium

Weak

Introduction of Latest Aircraft Programs by Industry OEMs featuring enhanced Fuel Efficiency & Capabilities Boosting Profitability & Enabling Airlines globally to Open New, Potentially Untapped Markets

Continued Strong Growth in the Commercial & Regional Aviation Segments driven by Significant & Sustained Air Traffic Growth projected through 2036

Sustained Crude Oil Price Environment boosting Profitability across Airlines

Anticipated Commercialization of Biofuels to Bolster Profitability across Airlines with Reduction in Fuel Costs likely over Medium Term

Development of New Business Models by Airlines & Significant Growth in the Low Cost Carrier Segment

Strong

Medium

Weak

## Restraining Forces

Volatility in Global Crude Oil Prices, Supply Concerns and Escalating Labour Costs Having Direct Impact on Airlines Profitability

Rapid Production Ramp-Ups by Boeing & Airbus and the Growing Trend for Vertical Integration across Key OEMs & Industry Players with In-Housing Strategy putting Pressure on Aerospace Supply Chain.

Rapid Production Ramp Up across OEMs Putting Pressure on Supplier Base with Limited Financial Resources & Risk Management Capabilities

Uncertain Global Macroeconomic Environment with Projected Slowdown of World Economy in 2019 & 2020, Trade Wars & Regional Issues like BREXIT

Low Crude Oil Price Environment over Recent Years has enabled some Airlines & Operators to bring back older, lesser fuel efficient aircrafts back into service impacting New Aircraft Sales

Tightening Environmental Regulations & Norms Requiring Industry Players to Invest significantly towards Technology Development, Eg. IATA 2020 & 2050 Guidelines

Note: The size of the arrow indicates relative strength of each active force.

# Market Outlook



## Part 8

# Strategic Market Outlook – Commercial Airplanes - 1

|      |                                                                 |      |                                                                                                                                                                                                                                                |
|------|-----------------------------------------------------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2018 | 42,730 New Commercial Airplane Deliveries worth \$6.3 Trillion  | 2037 | <b>Boeing Market Outlook</b><br><b>2018-2037</b><br>Air Traffic Growth: 4.7%<br>Fleet Growth: 3.5%<br><br><b>2017-2036</b><br>Air Traffic Growth: 4.7%<br>Cargo Traffic: 4.2%<br><b>2016-2035</b><br>Air Traffic: 4.8%<br>Airplane Fleet: 3.6% |
| 2017 | 41,030 New Commercial Airplane Deliveries worth \$6 Trillion    | 2036 |                                                                                                                                                                                                                                                |
| 2016 | 39,620 New Commercial Airplane Deliveries worth \$5,930 Billion | 2035 |                                                                                                                                                                                                                                                |

The Global Commercial Aircraft Market is in the middle of a significant growth phase driven primarily by significant growth in air traffic globally (projections indicate towards a 4.7% CAGR growth rate over the 2017-2036 period as against 4.8% projected for 2016-2035) led by emerging & traditional markets along with other strong, long-term growth drivers complemented perfectly by the introduction of new aircraft programs by almost all key industry OEMs featuring cutting-edge technologies along with commercialization of a range of incremental as well as radical innovations by key players in the global aerospace industry value chain which has been a key driver of major fleet renewals underway across airlines globally and is ultimately transpiring into significant growth avenues marked by burgeoning order backlogs for new aircraft programs across all key industry OEMs. The introduction of new wide-body aircraft programs by Boeing & Airbus; incorporating a range of performance enhancement features and delivering optimized operating economics through integration of new, highly fuel efficient engines; is enabling airlines globally to open new, untapped markets and pursue potential growth opportunities & avenues effectively.

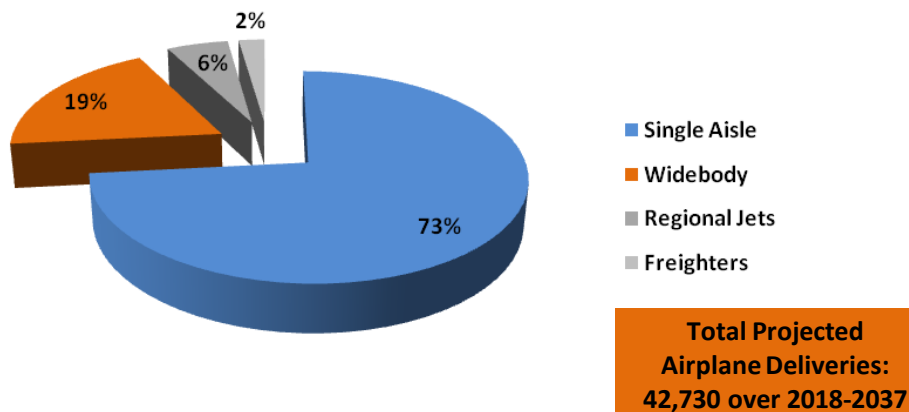
## **Steady Growth in Airline Traffic & Cargo Traffic projected by Boeing over the 2018-2037 Period with Demand for 42,730 new commercial airplanes worth \$6.3 Trillion**

The airline traffic is projected to grow at a CAGR of 4.7% as per Boeing's market outlook. Boeing further projects the global commercial passenger aircraft fleet to grow at 3.5% CAGR through 2037 with the addition of 42,730 new commercial airplanes worth \$6.3 trillion over the 2018-2037 period.

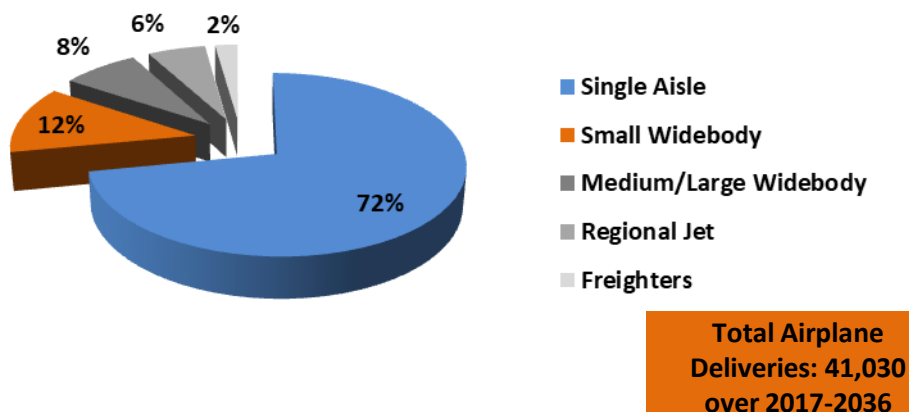
Source: Boeing Market Outlook 2018-2037

# Strategic Market Outlook – Commercial Airplanes - 2

Airplane Deliveries Split - 2018-2037



Airplane Deliveries Split in Volume Terms - 2017-2036 - Aircraft Units



Total airplane deliveries forecast for the 2018-2037 period is projected at 42,730 aircrafts, up from 41,030 for the 2017-2036 period. The single aisle/narrow body aircraft segment is projected to account for around 73% of airplane deliveries in volume terms for 2018-2037 as against 72% for 2017-2036 while the wide-body aircrafts are projected to account for around 19% of deliveries in volume terms over the 2018-2037 down from 20% projected for the 2017-2036 period.

Source: Boeing Market Outlook 2018-2037

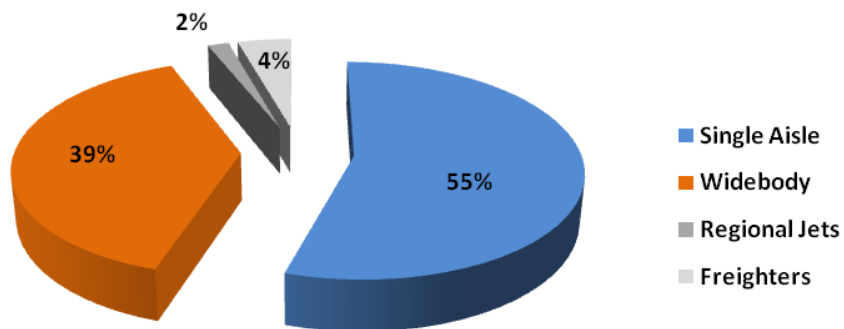
# Strategic Market Outlook – Commercial Airplanes - 3

## Narrow Body Aircraft Segment to Lead Airplane Deliveries in Volume Terms; Wide Body Segment to Lead in Value Terms

The overall value of airplane deliveries over the 2018-2037 period is projected to be \$6 trillion, a growth of \$300 billion over the previous forecast by Boeing which had put the value of total airplane deliveries for the 2017-2036 period at \$6 trillion in the previous year's forecast.

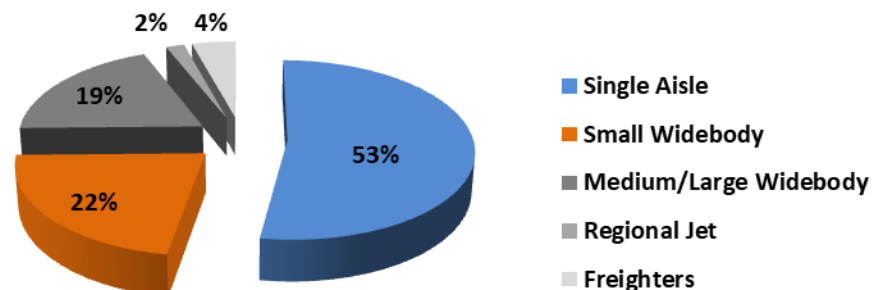
The single aisle/narrow body aircraft segment is projected to account for around 55% of airplane deliveries in value terms for the 2018-2037 period, up from 53% projected by Boeing in its 2017-2036 forecast. The wide-body aircrafts, on the contrary, are projected to account for around 39% of deliveries in value terms over the 2018-2037 period, down from 41% projected for the 2017-2036 period. The forecast for the share of regional jets and cargo aircrafts in total pie remains the same at 2% & 4% respectively.

Airplane Deliveries Split - Value - 2018-2037



Total Airplane Deliveries: 42,730 over 2018-2037 worth \$6.3 Trillion

Airplane Deliveries Split in Value Terms - 2017-2036



Total Airplane Deliveries: 41,030 over 2017-2036 worth \$6 Trillion

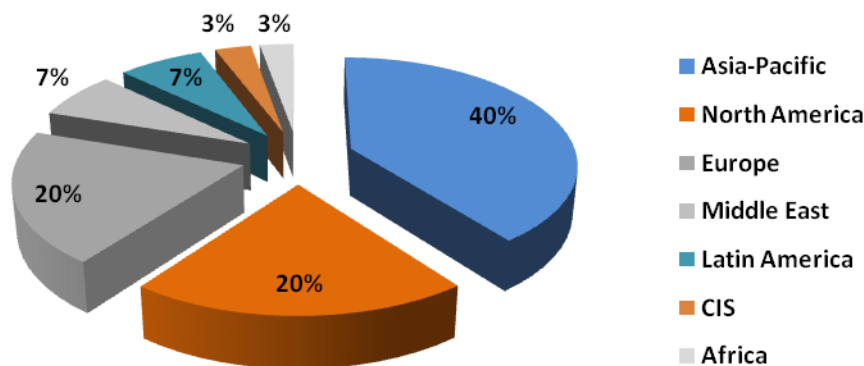
Source: Boeing Market Outlook 2018-2037

# Strategic Market Outlook – Commercial Airplanes - 4

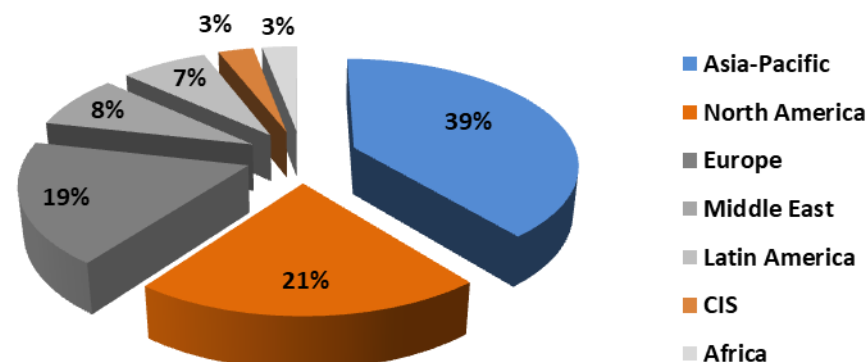
## Asia-Pacific Region to Lead Demand for Commercial Airplanes through 2037 in Both Volume & Value Terms - 1

The Asia-Pacific region is likely to be the pivot around which the Global Commercial Aircraft market is projected to revolve through 2037. The APAC region is projected to lead the demand for new commercial airplanes through 2037 with 40% of airplane deliveries estimated at 16,930 aircrafts (as against 16,050 aircraft deliveries projected in the 2017-2036 forecast). Around 20% of aircraft deliveries in volume terms are projected to be made to the North America region over the 2018-2037 period standing at 8,800 aircraft deliveries down from 21% projected in the 2017-2036 forecast. About 20% of the airplane deliveries in volume terms are projected to be made to Europe over the 2018-2037 period translating into 8490 aircraft deliveries over the forecast period. The APAC region is estimated to account for almost 37% of the global in-service fleet in 2037 as against its 30% share for 2017. North America's fleet share, on the contrary, is likely to decrease to 21% by 2037 as against 30% in 2017.

**Airplane Deliveries by Region - Units-2018-2037**



**Airplane Deliveries by Region - In Units - 2017-2036**



Source: Boeing Market Outlook 2018-2037

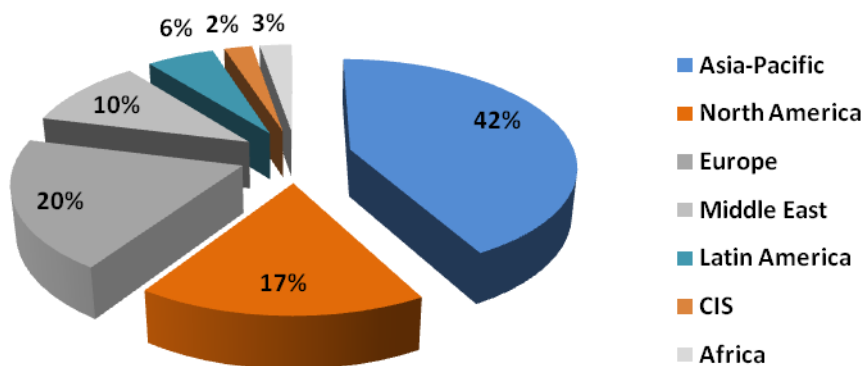


# Strategic Market Outlook – Commercial Airplanes - 5

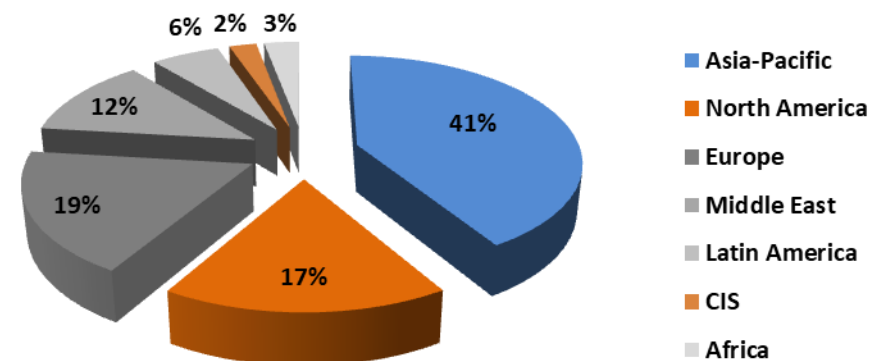
## Asia-Pacific Region to Lead Demand for Commercial Airplanes through 2037 in Both Volume & Value Terms - 2

The Asia-Pacific region is likely to command the largest share of the commercial aircraft deliveries in value terms over the 2018-2037 period, as per Boeing, with the region likely to account for almost 42% of the deliveries in value terms estimated at \$2.67 trillion. APAC is likely to be followed by Europe with a value share of 20% translating into \$1.24 trillion. North America region is projected to bag the third spot with its share of deliveries (17% value share) estimated to be worth \$1.1 trillion, less than half of the APAC's share.

Airplane Deliveries by Region - In \$Billion -  
2018-2037



Airplane Deliveries by Region - In \$Billion -  
2017-2036



Source: Boeing Market Outlook 2018-2037

# Strategic Market Outlook – Aviation Turbofan Engines - 1

A total of **84,021** new Commercial Aviation Turbofan Engines worth **\$1.24 Trillion** are likely to be produced by the industry over the 2018-2037 period, as per forecasts by the Japan Aircraft Development Corporation (JADC), a consortium of Japanese Aircraft Industries for the development of commercial airplanes.

**2018**

**84,021 New Aviation Turbofan Engines for  
Commercial Aviation to be Produced worth  
\$1.24 Trillion**

**2037**

Source:  
Japan Aircraft  
Development  
Corporation

**2017**

**83,167 New Aviation Turbofan Engines for  
Commercial Aviation to be Produced worth  
\$1.202 Trillion**

**2036**

Source:  
Japan Aircraft  
Development  
Corporation

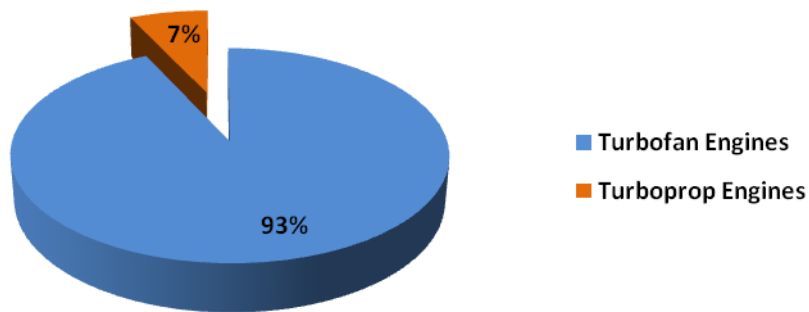
Turbofan engines are projected to account for almost 93% of the total demand for commercial aviation turboprop engines over the forecast period (in unit terms) while Turboprops are likely to account for 7% of the total demand, as per forecasts by JADC.

# Strategic Market Outlook – Aviation Turbofan Engines - 2

## Turbofan Engines Demand Projected to Lead in Units as well as Value Terms over the 2018-2037 Period - 1

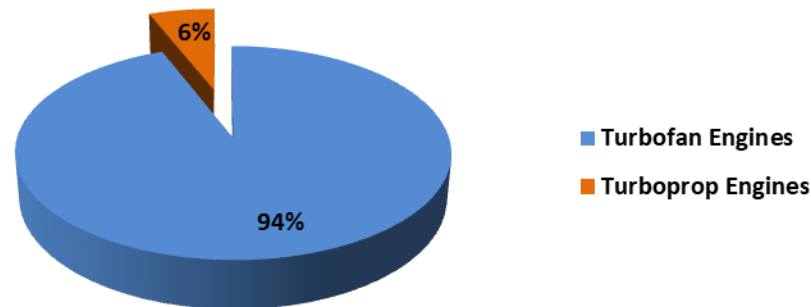
The Turbofan engine segment is the largest in the Global Aviation Gas Turbine Market and is projected to lead, in Units as well as Value terms, over the 2018-2037 period. The turbofans segment is projected to account for over 93% of the total demand for commercial aviation gas turbines over the forecast period, translating into demand for over 78,083 turbofan engines. The Turboprop segment is projected to require 5,938 turboprop engines over the 2018-2037 period, up from the forecast of 4,949 turboprop engines forecasted in 2017 for the 2017-2036 period. The value of turboprop engine deliveries is projected to be worth \$13 billion over the 2018-2037 period as against \$10.9 billion forecasted in 2017 for the 2017-2036 period. Turbofans are projected to account for around 98.3% of the total value of engines required by the commercial aviation sector over the forecast period translating into a whopping \$1.22 Trillion.

Global Commercial Aviation Engine Demand  
Forecast - 2018-2037 - Unit Share



Total Forecast for Engines  
Demand – In Units: **84,021 Units**  
(2018-2037)

Global Commercial Aviation Engine Demand  
Forecast - 2017-2036 - Unit Share



Total Forecast for Engines  
Demand – In Units: **83,167 Units**  
(2017-2036)

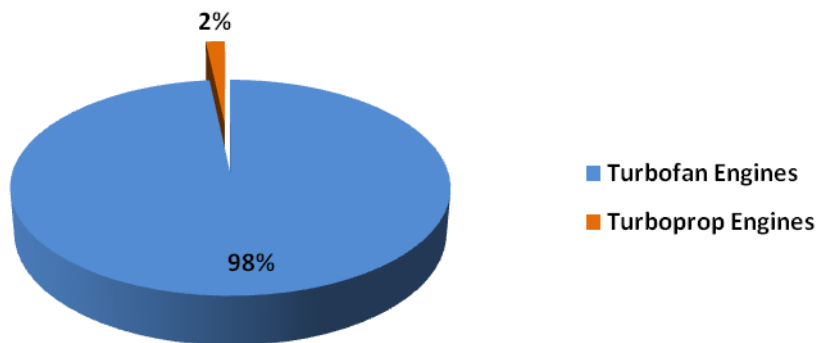
Data source: JADC  
Noealt Analysis

# Strategic Market Outlook – Aviation Turbofan Engines - 3

## Turbofan Engines Demand Projected to Lead in Units as well as Value Terms over the 2018-2037 Period - 2

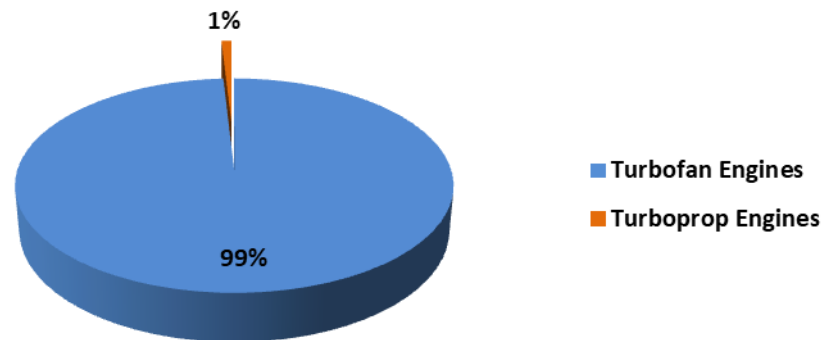
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Global Commercial Aviation Engine Demand  
Forecast - 2018-2037 - Value Share



Total Forecast for Engines  
Demand – In Value: **\$1.24 Trillion**

Global Commercial Aviation Engine Demand  
Forecast - 2017-2036 - Value Share



Total Forecast for Engines  
Demand - in Value: **\$1.202 Trillion**

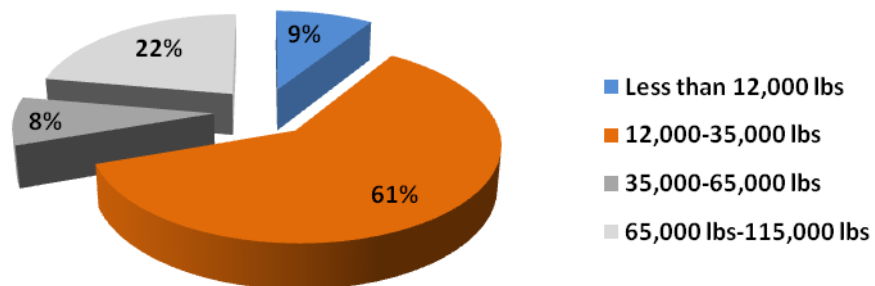
Data source: JADC  
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# Strategic Market Outlook – Aviation Turbofan Engines - 4

## Commercial Aviation Turbofan Engines Demand by Market Segments – Engine Thrust Class – In Engine Units - 1

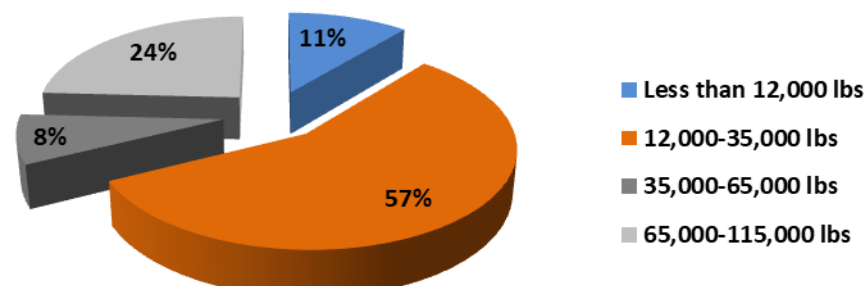
The 12,000-35,000 lbs thrust segment; which powers most narrow body aircraft programs, most noticeably the Airbus A320 and Boeing 737 aircraft families & the Bombardier C-Series (now Airbus A220); is projected to account for over 61%(46,977 engines) of new commercial aviation turbofan engine deliveries (in unit terms) over the 2018-2037 period as against 57% (44,298 engines) forecasted in 2017 for the 2017-2036 period. The 65,000 to 115,000 lbs. thrust class; which powers the medium/large wide body aircraft segment; is projected to account for the second largest number of turbofan engine deliveries at 22% (17,398 engines) for the 2018-2037 period, in unit terms, as against 24% (18,908 engines) forecasted in 2017 for the 2017-2036 period.

**Turbofan Engines Demand Forecast for Commercial Aviation by Thrust Class - 2018-2037 - Units**



Total Forecast for Turbofan Engines Demand – In Units:  
**78,083 Units**

**Turbofan Engines Demand Forecast for Commercial Aviation by Thrust Class - 2017-2036 - In Units**



Total Forecast for Turbofan Engines Demand – In Units:  
**78,218 Units**

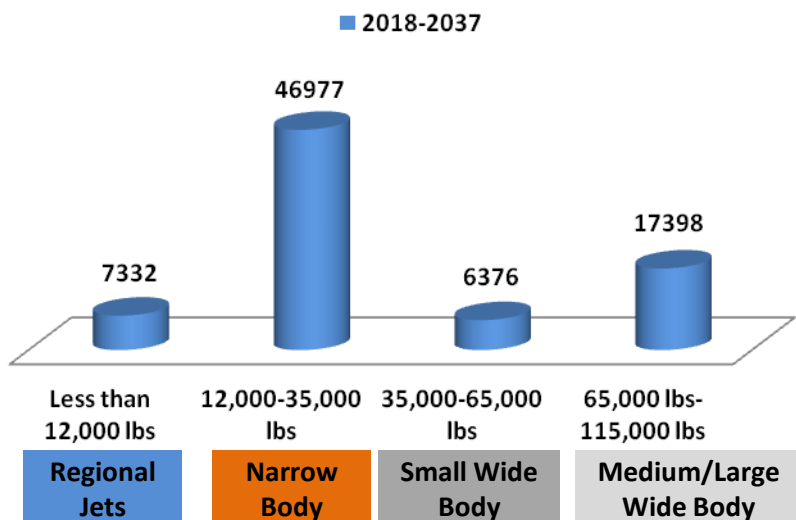
Data source: JADC  
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# Strategic Market Outlook – Aviation Turbopfan Engines - 5

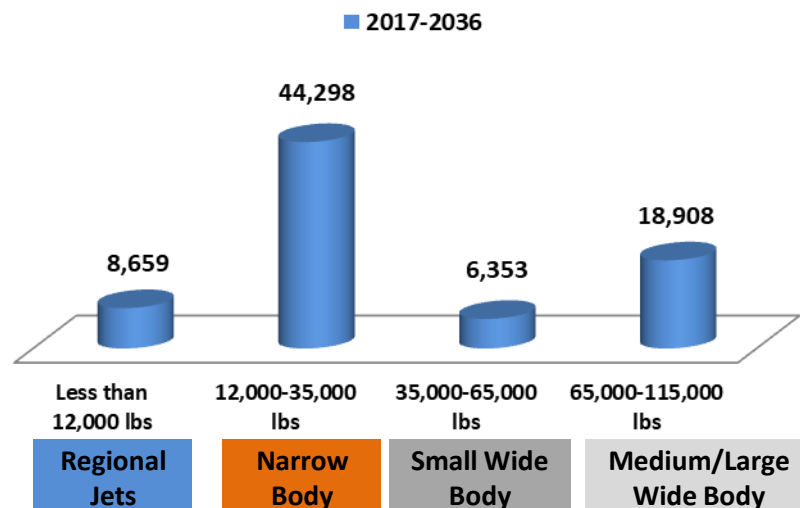
## Commercial Aviation Turbopfan Engines Demand by Market Segments – Engine Thrust Class – In Engine Units - 2

The 12,000-35,000 lbs thrust segment; which powers most narrow body aircraft programs, most noticeably the Airbus A320 and Boeing 737 aircraft families & the Bombardier C-Series; is projected to account for over 57% (44,298 engines) of new commercial aviation turbopfan engine deliveries (in unit terms) over the 2017-2036 period. The 65,000 to 115,000 lbs. thrust class; which powers the medium/large wide body aircraft segment; is projected to account for the second largest number of turbopfan engine deliveries at 24% (18,908 engines), in unit terms, over the forecast period. Collectively, the narrow body and medium/large wide body aircraft segments are estimated to account for around 81% of total turbopfan engine deliveries in units over the 2017-2036 period.

**Turbopfan Engines Demand Forecast for Commercial Aviation by Thrust Class - 2018-2037 - Units**



**Turbopfan Engines Demand Forecast for Commercial Aviation by Thrust Class - 2017-2036 - In Units**



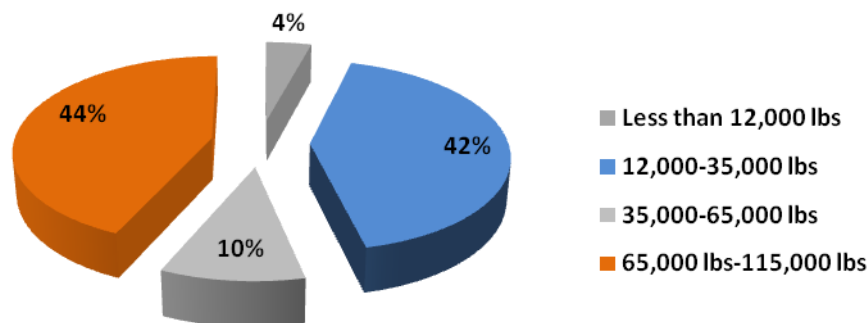
Data source: JADC  
Noealt Analysis

# Strategic Market Outlook – Aviation Turbofan Engines - 6

## Commercial Aviation Turbofan Engines Demand by Market Segments – Engine Thrust Class – In Value Terms (\$ Billion) - 1

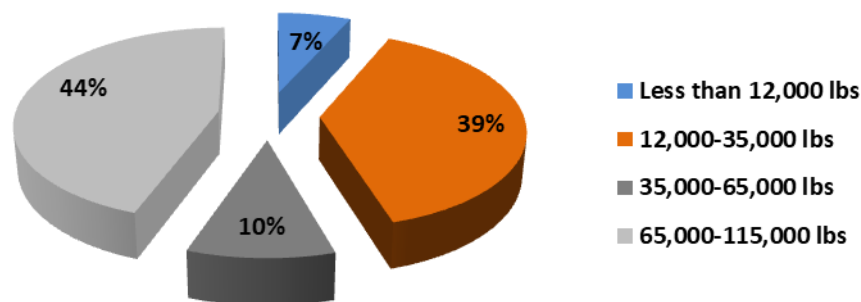
When looked in value term, the 65,000-115,000 lbs thrust segment; which powers most medium/large wide body aircraft program, is likely to be the largest & most significant turbofan engines segment projected to account for almost 44% of new commercial aviation turbofan engine deliveries (in value terms) worth \$531 billion over the 2018-2037 as well as 2017-2036 periods. The 12,000 to 35,000 lbs. thrust class; which powers the narrow body aircraft segment; is projected to account for the second largest number of turbofan engine deliveries in value terms at 42% for the 2018-2037 period worth \$517 billion up from 39% worth \$461 billion forecasted for the 2017-2036 period. Collectively, the narrow body and medium/large wide body aircraft segments are estimated to account for around 86% of total turbofan engine deliveries, in value terms, worth \$1048 billion for the 2018-2037 period as against 83% and worth \$992 billion forecasted for the 2017-2036 period.

**Turbofan Engines Demand Forecast for Commercial Aviation by Thrust Class - 2018-2037 - In Value Terms**



**Total Forecast for Turbofan Engines Demand – In Value: \$1.237 Trillion**

**Turbofan Engines Demand Forecast for Commercial Aviation by Thrust Class - 2017-2036 - In Value Terms**



**Total Forecast for Turbofan Engines Demand – In Value: \$1.202 Trillion**

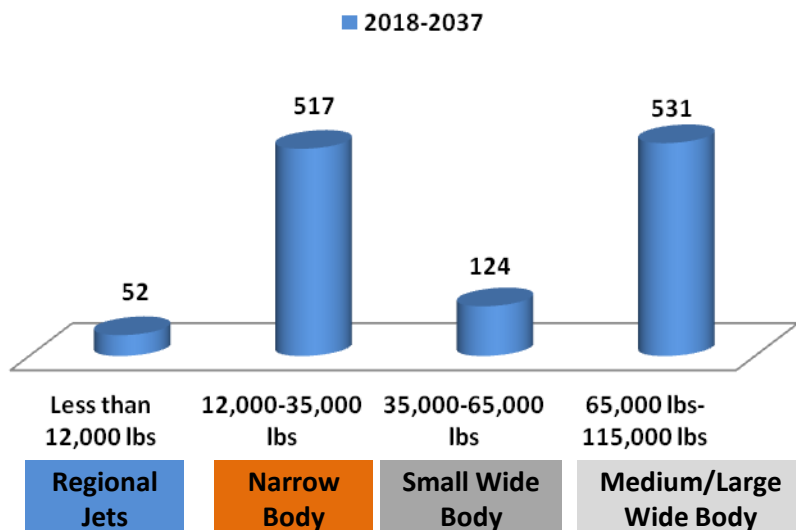


# Strategic Market Outlook – Aviation Turbopfan Engines - 7

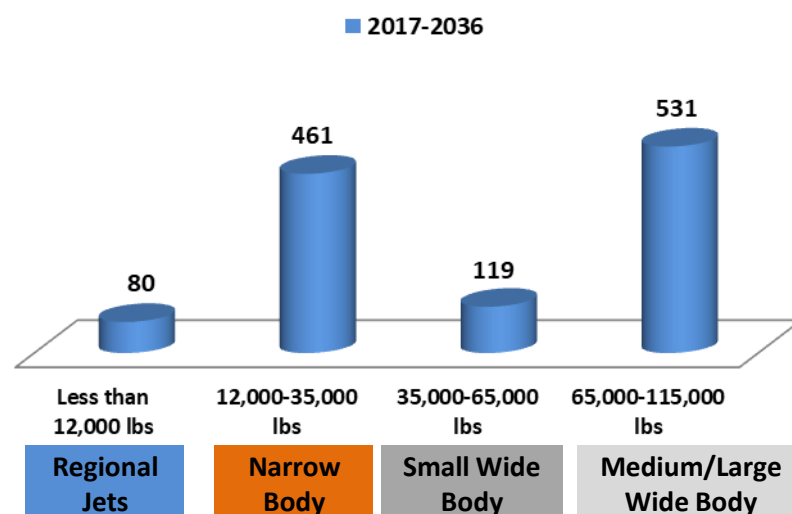
## Commercial Aviation Turbopfan Engines Demand by Market Segments – Engine Thrust Class – In Value Terms (\$ Billion) - 2

When looked in value term, the 65,000-115,000 lbs thrust segment; which powers most medium/large wide body aircraft program, is likely to be the largest & most significant turbopfan engines segment projected to account for almost 44% of new commercial aviation turbopfan engine deliveries (in value terms) worth \$531 billion over the 2018-2037 as well as 2017-2036 periods. The 12,000 to 35,000 lbs. thrust class; which powers the narrow body aircraft segment; is projected to account for the second largest number of turbopfan engine deliveries in value terms at 42% for the 2018-2037 period worth \$517 billion up from 39% worth \$461 billion forecasted for the 2017-2036 period. Collectively, the narrow body and medium/large wide body aircraft segments are estimated to account for around 86% of total turbopfan engine deliveries, in value terms, worth \$1048 billion for the 2018-2037 period as against 83% and worth \$992 billion forecasted for the 2017-2036 period.

**Turbopfan Engines Demand Forecast for Commercial Aviation by Thrust Class - 2018-2037 - In \$Billion**



**Turbopfan Engines Demand Forecast for Commercial Aviation by Thrust Class - 2017-2036 - In \$Billion**

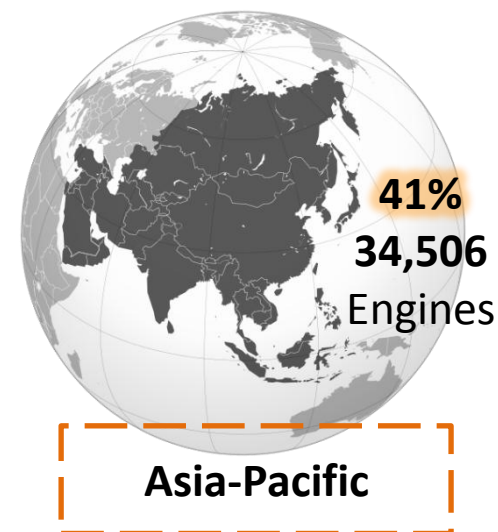
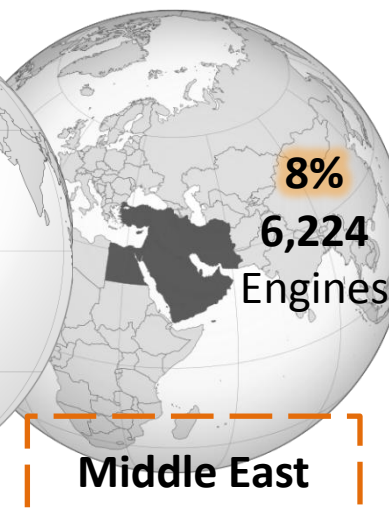
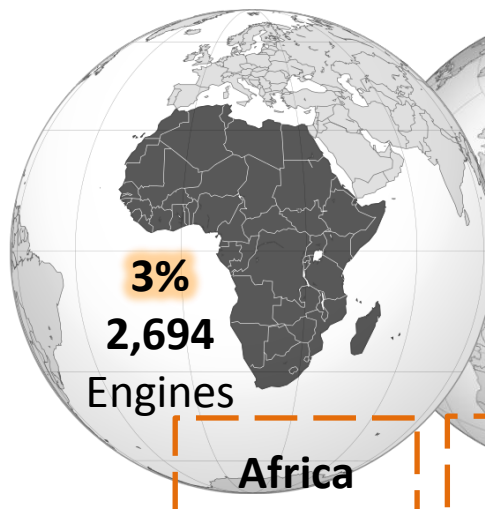
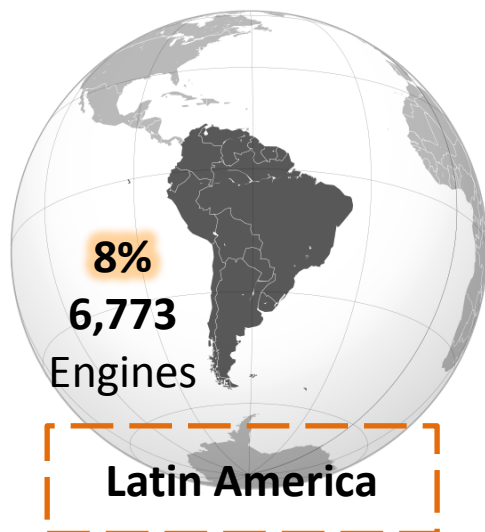
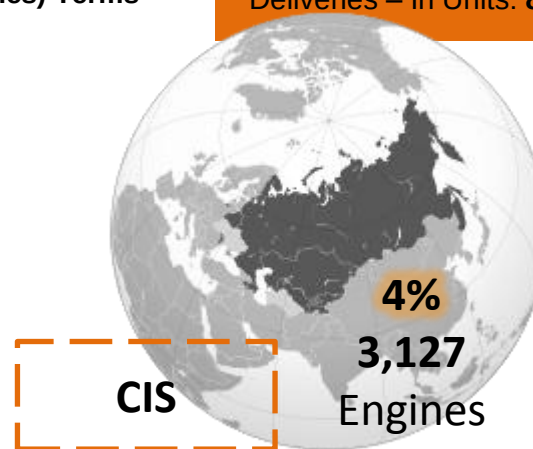
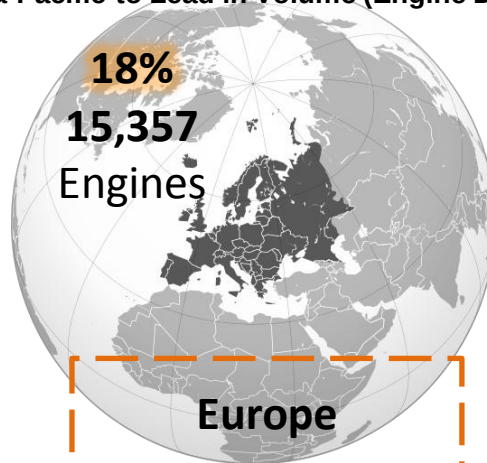
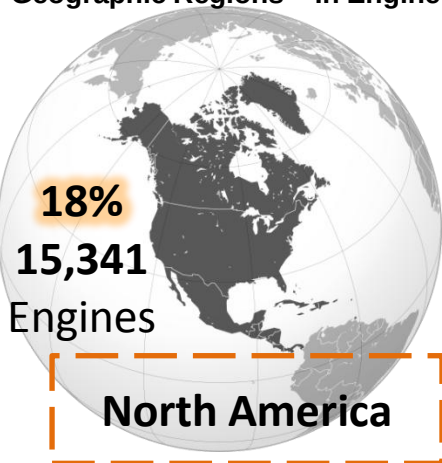


# Strategic Market Outlook – Aviation Turbopfan Engines – 8

## Engine Delivery Forecasts by Region – Units – 2018-2037

Commercial Aviation Engine Deliveries(including Turbopfans & Turboprops) Forecast Split by Geographic Regions – In Engine Units – Asia-Pacific to Lead in Volume (Engine Deliveries) Terms

Total Forecast for Engines Deliveries – In Units: **84,021 Units**



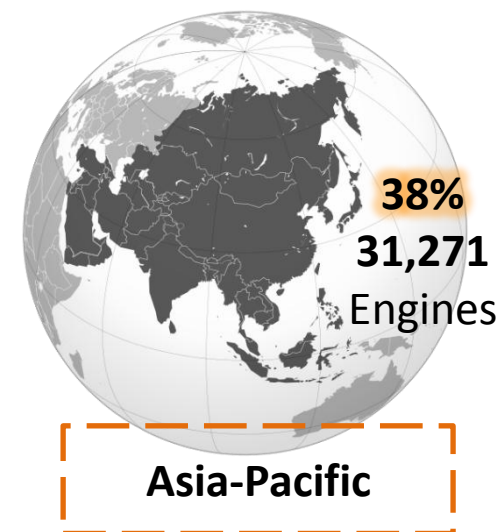
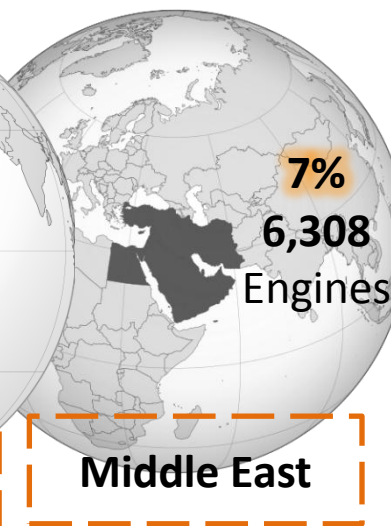
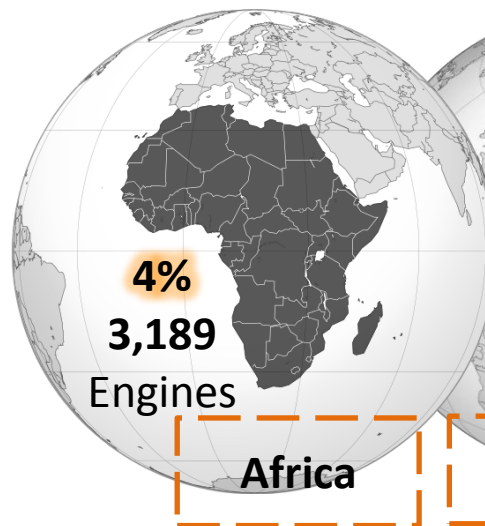
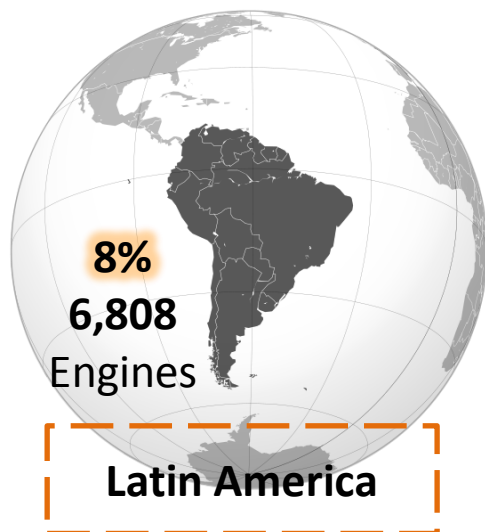
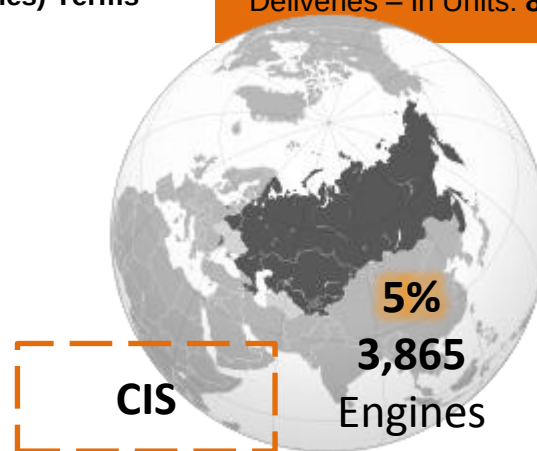
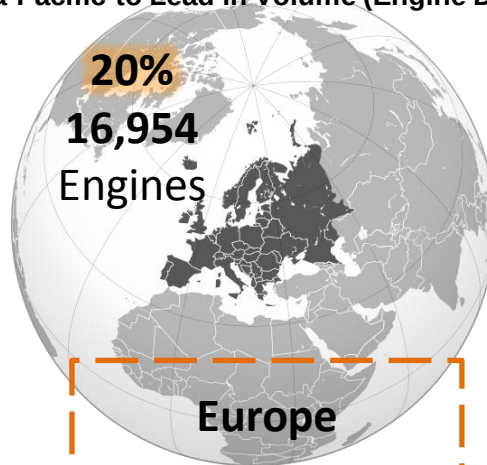
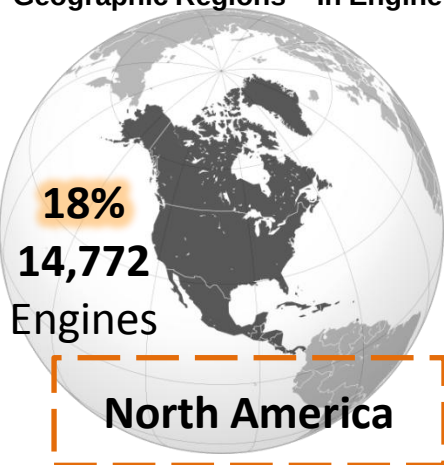
Data source: JADC  
Noealt Analysis

# Strategic Market Outlook – Aviation Turbofan Engines – 9

## Engine Delivery Forecasts by Region – Units – 2017-2036

Commercial Aviation Engine Deliveries(including Turbofans & Turboprops) Forecast Split by Geographic Regions – In Engine Units – Asia-Pacific to Lead in Volume (Engine Deliveries) Terms

Total Forecast for Engines Deliveries – In Units: **83,167 Units**



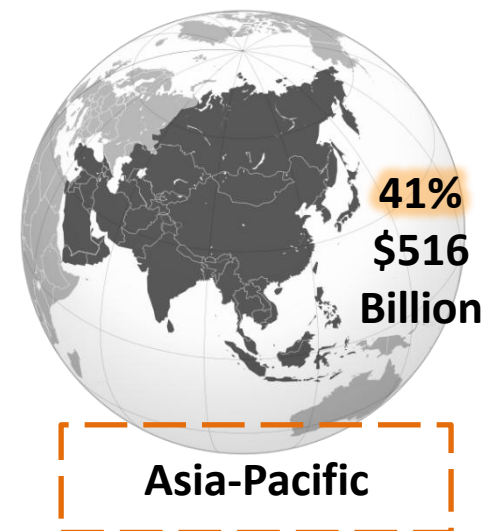
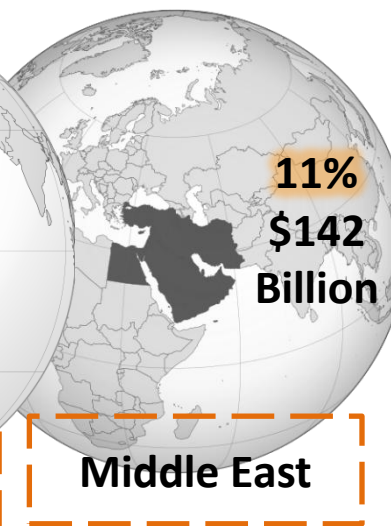
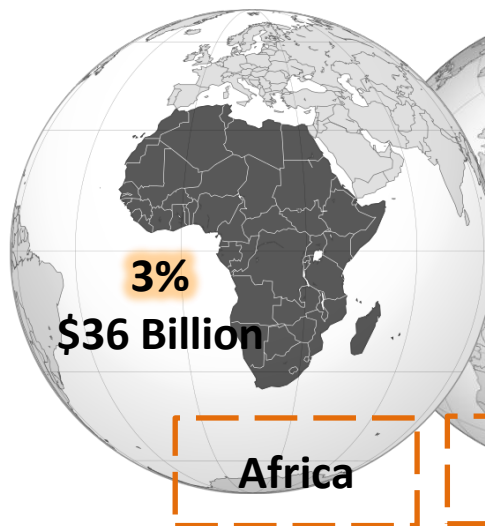
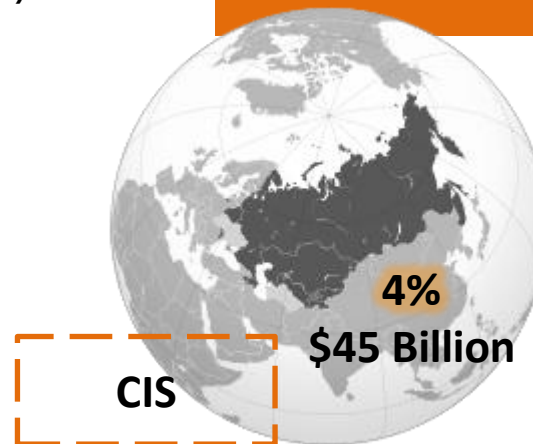
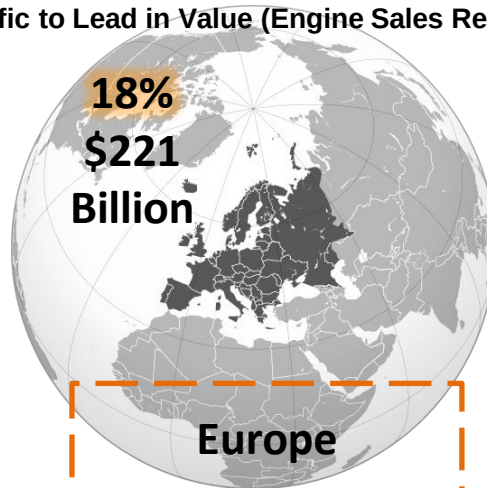
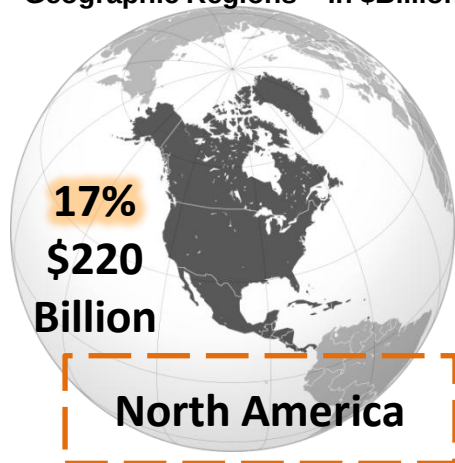
Data source: JADC  
Noealt Analysis

# Strategic Market Outlook – Aviation Turbofan Engines – 10

## Engine Delivery Forecasts by Region – Value - \$Billion – 2018-2037

Commercial Aviation Engine Deliveries(including Turbofans & Turboprops) Forecast Split by Geographic Regions – In \$Billion - Asia-Pacific to Lead in Value (Engine Sales Revenues) Terms

Total Forecast for Engines Deliveries – In Value: **\$1.237 Trillion**



Data source: JADC  
Noealt Analysis

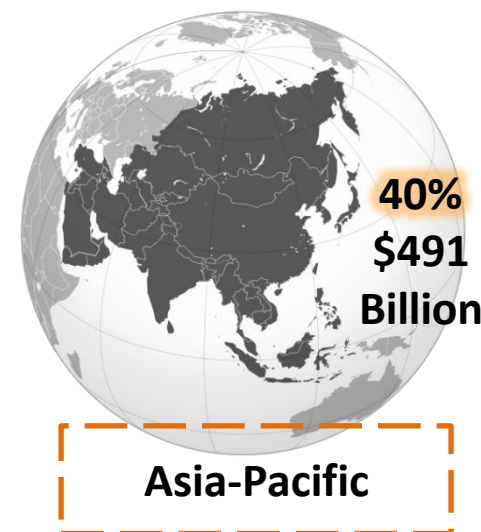
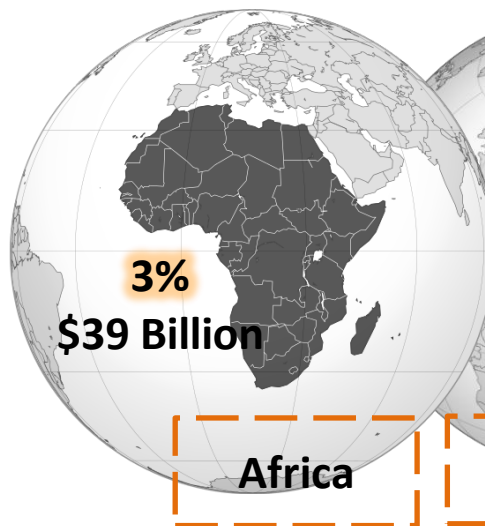
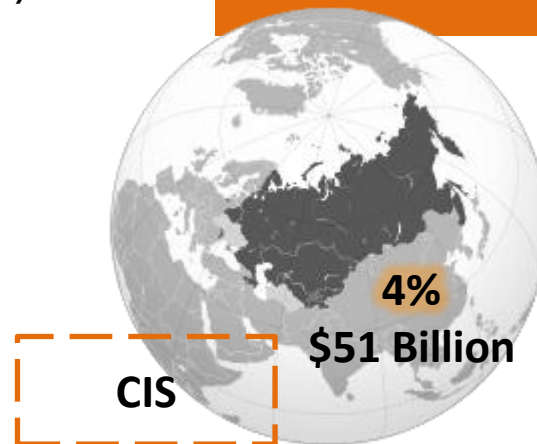
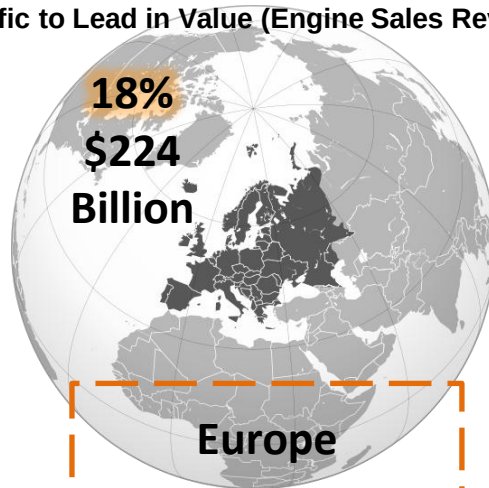
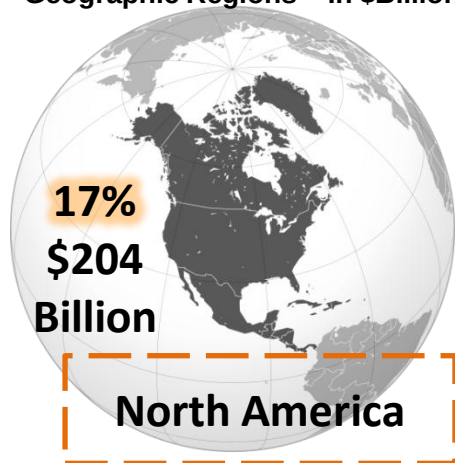


# Strategic Market Outlook – Aviation Turbofan Engines – 11

## Engine Delivery Forecasts by Region – Value - \$Billion – 2017-2036

Commercial Aviation Engine Deliveries(including Turbofans & Turboprops) Forecast Split by Geographic Regions – In \$Billion - Asia-Pacific to Lead in Value (Engine Sales Revenues) Terms

Total Forecast for Engines Deliveries – In Value: **\$1.202 Trillion**



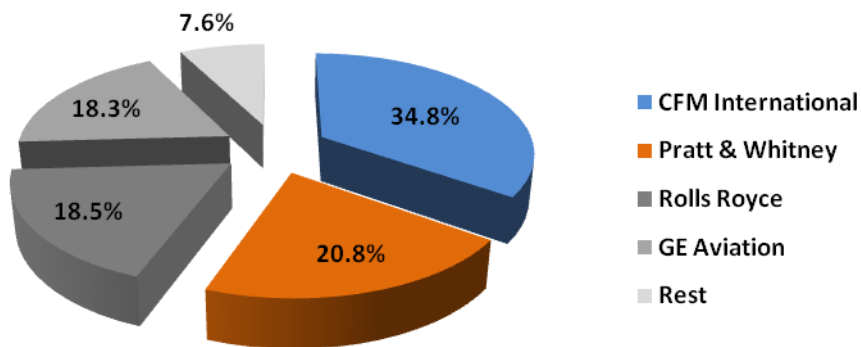
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Noealt Analysis

# Strategic Market Outlook – Aviation Turbofan Engines - 12

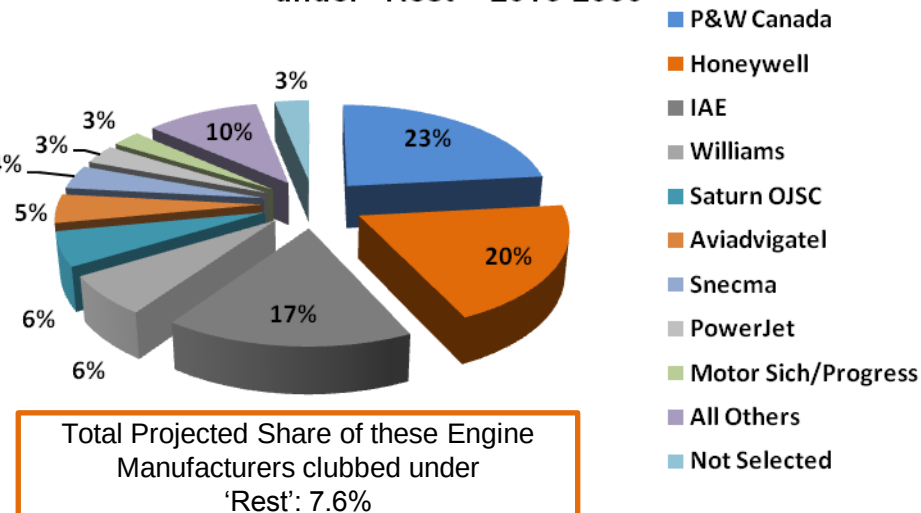
## Projected Market Shares for Aviation Turbofan Engine Manufacturers for the 2016-2030 Period (Value of Engines Production)

The Global Aviation Turbofan Engines Market is projected to be dominated by CFM International and Pratt & Whitney over the 2016-2030 period with the two engine manufacturers projected to collectively account for over 55% of the market in value terms over the forecast period. CFM is projected to clearly lead the market with a 35% production value share given strong positioning on a number of leading aircraft programs. Pratt & Whitney is projected to follow suit with a 21% production value share with the PW1000G Geared Turbofan Engine family (GTF) gaining ground & making inroads with presence over a number of key, upcoming aircraft programs in the narrow body & regional aircraft segments. Rolls Royce and GE Aviation are projected to collectively account for over 37% of engine production value share split almost equally amongst these two industry giants. Collectively, these 4 players are projected to account for over 92% of the engine production value share over the forecast period.

Aviation Turbofan Engines - Projected Production Value Share by Manufacturers - 2016-2030



Share of Remaining Manufacturers Clubbed under "Rest"- 2016-2030



## About Noealt Corporate Services

Noealt Corporate Services provides cutting-edge strategic insights as well as performance & growth solutions in the twin domains of Integrated Intelligence & Research as well as Organizational Development & Intervention to a client roster comprising of Global Fortune 500 Organizations, Key Global & Regional Players, Big 4 Consulting Firms, Key Global B-Schools, Universities & Educational Institutions and SMEs across a broad range of Key Global Industries & Sectors. By extending an array of products & services solutions configured on a perfect blend of proven & innovative as well as cutting-edge capabilities and practices; Noealt enables players across these industries & clients to maintain as well as enhance their lead & competitiveness over peers & competitors while simultaneously focusing on multi-dimensional growth ultimately transpiring into tangible business results & effectively converging with the overall business growth trajectory.

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**Integrated Intelligence & Research Solutions:** Syndicated Competitive & Market Intelligence as well as Business & Market Intelligence Solutions and Publications in addition to Custom Research Services & Solutions.

- Business & Industry Intelligence
- Market Intelligence & Research
- Competitive Intelligence

**Organizational Development Interventions** through People, Process, Potential and Performance Solutions.



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